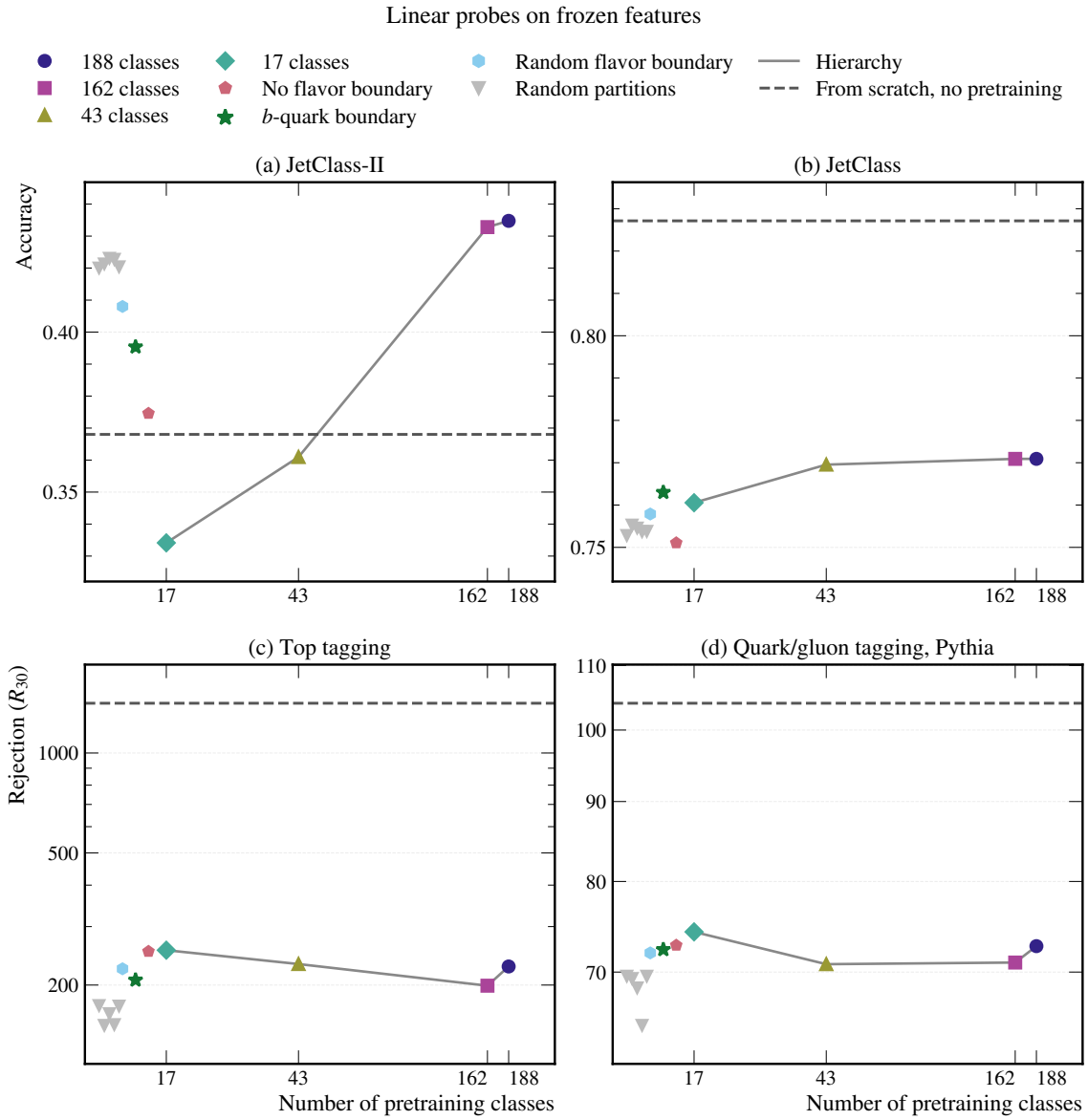


Figure 1: Linear probes on the frozen features against the number of pretraining classes: accuracy on class-balanced JetClass-II and on JetClass, and background rejection R_{30} at 30% signal efficiency on top and quark/gluon (Pythia) tagging, log axis. Markers are means over runs and the grey line (hierarchy) joins the four label sets. The 17-class random partitions, three with controlled flavor boundaries and five plain, are drawn just left of 17. The dashed line is the network trained from scratch.

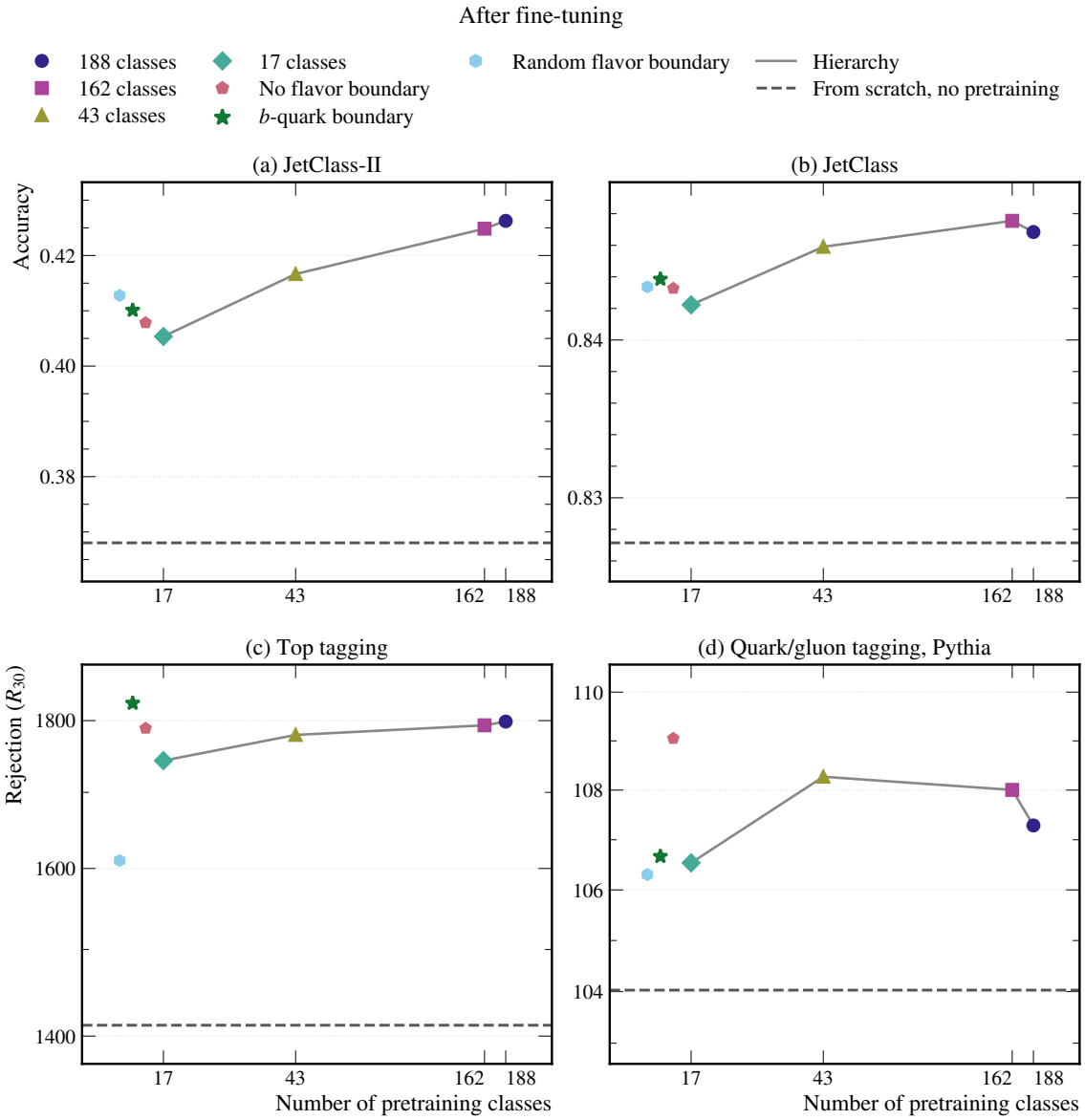


Figure 2: Fine-tuned models against the number of pretraining classes: accuracy on class-balanced JetClass-II and on JetClass, and background rejection R_{30} at 30% signal efficiency on top and quark/gluon tagging. Log axes on (c, d). Markers are means over runs and the grey line (hierarchy) joins the four label sets. The three 17-class partitions with controlled flavor boundaries are drawn just left of 17. The dashed line is the network trained from scratch.

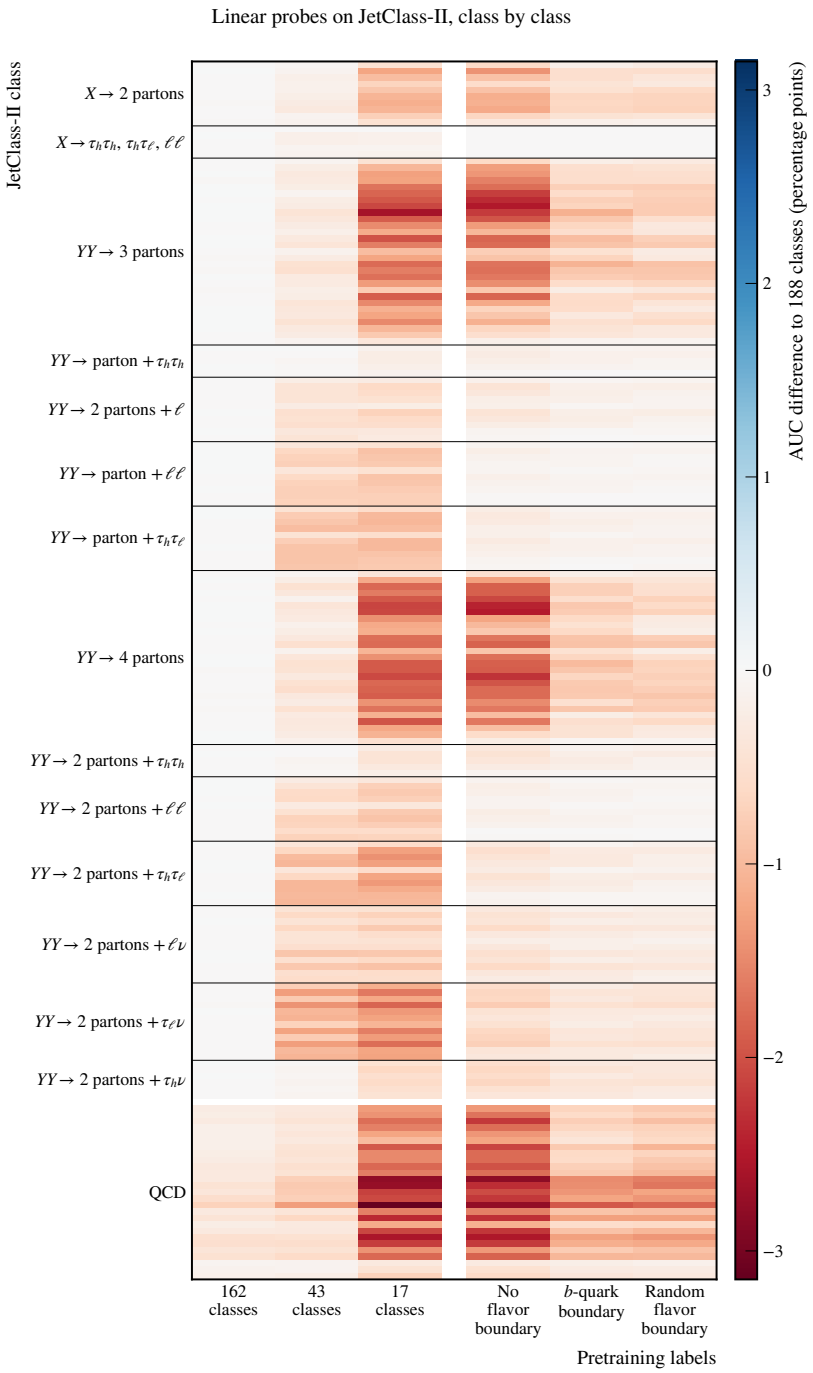


Figure 3: Difference in per-class AUC between each model and the 188-class model, in percentage points, for the 188 classes of class-balanced JetClass-II, from linear probes. Columns are the 162-, 43- and 17-class label sets and the three random partitions with controlled flavor boundaries. Rows are grouped by decay topology, with the QCD classes at the bottom. Values are means over runs; red marks an AUC below that of the 188-class model. Every class's values are in the data release.

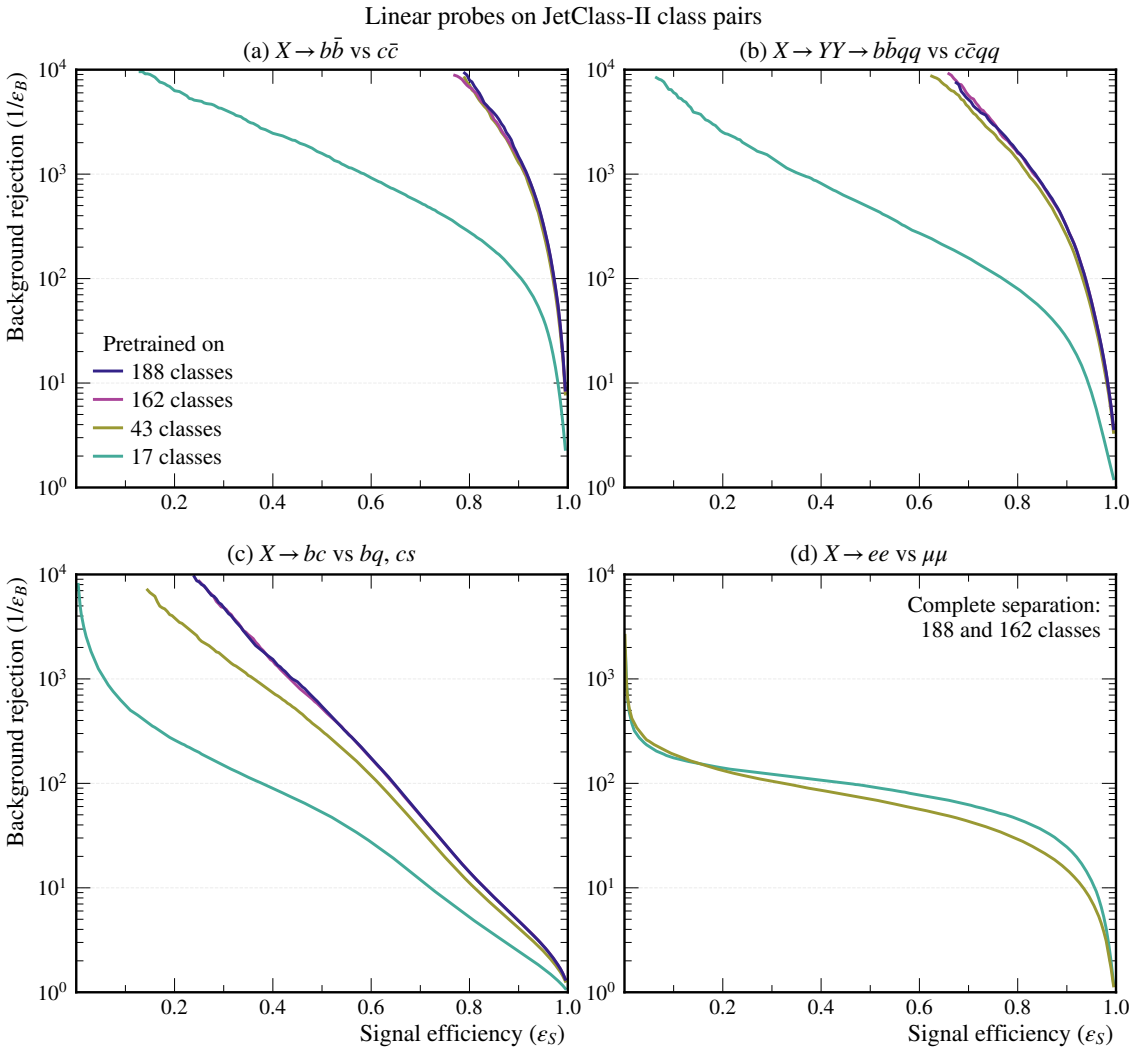


Figure 4: Background rejection against signal efficiency for four pairs of JetClass-II classes that differ only in flavor, from linear probes on networks pretrained with 188, 162, 43 and 17 classes. The first class named is the signal. Curves are means over three runs and end where fewer than 20 background jets remain; models with fewer at every efficiency are named in the panel. In (a, b) the three finest label sets nearly coincide.

Table 1: JetClass, in the form of ParT's Table 1: accuracy, AUC averaged over all pairs of classes (one against one, ParT's convention; on JetClass's equal class sizes it equals the one-against-rest average, checked), and for each signal class the rejection of QCD jets at 50% signal efficiency (99% for $H \to \ell\nu qq'$, 99.5% for $t \to b\ell\nu$) on $D_S = g_S/(g_S + g_{\mathrm{QCD}})$. Published rows: trained from scratch on 2 million jets and scored on ParT's 20-million-jet test set (ParT on 100 million jets reaches an accuracy of 0.861 and an AUC of 0.9877); ours fine-tuned on one million jets and scored on two million (the AUC on 500,000 of them). Mean ± standard deviation over the three pretraining runs. From scratch: three seeds. Bold: the best value of each column, and every value within one standard deviation of it (the larger of the two; two runs count with their mean). †: fewer than 20 QCD jets pass in at least one run, so that run's rejection has a counting uncertainty above 22%; these columns carry no claim.

	jets	accuracy	AUC	rejection of QCD
				$H \to b\bar{b}$ R_{50}
ParticleNet	2 M	0.828	0.9820	5,540
ParT	2 M	0.836	0.9834	5,587
this work (Sophon's network, fine-tuned)				
from scratch	1 M	0.8271 ± 0.0012	0.9819 ± 0.0002	$4{,}920 \pm 190$
188 classes	1 M	$\mathbf{0.8468 \pm 0.0006}$	0.9854 ± 0.0001	$\mathbf{6{,}600 \pm 330}$
162 classes	1 M	$\mathbf{0.8475 \pm 0.0009}$	$\mathbf{0.9856 \pm <0.0001}$	$6{,}060 \pm 0$
43 classes	1 M	0.8459 ± 0.0002	0.9853 ± 0.0001	$\mathbf{6{,}500 \pm 500}$
17 classes	1 M	0.8422 ± 0.0005	0.9847 ± 0.0001	$\mathbf{6{,}670 \pm 220}$

	jets	rejection of QCD			
		$H \to c\bar{c}$ R_{50}	$H \to gg$ R_{50}	$H \to 4q$ R_{50}	$H \to \ell\nu qq'$ R_{99}
ParticleNet	2 M	1,681	90	662	1,654
ParT	2 M	1,982	93	761	1,609
this work (Sophon's network, fine-tuned)					
from scratch	1 M	$1{,}580 \pm 60$	88.3 ± 3.2	670 ± 40	$1{,}270 \pm 100$
188 classes	1 M	$\mathbf{2{,}650 \pm 260}$	$\mathbf{104 \pm 4}$	$\mathbf{1{,}091 \pm 15}$	$\mathbf{3{,}100 \pm 400}$
162 classes	1 M	$\mathbf{2{,}820 \pm 40}$	$\mathbf{101.8 \pm 1.5}$	$\mathbf{1{,}150 \pm 70}$	$\mathbf{3{,}070 \pm 330}$
43 classes	1 M	$2{,}610 \pm 160$	$\mathbf{103.5 \pm 1.3}$	$1{,}030 \pm 40$	$\mathbf{3{,}160 \pm 280}$
17 classes	1 M	$2{,}390 \pm 40$	$\mathbf{101 \pm 6}$	$1{,}049 \pm 17$	$2{,}800 \pm 200$

		rejection of QCD			
	jets	$t \to bqq'$ R_{50}	$t \to b\ell\nu$ $R_{99.5}$	$W \to qq'$ R_{50}	$Z \to q\bar{q}$ R_{50}
ParticleNet	2 M	4,049	4,673	260	215
ParT	2 M	6,061	4,474	307	236
this work (Sophon's network, fine-tuned)					
from scratch	1 M	$6{,}100 \pm 600$	$3{,}940 \pm 320$	272 ± 6	215 ± 8
188 classes	1 M	$12{,}900 \pm 1{,}600^{\dagger}$	$9{,}400 \pm 1{,}500^{\dagger}$	374 ± 8	$\mathbf{312 \pm 6}$
162 classes	1 M	$14{,}100 \pm 3{,}500^{\dagger}$	$7{,}800 \pm 340$	$\mathbf{389 \pm 9}$	$\mathbf{307.7 \pm 1.2}$
43 classes	1 M	$14{,}200 \pm 2{,}100^{\dagger}$	$8{,}450 \pm 210$	376 ± 6	$\mathbf{307 \pm 7}$
17 classes	1 M	$10{,}600 \pm 2{,}700^{\dagger}$	$8{,}600 \pm 1{,}300$	$\mathbf{378 \pm 22}$	285 ± 11

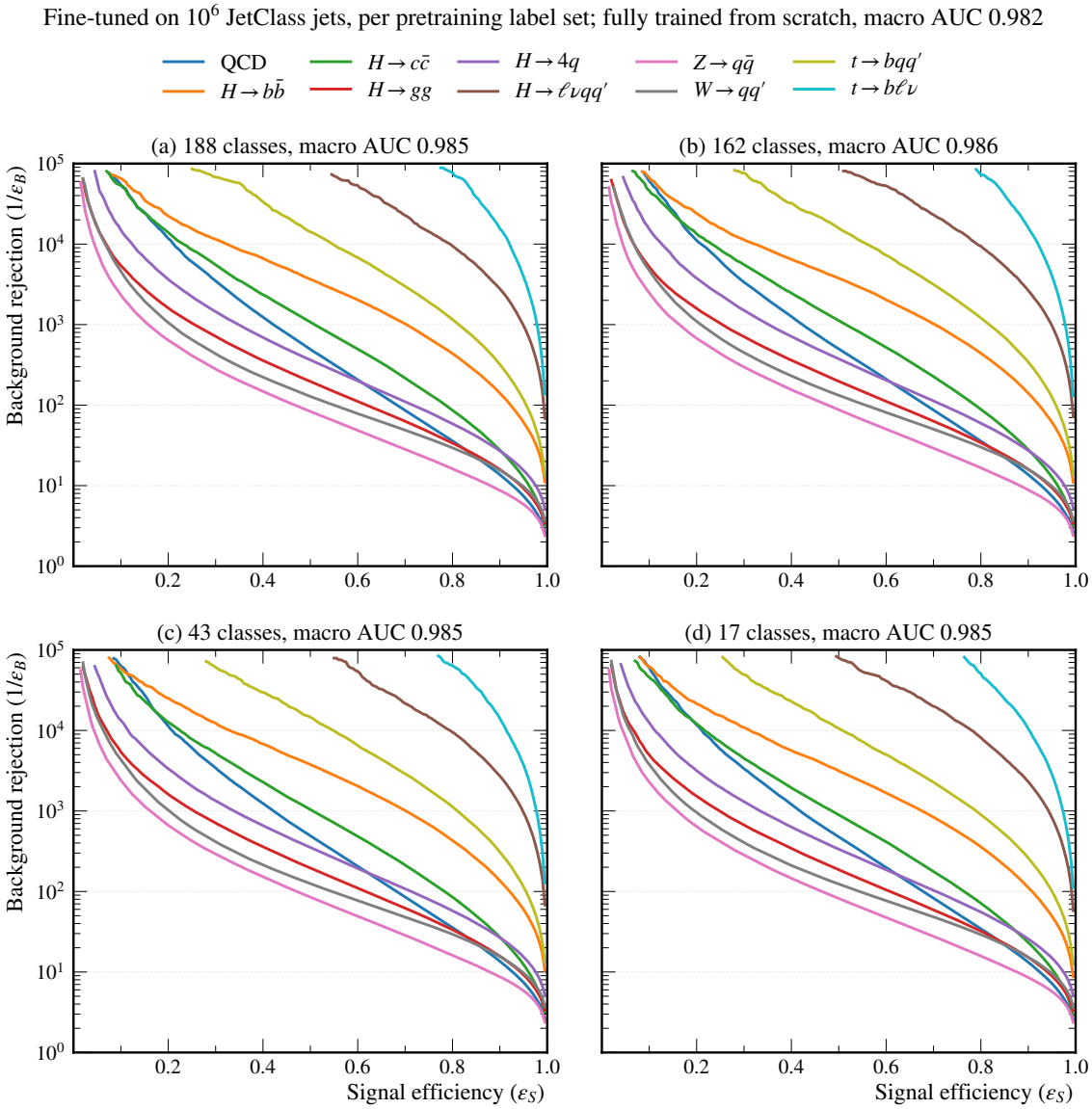


Figure 5: Background rejection against signal efficiency on JetClass after fine-tuning on one million jets. Each curve takes one of the ten JetClass classes as signal and the other nine as background, and is the mean over three pretraining runs. Curves begin where 20 background jets remain. One panel per pretraining label set; its title gives the mean area under the curve over the ten classes, and the heading that of the network trained from scratch.

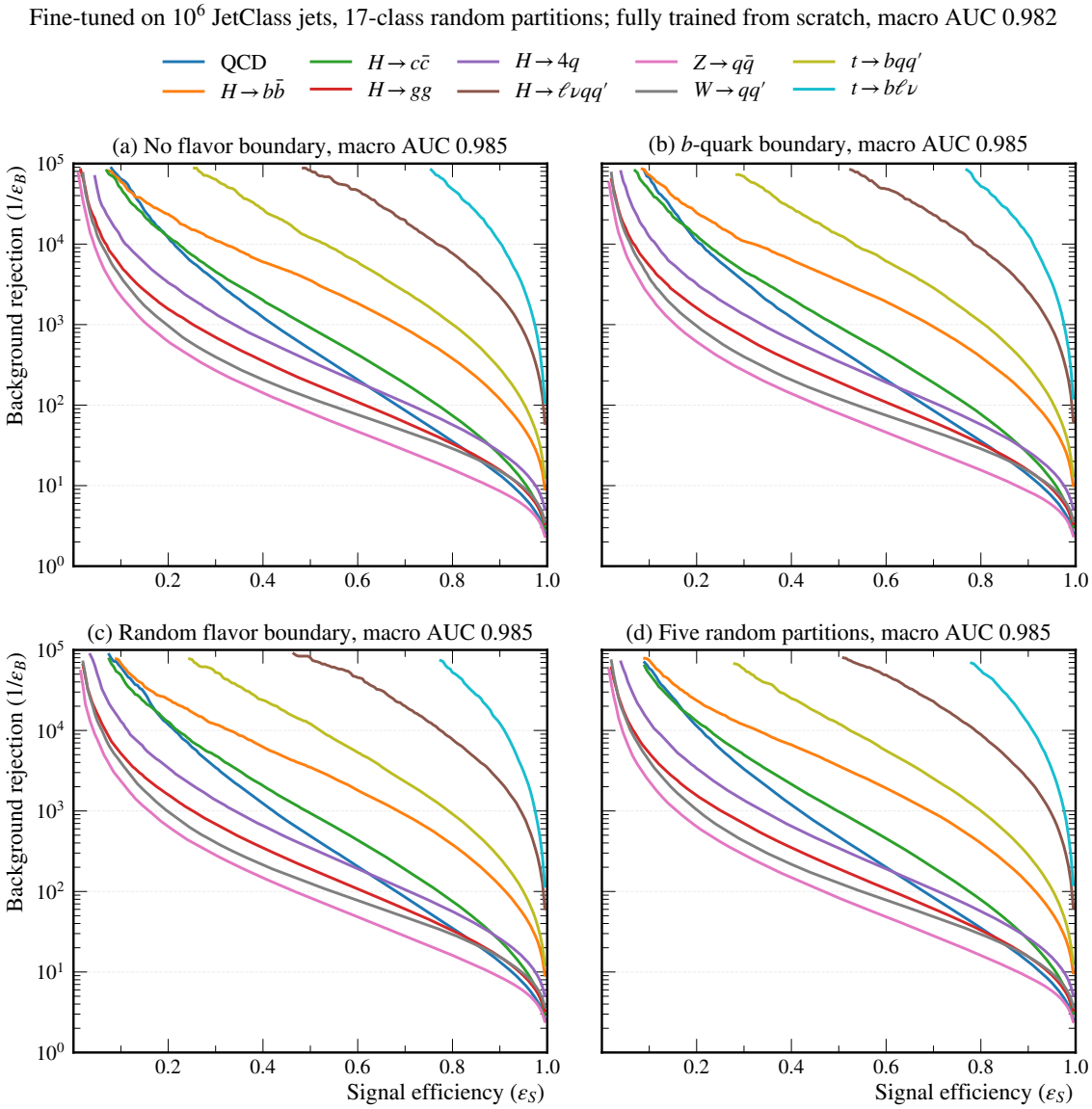


Figure 6: Background rejection against signal efficiency on JetClass after fine-tuning on one million jets, for networks pretrained on 17-class random partitions: with no flavor boundary, a b-quark boundary or a random flavor boundary (two runs each), and five plain random partitions (averaged). Each curve takes one class as signal against the other nine and begins where 20 background jets remain. Titles give the mean area under the curve; the heading, from scratch.

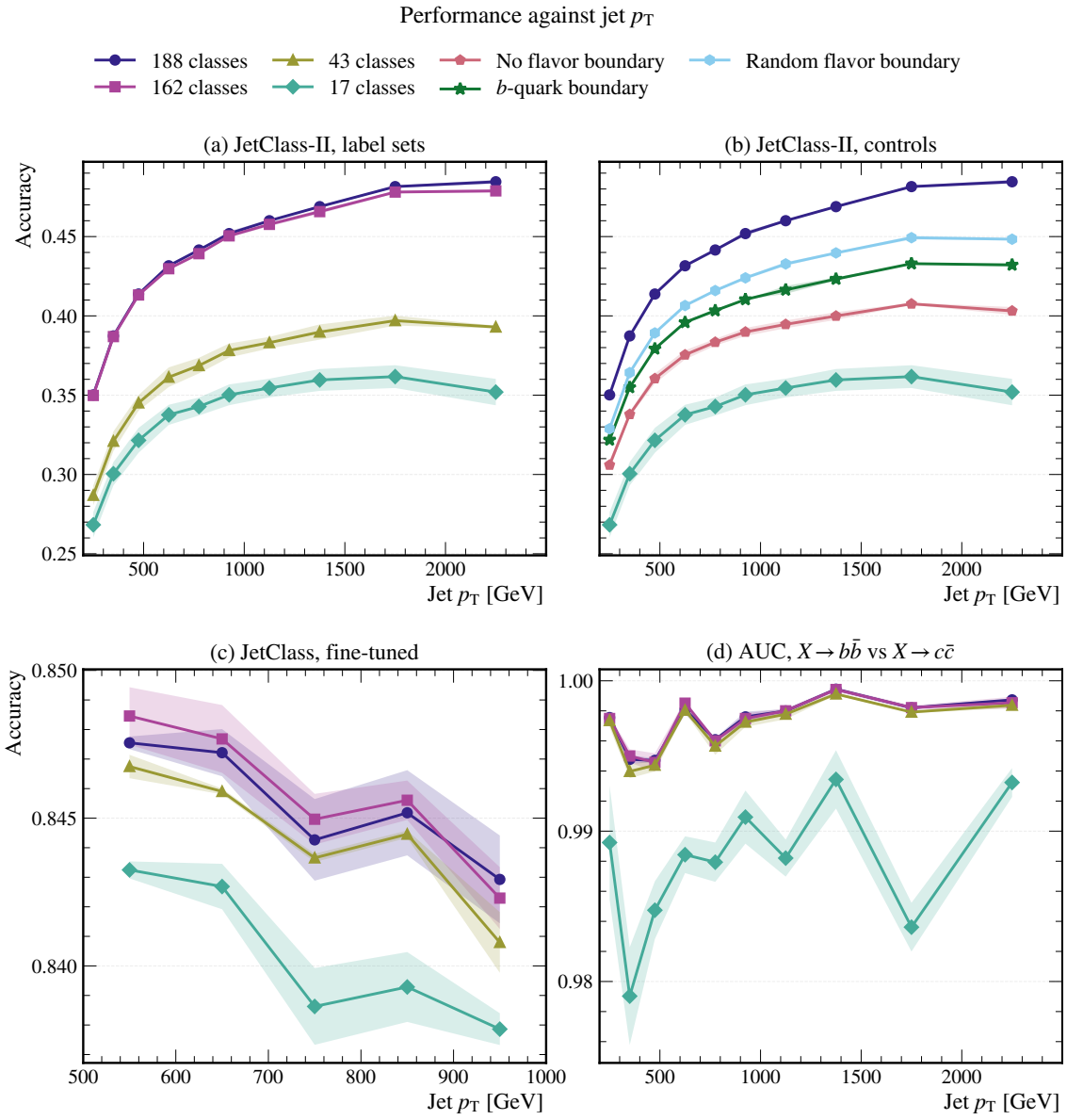


Figure 7: Performance in bins of jet transverse momentum: (a) accuracy of linear probes on class-balanced JetClass-II test jets per pretraining label set; (b) the same for the 17-class random partitions with controlled flavor boundaries, with the 188- and 17-class models for reference; (c) accuracy on JetClass after fine-tuning on one million jets; (d) the probes' area under the curve of $X \to b\bar{b}$ against $X \to c\bar{c}$. Points are means over three runs (two per partition), bands the standard deviation.

Table 2: Top tagging, fine-tuned on the full training set (1.2 million jets): accuracy, AUC and background rejection $1/\varepsilon_B$ at 50% and 30% signal efficiency. Published rows from the tables cited; f.t. marks a model pretrained on JetClass and fine-tuned. Our pretrained models see the 10 inputs the top data lack set to zero; the model with kinematic inputs only, a diagnostic of that mismatch, has none missing (one run). Uncertainties: ours, the standard deviation over the pretraining runs, one fine-tuning each, and over three seeds from scratch; ParT, the spread over 9 fine-tunings of one pretrained model with different output-layer initializations, quoted for the run of median accuracy; L-GATr and OmniLearned, the standard deviation of five trainings; the other published uncertainties as given there. Bold: the best value of each column, and every value within one standard deviation of it (the larger of the two; two runs count with their mean).

	accuracy	AUC	R_{50}	R_{30}
ParticleNet	0.940	0.9858	397 ± 7	$1{,}615 \pm 93$
ParT	0.940	0.9858	413 ± 16	$1{,}602 \pm 81$
ParT-f.t.	0.944	0.9877	$\mathbf{691 \pm 15}$	$2{,}766 \pm 130$
MIParT-L-f.t.	0.944	0.9878	640 ± 10	$2{,}789 \pm 133$
L-GATr-f.t.	**0.9446**	0.9879	651 ± 11	$2{,}894 \pm 84$
OmniLearned-l	0.944	**0.9880**	$\mathbf{688 \pm 9}$	$\mathbf{3{,}486 \pm 157}$
this work (Sophon's network)				
from scratch	0.9387 ± 0.0002	0.9852 ± 0.0001	358 ± 11	$1{,}412 \pm 20$
188 classes	0.9400 ± 0.0002	0.9861 ± 0.0001	461 ± 4	$1{,}800 \pm 60$
162 classes	0.9401 ± 0.0002	0.9861 ± 0.0001	443 ± 17	$1{,}790 \pm 60$
43 classes	0.9402 ± 0.0004	0.9861 ± 0.0001	450 ± 13	$1{,}780 \pm 150$
17 classes	0.9398 ± 0.0001	0.9859 ± 0.0001	429 ± 10	$1{,}740 \pm 140$
188 classes, kinematic inputs only	0.941	0.9862	443	1,803

ParT's top-tagging training script, unchanged, on our cluster gives R_{30} of 1,711 (published $1{,}602 \pm 81$); with Sophon's output head, 1,577; with the head and Sophon's jet-p_{T} rescaling, 1,224 (one run each; our network from scratch, which uses the rescaling, $1{,}412 \pm 20$); table 12.

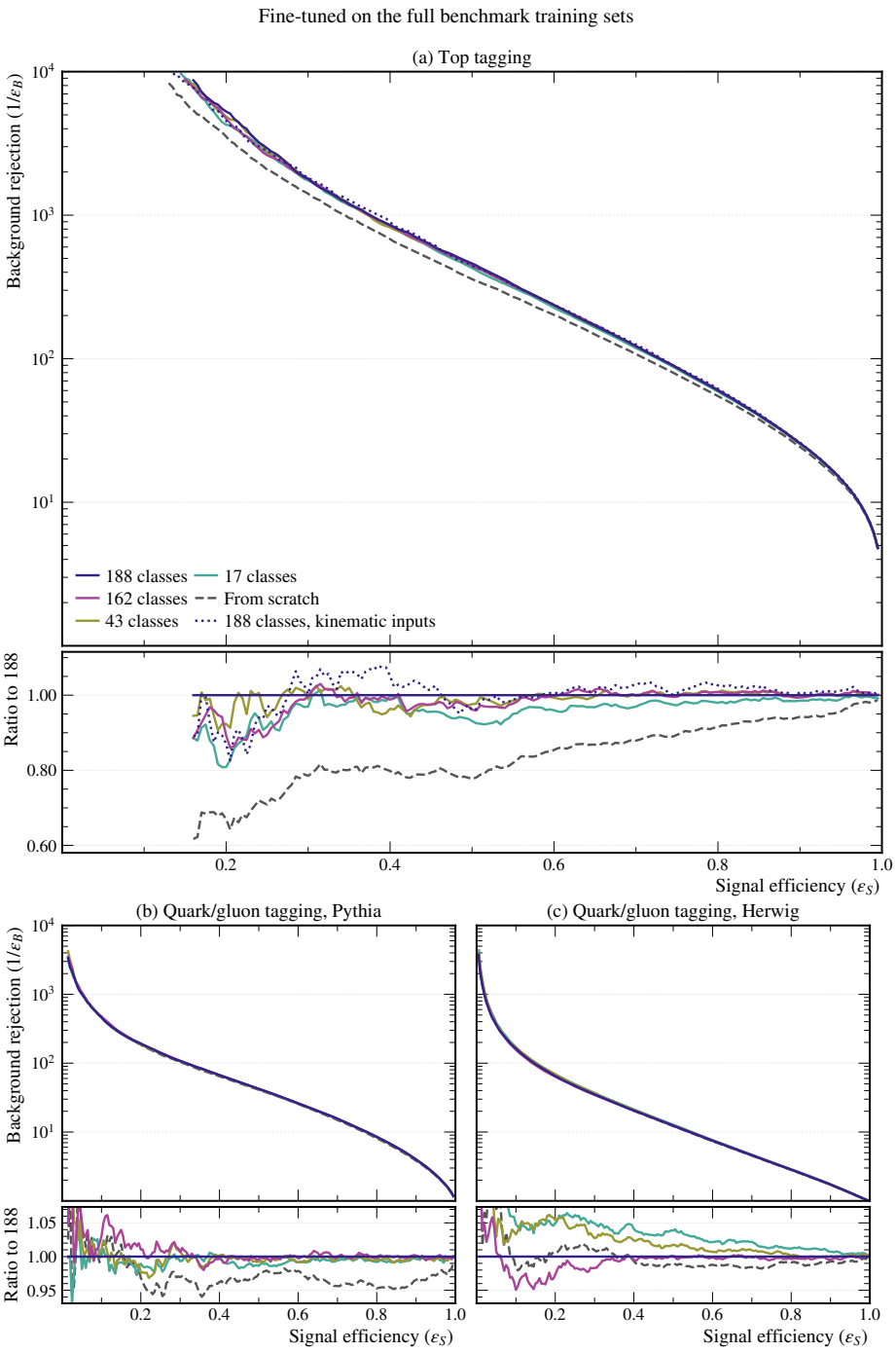


Figure 8: Background rejection against signal efficiency after fine-tuning on the full training sets: top tagging, and quark/gluon separation tested on Pythia and Herwig jets. Curves are networks pretrained with 188, 162, 43 and 17 classes and the network trained from scratch, each the mean over three runs; in (a), dotted, one run of the 188-class model with kinematic inputs only. Curves end where fewer than 20 background jets remain. Lower panels: ratio to 188 classes.

Table 3: Linear probes of the four label sets with the missing inputs set to zero, to the training mean or to random draws of a real particle, at the largest training set. Upper part: relative drop in JetClass-II accuracy (per cent) when the slots the top or the quark/gluon data lack are substituted, against the same model with every input. Lower part: R_{30} on top and quark/gluon tagging. Mean and standard deviation over three runs. The best substitution of each row and data set is in bold. Every configuration is in table 7.

	JetClass-II, top slots		
pretraining	zero	mean	draws
188 classes	$88.9 \pm 1.0\%$	$87.9 \pm 1.1\%$	$\mathbf{85.98 \pm 0.07\%}$
162 classes	$89.0 \pm 0.9\%$	$87.8 \pm 0.7\%$	$\mathbf{85.89 \pm 0.11\%}$
43 classes	$87.81 \pm 0.28\%$	$86.7 \pm 0.9\%$	$\mathbf{83.17 \pm 0.29\%}$
17 classes	$86.7 \pm 0.7\%$	$86.1 \pm 0.8\%$	$\mathbf{81.4 \pm 0.5\%}$

	JetClass-II, quark/gluon slots		
pretraining	zero	mean	draws
188 classes	$61.2 \pm 3.0\%$	$\mathbf{48.8 \pm 1.1\%}$	$54.30 \pm 0.21\%$
162 classes	$64 \pm 5\%$	$\mathbf{49.5 \pm 0.5\%}$	$54.6 \pm 0.6\%$
43 classes	$68 \pm 5\%$	$\mathbf{47.6 \pm 1.6\%}$	$53.5 \pm 1.4\%$
17 classes	$70.9 \pm 2.3\%$	$\mathbf{50.0 \pm 1.0\%}$	$51.7 \pm 0.4\%$

	top, R_{30}		
pretraining	zero	mean	draws
188 classes	60 ± 12	73 ± 15	$\mathbf{228 \pm 25}$
162 classes	76 ± 17	64 ± 11	$\mathbf{199 \pm 25}$
43 classes	57 ± 13	58 ± 14	$\mathbf{231 \pm 8}$
17 classes	55 ± 5	55 ± 5	$\mathbf{255 \pm 17}$
188 classes, kinematic inputs only	231 (nothing missing)		

	quark/gluon, R_{30}		
pretraining	zero	mean	draws
188 classes	66 ± 7	$\mathbf{72.7 \pm 2.4}$	68.1 ± 1.8
162 classes	58 ± 9	$\mathbf{70.99 \pm 0.31}$	66.8 ± 0.4
43 classes	61 ± 9	$\mathbf{70.8 \pm 1.8}$	69.0 ± 1.1
17 classes	62 ± 7	$\mathbf{74.3 \pm 1.9}$	71.6 ± 2.8

Table 4: Quark/gluon discrimination with the particle type as five flags and the charge (ParT's exp inputs), fine-tuned on the full training set (1.6 million Pythia jets) and scored on Pythia and on Herwig jets: accuracy, AUC and $1/\varepsilon_B$ at 50% and 30% quark efficiency. Our models see the 4 displacement inputs set to zero. Published rows (exp inputs) from the tables cited; PFN-Ex's uncertainty is half the interquartile range of ten trainings, the others as in table 2, with nine trainings for OmniLearned. Bold as in table 2.

	Pythia	
	accuracy	AUC
PFN-Ex	–	0.9005
ParticleNet	0.840	0.9116
ParT	0.840	0.9121
ParticleNet-f.t.	0.839	0.9115
ParT-f.t.	0.843	0.9151
OmniLearned-m	**0.845**	**0.9162**
this work (Sophon's network)		
from scratch	0.8397 ± 0.0007	0.9121 ± 0.0002
188 classes	0.8433 ± 0.0002	0.9148 ± 0.0001
162 classes	$0.8432 \pm <0.0001$	0.9148 ± 0.0001
43 classes	0.8430 ± 0.0005	0.9146 ± 0.0003
17 classes	0.8427 ± 0.0003	$0.9144 \pm <0.0001$

	Pythia		Herwig
	R_{50}	R_{30}	accuracy
PFN-Ex	34.7 ± 0.4	–	–
ParticleNet	39.8 ± 0.2	98.6 ± 1.3	–
ParT	41.3 ± 0.3	101.2 ± 1.1	–
ParticleNet-f.t.	40.1 ± 0.2	100.3 ± 1.0	–
ParT-f.t.	42.4 ± 0.2	107.9 ± 0.5	–
OmniLearned-m	$\mathbf{43.2 \pm 0.1}$	$\mathbf{111.2 \pm 1.5}$	–
this work (Sophon's network)			
from scratch	41.47 ± 0.29	104.0 ± 1.0	$\mathbf{0.7153 \pm 0.0024}$
188 classes	42.56 ± 0.10	107.3 ± 2.2	0.712 ± 0.005
162 classes	42.39 ± 0.22	108.0 ± 1.4	0.7100 ± 0.0017
43 classes	42.2 ± 0.8	108.3 ± 1.0	$\mathbf{0.713 \pm 0.006}$
17 classes	41.97 ± 0.25	106.5 ± 0.8	$\mathbf{0.7175 \pm 0.0033}$

	Herwig		
	AUC	R_{50}	R_{30}
this work (Sophon's network)			
from scratch	0.8061 ± 0.0007	12.18 ± 0.11	35.5 ± 1.2
188 classes	0.8078 ± 0.0023	12.3 ± 0.4	$\mathbf{35.2 \pm 2.1}$
162 classes	0.8074 ± 0.0010	12.33 ± 0.28	34.6 ± 2.0
43 classes	$\mathbf{0.809 \pm 0.005}$	$\mathbf{12.5 \pm 0.6}$	$\mathbf{36.1 \pm 2.7}$
17 classes	$\mathbf{0.8103 \pm 0.0018}$	$\mathbf{12.79 \pm 0.20}$	$\mathbf{36.9 \pm 0.9}$

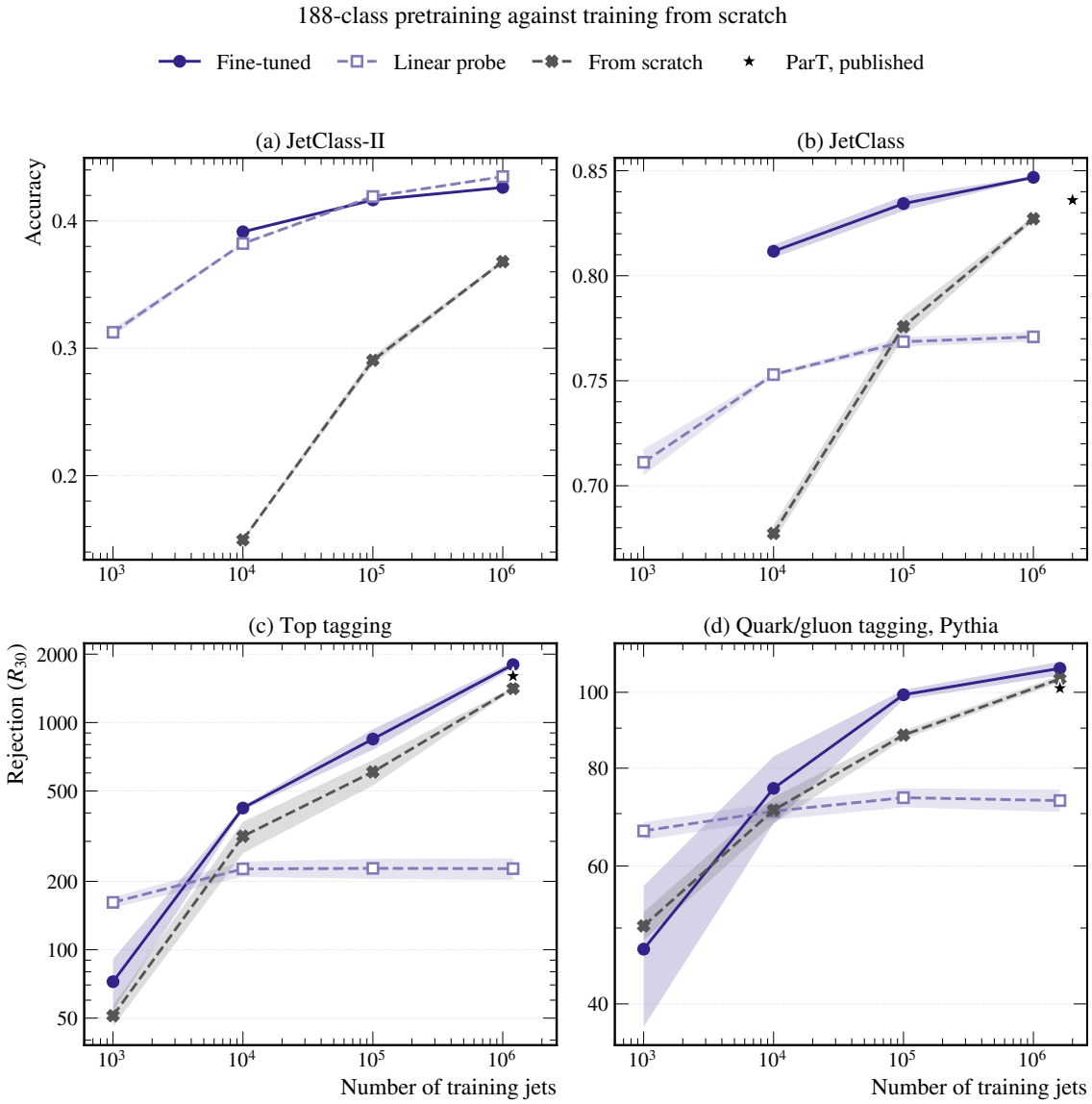


Figure 9: Accuracy on class-balanced JetClass-II and on JetClass, and rejection R_{30} at 30% signal efficiency on top and quark/gluon tagging, against the number of training jets: the 188-class model fine-tuned and with a linear probe, and a network trained from scratch. Points are means; bands the standard deviation over three runs. Fine-tuning on JetClass-II and JetClass starts at 10^4 jets. Stars are published ParT: same inputs on JetClass, kinematic only on top, kinematic and particle type on quark/gluon.

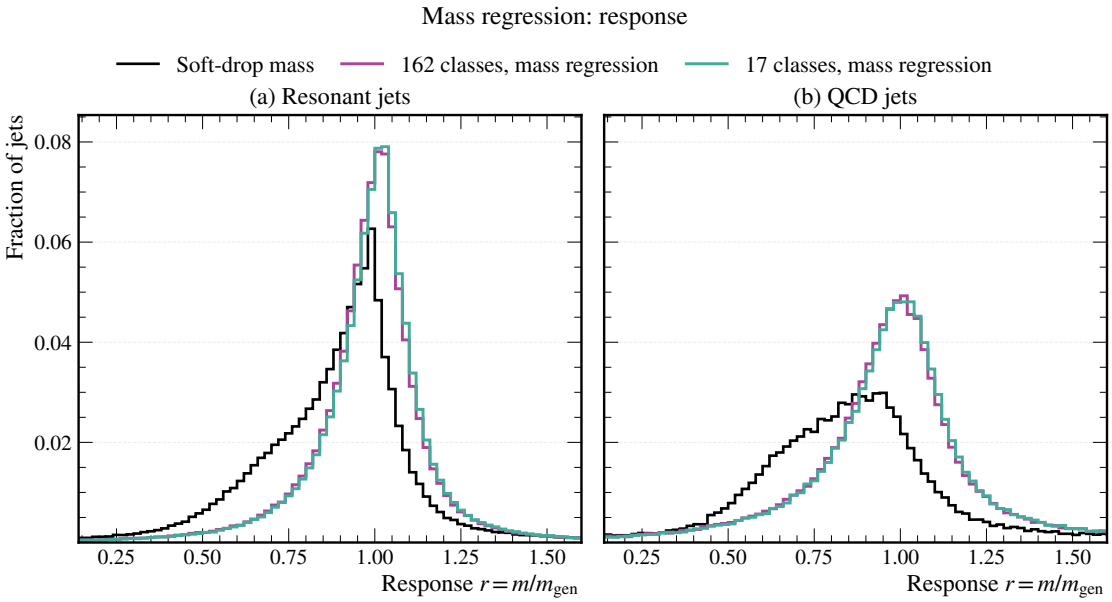


Figure 10: Distribution of the jet mass response $r = m/m_{\text{gen}}$, a jet's mass over its generator-level mass, for (a) resonant and (b) QCD jets of the JetClass-II test set: the soft-drop mass (solid black) and the mass regression outputs of the models pretrained on 162 and 17 classes, each the mean of two runs, which nearly coincide. Both panels share their axes.

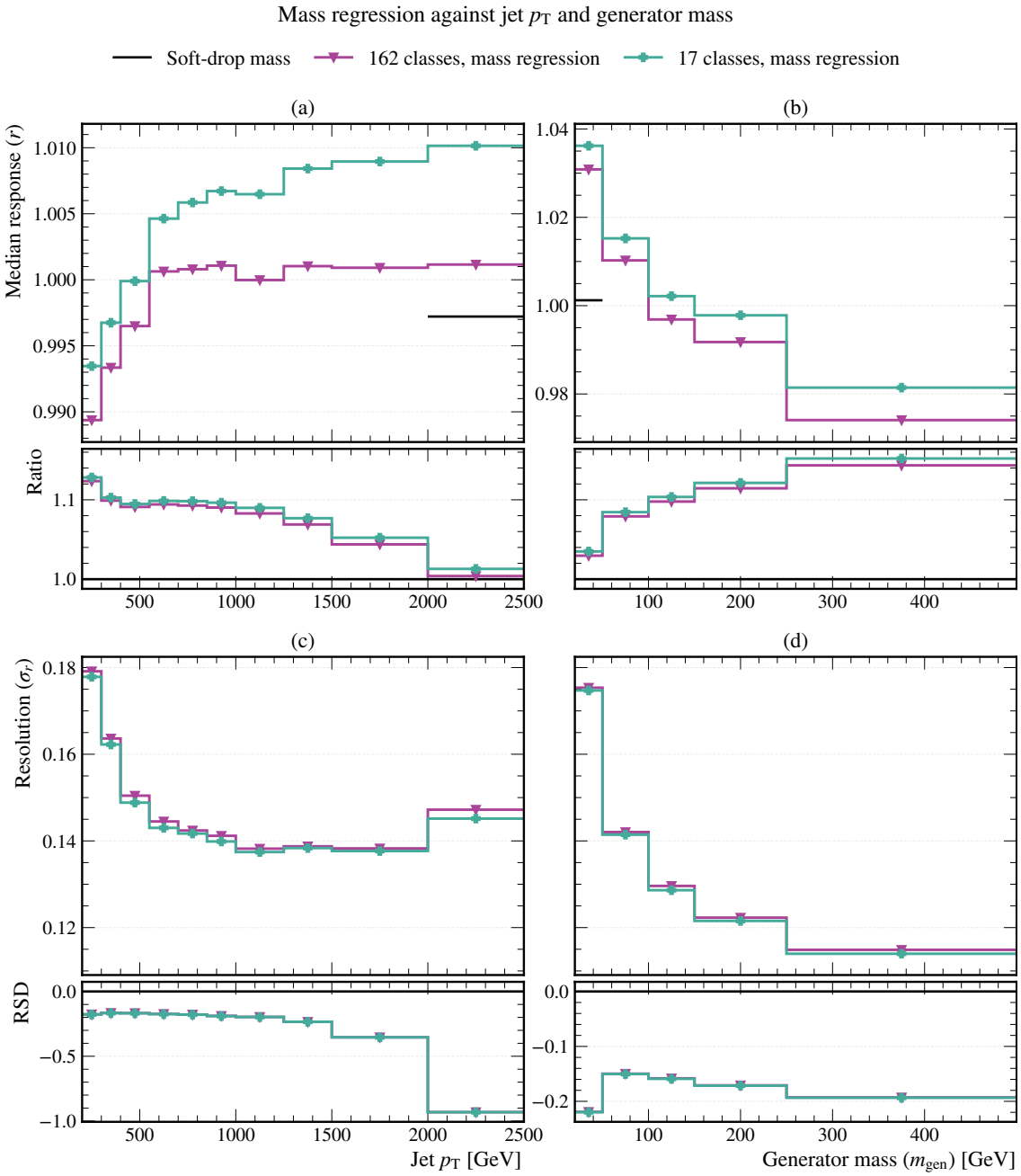


Figure 11: Median mass response $r = m/m_{\text{gen}}$ (a, b) and resolution σ_r, half the 16–84% quantile range of r over its median (c, d), of JetClass-II test jets against transverse momentum (a, c; all jets) and generator mass (b, d; 20–500 GeV): mass regression of the models pretrained on 162 and 17 classes (two runs). Lower panels: ratio to the soft-drop mass (black where in range) and signed root square difference (RSD), negative where better. Unlabelled axes share the left ones.

Table 5: Jet mass from frozen features: a multilayer perceptron regresses the mass from the frozen features of each configuration; the soft-drop mass is the reference. Resolution: σ_r, half the 15.9–84.1% inter-quantile range of the response $r = m/m_{\text{gen}}$, a jet's mass over its generator-level mass, divided by its median; and the median response, over all JetClass-II test jets with a generator match (resonant and QCD jets separately in table 24); lower resolution is better. The regression target is centred on the training median of each of the 188 classes, a truth-label input the soft-drop-mass row does not have. A multilayer perceptron is used because the mass is not linear in the features (ridge regression in table 24). The randomly initialized network shows what a network that never trained exposes. The mass regression in pretraining has the weight $\lambda = 5$ at both label sets, a share of the training loss of 13.78 and 13.87% (162 classes) and 28.65 and 28.67% (17 classes). Mean ± standard deviation over the three pretraining runs; the models with the mass regression have two runs, and both values are given.

	resolution	median response
soft-drop mass	0.245	0.929
MLP regression on frozen features		
randomly initialized network	0.2421 ± 0.0018	0.983 ± 0.005
188 classes	0.2200 ± 0.0014	0.991 ± 0.005
162 classes	0.2195 ± 0.0017	0.9876 ± 0.0014
43 classes	0.2158 ± 0.0029	0.991 ± 0.005
17 classes	0.2098 ± 0.0031	0.991 ± 0.004
162 classes, mass regression	0.1591, 0.1586	1.004, 1.006
17 classes, mass regression	0.1583, 0.1588	1.009, 1.006

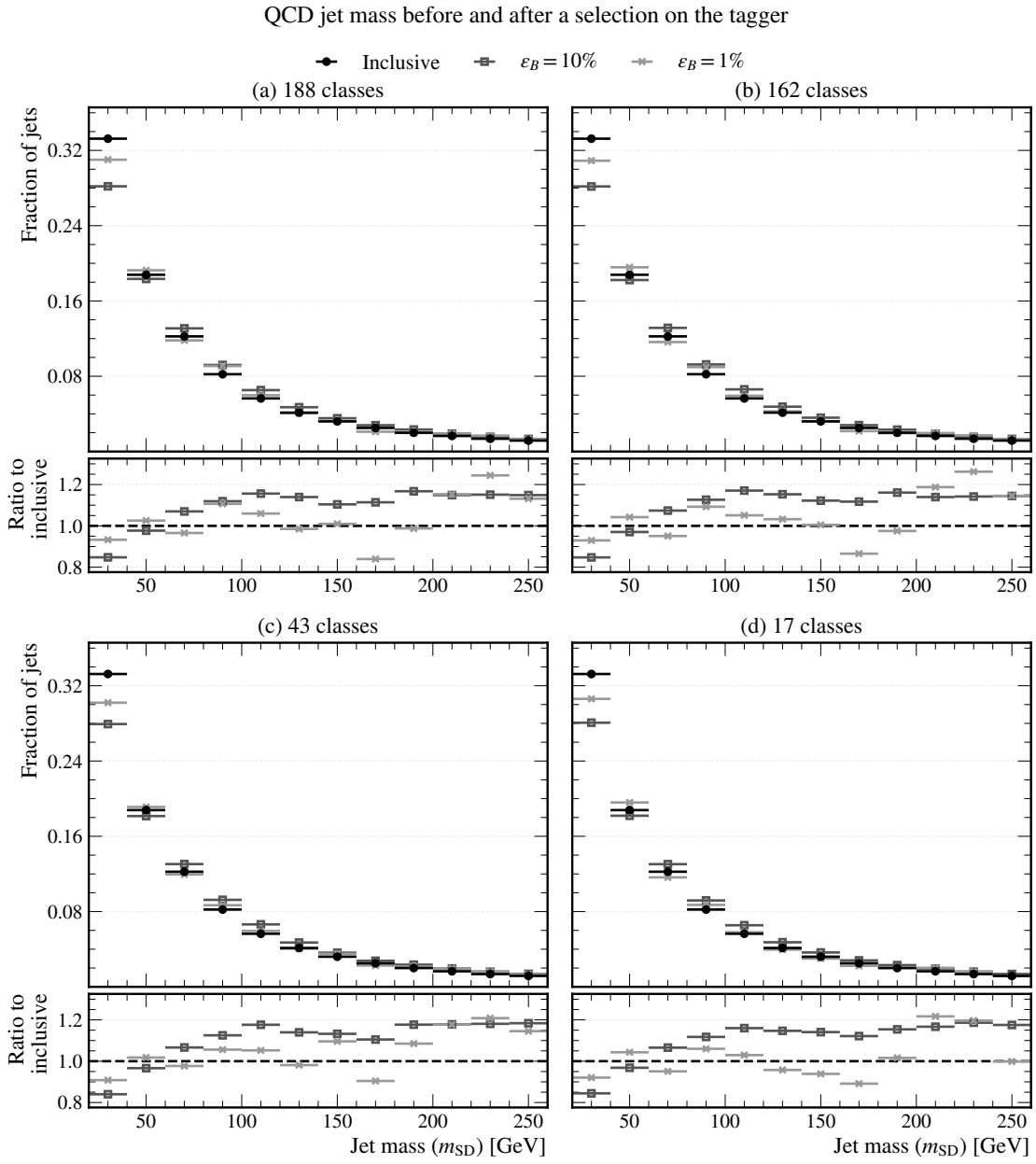


Figure 12: Soft-drop mass distribution of QCD test jets of JetClass-II before any selection (inclusive) and after selections on $1 - g_{\mathrm{QCD}}$, one minus the summed QCD score, at QCD efficiencies ε_B of 10% and 1%, for the models pretrained on 188, 162, 43 and 17 classes (mean of three runs). Bars show the bin width; the axis stops at 260 GeV. Below each panel, the ratio to the inclusive distribution.

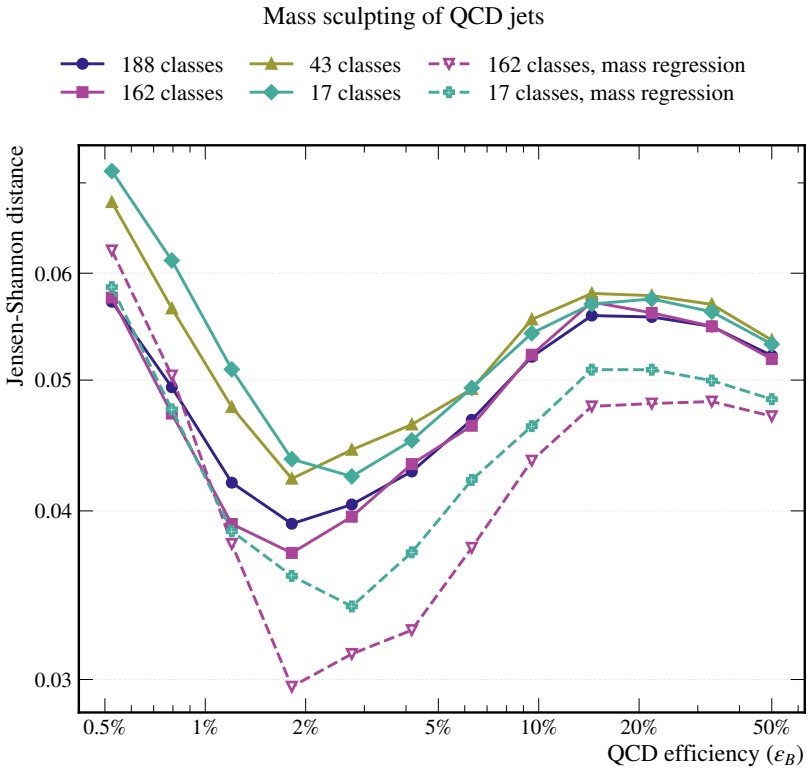


Figure 13: Jensen-Shannon distance (base 2) between the soft-drop mass distributions of QCD test jets of JetClass-II before and after a selection on $1 - g_{\mathrm{QCD}}$, one minus the summed QCD score, against the QCD efficiency ε_B of the selection (tighter to the left), for the models pretrained on 188, 162, 43 and 17 classes (solid, mean of three runs) and on 162 and 17 classes with mass regression (dashed, mean of two).

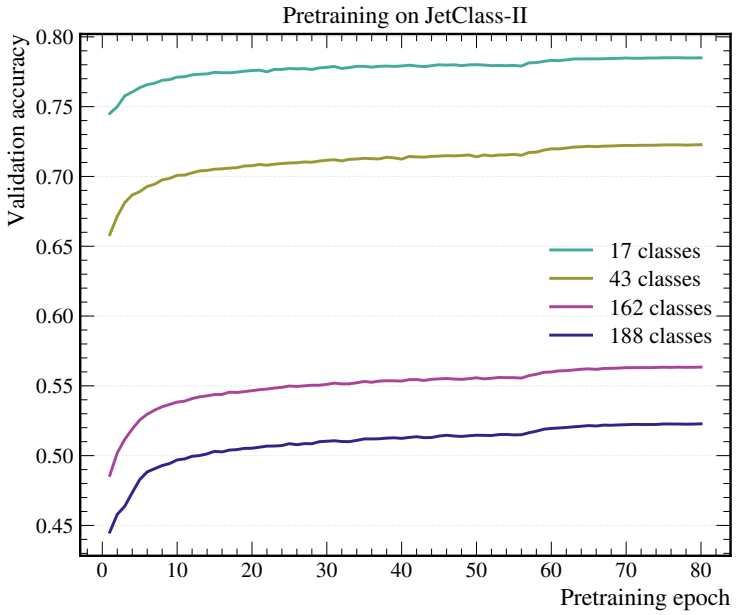


Figure 14: Validation accuracy of the pretraining against epoch for the 188-, 162-, 43- and 17-class label sets, each on its own classes, as the mean over three runs. A coarser label set has fewer classes to tell apart, so its accuracy is not comparable with a finer set's. The step near epoch 56 is where the learning rate starts to decay.

Table 6: Learning-rate choice at 10^4 training jets: the best validation accuracy of each rate on top and quark/gluon and their sum. Pretrained, backbone: the rate of the pretrained layers, with the new output head at 50 times that rate, at most 5×10^{-3}. *: the rate chosen, on the sum for the pretrained weights and on each task's accuracy for the networks trained from scratch; the test jets play no part.

network	learning rate	top	quark/gluon	sum
pretrained, backbone	10^{-4}	0.9204	0.8204	1.741
pretrained, backbone	3×10^{-4}	0.9192	0.8202	1.739
pretrained, backbone	10^{-3}	0.9212	0.8211	1.742*
from scratch	3×10^{-4}	0.9152	0.8052	1.720
from scratch	10^{-3}	0.9188*	0.8029	1.722
from scratch	3×10^{-3}	0.9177	0.8059*	1.724

Table 7: The three substitutions of the missing inputs (zero, the training mean, random draws of a real particle) for every configuration, at the largest training set. Upper part: relative drop in JetClass-II accuracy (per cent) when the slots the top (10) or the quark/gluon (4) data lack are substituted, against the same model with every input. Lower part: R_{30} of the linear probe on top and quark/gluon jets under each substitution; the model with kinematic inputs only has nothing missing. The substitution chosen on validation is random draws for top and the training mean for quark/gluon. The best substitution of each row and data set is in bold. Mean ± standard deviation over the three pretraining runs of each level of the class hierarchy; configurations with two runs give both values, and a random partition has a single run.

	JetClass-II, top slots		
pretraining	zero	mean	draws
188 classes	$88.9 \pm 1.0\%$	$87.9 \pm 1.1\%$	$\mathbf{85.98 \pm 0.07\%}$
162 classes	$89.0 \pm 0.9\%$	$87.8 \pm 0.7\%$	$\mathbf{85.89 \pm 0.11\%}$
43 classes	$87.81 \pm 0.28\%$	$86.7 \pm 0.9\%$	$\mathbf{83.17 \pm 0.29\%}$
17 classes	$86.7 \pm 0.7\%$	$86.1 \pm 0.8\%$	$\mathbf{81.4 \pm 0.5\%}$
162 classes, mass regression	89.20, 88.29%	88.47, 87.78%	**85.35, 85.37%**
17 classes, mass regression	84.8, 87.4%	85.93, 86.89%	**79.1, 81.7%**
random partition 1, seed 1	90.2%	88.6%	**85.7%**
random partition 2, seed 2	90.4%	88.8%	**85.9%**
random partition 3, seed 1	90.8%	89.5%	**86.2%**
random partition 4, seed 2	89.9%	88.7%	**87.1%**
random partition 5, seed 2	89.5%	89.7%	**86.1%**
random partition, no flavor boundary	88.88, 88.46%	88.3, 87.0%	**83.26, 83.89%**
random partition, b-quark boundary	87.80, 87.65%	88.9, 86.9%	**84.44, 85.21%**
random partition, random flavor boundary	89.58, 90.28%	89.44, 89.66%	**84.94, 85.58%**
randomly initialized network, BatchNorm recomputed	$\mathbf{56.4 \pm 1.4\%}$	$59.8 \pm 0.8\%$	$59.8 \pm 1.2\%$

	JetClass-II, quark/gluon slots		
pretraining	zero	mean	draws
188 classes	$61.2 \pm 3.0\%$	$\mathbf{48.8 \pm 1.1\%}$	$54.30 \pm 0.21\%$
162 classes	$64 \pm 5\%$	$\mathbf{49.5 \pm 0.5\%}$	$54.6 \pm 0.6\%$
43 classes	$68 \pm 5\%$	$\mathbf{47.6 \pm 1.6\%}$	$53.5 \pm 1.4\%$
17 classes	$70.9 \pm 2.3\%$	$\mathbf{50.0 \pm 1.0\%}$	$51.7 \pm 0.4\%$
162 classes, mass regression	61.8, 67.9%	**49.7, 52.2%**	54.34, 55.13%
17 classes, mass regression	66.7, 65.4%	**49.04, 49.64%**	50.77, 51.07%
random partition 1, seed 1	66.5%	**51.6%**	56.3%
random partition 2, seed 2	69.5%	**52.4%**	56.3%
random partition 3, seed 1	63.7%	**52.8%**	57.8%
random partition 4, seed 2	72.9%	**53.4%**	55.5%
random partition 5, seed 2	69.5%	**53.9%**	56.3%
random partition, no flavor boundary	62.72, 62.24%	**51.87, 51.28%**	53.92, 53.44%
random partition, *b*-quark boundary	66.2, 62.8%	**52.67, 52.80%**	55.58, 56.49%
random partition, random flavor boundary	67.2, 69.0%	**52.9, 51.8%**	57.2, 55.4%
randomly initialized network, BatchNorm recomputed	$\mathbf{8.7 \pm 2.6\%}$	$10.4 \pm 1.4\%$	$11.4 \pm 2.1\%$

	top, R_{30}		
pretraining	zero	mean	draws
188 classes	60 ± 12	73 ± 15	$\mathbf{228 \pm 25}$
162 classes	76 ± 17	64 ± 11	$\mathbf{199 \pm 25}$
43 classes	57 ± 13	58 ± 14	$\mathbf{231 \pm 8}$
17 classes	55 ± 5	55 ± 5	$\mathbf{255 \pm 17}$
162 classes, mass regression	77, 95	57, 79	**260, 306**
17 classes, mass regression	103, 69	55, 81	**306, 378**
random partition 1, seed 1	44.2	55.5	**174**
random partition 2, seed 2	60.5	53.6	**151**
random partition 3, seed 1	50.0	42.9	**164**
random partition 4, seed 2	30.2	56.5	**152**
random partition 5, seed 2	73.9	45.5	**173**
random partition, no flavor boundary	69.65, 69.34	44, 65	**262, 243**
random partition, *b*-quark boundary	73.7, 80.7	45, 68	**202.5, 211.6**
random partition, random flavor boundary	65, 41	40.9, 33.2	**214, 233**
randomly initialized network, BatchNorm recomputed	58 ± 13	$\mathbf{60.4 \pm 2.4}$	48.6 ± 3.0
188 classes, kinematic inputs only	231 (nothing missing)		

	quark/gluon, R_{30}		
pretraining	zero	mean	draws
188 classes	66 ± 7	$\mathbf{72.7 \pm 2.4}$	68.1 ± 1.8
162 classes	58 ± 9	$\mathbf{70.99 \pm 0.31}$	66.8 ± 0.4
43 classes	61 ± 9	$\mathbf{70.8 \pm 1.8}$	69.0 ± 1.1
17 classes	62 ± 7	$\mathbf{74.3 \pm 1.9}$	71.6 ± 2.8
162 classes, mass regression	65.4, 61.1	**73.9, 70.7**	70.4, 72.1
17 classes, mass regression	66.01, 65.57	68.4, 72.2	**71.3, 75.3**
random partition 1, seed 1	65.9	**69.6**	66.9
random partition 2, seed 2	59.0	**69.3**	64.6
random partition 3, seed 1	65.2	**68.4**	62.3
random partition 4, seed 2	51.0	**64.7**	61.6
random partition 5, seed 2	54.9	**69.6**	68.2
random partition, no flavor boundary	63.1, 66.4	**73.9, 71.7**	71.07, 71.48
random partition, b-quark boundary	54.6, 59.4	**72.52, 72.20**	70.87, 70.47
random partition, random flavor boundary	66, 54	**71.2, 72.8**	71.2, 69.8
randomly initialized network, BatchNorm recomputed	$\mathbf{54.2 \pm 2.4}$	52.2 ± 0.8	52.4 ± 0.4

Table 8: Nonlinear probes (a multilayer perceptron with one hidden layer of 256 units, three seeds averaged) at the largest training set: class-balanced JetClass-II AUC (all, resonant and QCD classes) with the ratio ρ of $1 - \text{AUC}$ to the 188-class run, JetClass AUC, and AUC and R_{30} on top, quark/gluon and Herwig jets (missing inputs set to zero); then the class pairs of table 23, class-balanced, except $X \to ee$ vs $\mu\mu$, which has no nonlinear probe. The models with the mass regression and the numbered random partitions have no nonlinear probe on the class-balanced test set, so they are left out of the first blocks and the pair blocks; the next block gives them on the natural-composition test set beside the levels of the hierarchy on that set, and no ρ crosses the two test sets. Last, the linear probe on the 512-unit hidden layer of the 188-class models' pretraining head beside the one on the class token, with the missing inputs also set to zero. Mean $\pm$ standard deviation over the three pretraining runs of each level of the class hierarchy; configurations with two runs give both values, and a random partition has a single run. The 188-class model with kinematic inputs only, a diagnostic of the missing inputs, has one run and no input missing, and is evaluated on top tagging only.

	JetClass-II	
pretraining	AUC	resonant
188 classes	$0.9806 \pm <0.0001$	$0.9834 \pm <0.0001$
162 classes	0.9802 ± 0.0001	0.9833 ± 0.0001
43 classes	0.9773 ± 0.0001	0.9802 ± 0.0002
17 classes	0.9731 ± 0.0004	0.9766 ± 0.0004
random partition, no flavor boundary	0.9749, 0.9746	0.9786, 0.9783
random partition, b-quark boundary	0.9774, 0.9775	0.9806, 0.9807
random partition, random flavor boundary	0.9776, 0.9777	0.9810, 0.9811
randomly initialized network	0.8648 ± 0.0014	0.8694 ± 0.0017
randomly initialized network, BatchNorm recomputed	0.8974 ± 0.0018	0.9035 ± 0.0020

	JetClass-II		JetClass
pretraining	QCD	ρ	AUC
188 classes	0.9643 ± 0.0001	1.00	0.9786 ± 0.0004
162 classes	0.9618 ± 0.0003	1.019 ± 0.007	0.9786 ± 0.0002
43 classes	0.9605 ± 0.0001	1.170 ± 0.006	0.9782 ± 0.0003
17 classes	0.9524 ± 0.0004	1.388 ± 0.021	0.9776 ± 0.0005
162 classes, mass regression	–	–	0.9794, 0.9799
17 classes, mass regression	–	–	0.9787, 0.9785
random partition 1, seed 1	–	–	0.9739
random partition 2, seed 2	–	–	0.9750
random partition 3, seed 1	–	–	0.9744
random partition 4, seed 2	–	–	0.9742
random partition 5, seed 2	–	–	0.9752
random partition, no flavor boundary	0.9528, 0.9522	1.294, 1.310	0.9764, 0.9763
random partition, b-quark boundary	0.9582, 0.9584	1.168, 1.159	0.9767, 0.9776
random partition, random flavor boundary	0.9571, 0.9576	1.158, 1.150	0.9767, 0.9763
randomly initialized network	0.8372 ± 0.0015	–	0.8859 ± 0.0021
randomly initialized network, BatchNorm recomputed	0.8610 ± 0.0008	–	0.9146 ± 0.0012

	top		quark/gluon
pretraining	AUC	R_{30}	AUC
188 classes	0.9713 ± 0.0034	290 ± 60	0.894 ± 0.004
162 classes	0.971 ± 0.004	280 ± 90	0.889 ± 0.005
43 classes	0.9688 ± 0.0029	250 ± 60	0.8904 ± 0.0031
17 classes	0.9688 ± 0.0018	224 ± 24	0.887 ± 0.005
162 classes, mass regression	0.9731, 0.9754	317, 376	0.8932, 0.8914
17 classes, mass regression	0.9747, 0.9704	356, 261	0.8933, 0.8927
random partition 1, seed 1	0.9635	161	0.8927
random partition 2, seed 2	0.9676	210	0.8882
random partition 3, seed 1	0.9667	190	0.8938
random partition 4, seed 2	0.9587	125	0.8811
random partition 5, seed 2	0.9701	251	0.8858
random partition, no flavor boundary	0.9668, 0.9682	210.1, 218.3	0.8912, 0.8921
random partition, b-quark boundary	0.9718, 0.9724	304.1, 313.5	0.8887, 0.8911
random partition, random flavor boundary	0.9670, 0.9626	215, 160	0.8906, 0.8870
randomly initialized network	0.9667 ± 0.0006	210 ± 4	0.8844 ± 0.0001
randomly initialized network, BatchNorm recomputed	0.9712 ± 0.0024	300 ± 50	0.8831 ± 0.0003
188 classes, kinematic inputs only	0.9811	716	–

	quark/gluon	Herwig	
pretraining	R_{30}	AUC	R_{30}
188 classes	82.6 ± 3.5	0.8077 ± 0.0006	36.64 ± 0.25
162 classes	77.0 ± 2.2	0.8060 ± 0.0014	35.6 ± 0.6
43 classes	81 ± 4	0.8071 ± 0.0007	35.67 ± 0.28
17 classes	78.9 ± 1.7	0.8062 ± 0.0024	36.2 ± 0.5
162 classes, mass regression	81.8, 78.4	0.8094, 0.8071	37.64, 36.83
17 classes, mass regression	80.91, 81.90	0.8079, 0.8084	35.95, 36.47
random partition 1, seed 1	80.9	0.8087	36.6
random partition 2, seed 2	79.0	0.8057	35.2
random partition 3, seed 1	80.5	0.8059	35.2
random partition 4, seed 2	72.0	0.8062	35.5
random partition 5, seed 2	75.5	0.8049	35.4
random partition, no flavor boundary	77.5, 83.1	0.8073, 0.8071	35.27, 34.98
random partition, b-quark boundary	80.8, 78.6	0.8069, 0.8067	36.08, 35.57
random partition, random flavor boundary	80.6, 77.9	0.8074, 0.8052	35.31, 34.53
randomly initialized network	76.6 ± 0.8	0.8074 ± 0.0007	38.2 ± 0.4
randomly initialized network, BatchNorm recomputed	76.3 ± 1.9	0.8087 ± 0.0009	37.8 ± 1.5

pretraining	$X \to bc$ vs bq, cs		$X \to b\bar{b}$ vs $c\bar{c}$
	AUC	ρ	AUC
188 classes	0.9434 ± 0.0001	1.00	$0.9978 \pm <0.0001$
162 classes	0.9432 ± 0.0002	1.002 ± 0.006	$0.9977 \pm <0.0001$
43 classes	0.9375 ± 0.0008	1.104 ± 0.017	$0.9975 \pm <0.0001$
17 classes	0.9080 ± 0.0013	1.624 ± 0.022	0.9932 ± 0.0002
random partition, no flavor boundary	0.9077, 0.9046	1.633, 1.679	0.9926, 0.9902
random partition, b-quark boundary	0.9216, 0.9227	1.385, 1.360	0.9961, 0.9963
random partition, random flavor boundary	0.9231, 0.9242	1.359, 1.335	0.9951, 0.9946
randomly initialized network	0.630 ± 0.004	–	0.759 ± 0.009
randomly initialized network, BatchNorm recomputed	0.6865 ± 0.0030	–	0.866 ± 0.008

pretraining	$X \to b\bar{b}$ vs $c\bar{c}$	$X \to YY \to b\bar{b}qq$ vs $c\bar{c}qq$	
	ρ	AUC	ρ
188 classes	1.00	0.9941 ± 0.0001	1.00
162 classes	1.023 ± 0.021	0.9941 ± 0.0001	1.011 ± 0.030
43 classes	1.125 ± 0.011	$0.9935 \pm <0.0001$	1.112 ± 0.006
17 classes	3.04 ± 0.09	0.9828 ± 0.0011	2.93 ± 0.17
random partition, no flavor boundary	3.3, 4.3	0.9813, 0.9776	3.20, 3.78
random partition, b-quark boundary	1.74, 1.64	0.9916, 0.9919	1.442, 1.372
random partition, random flavor boundary	2.21, 2.37	0.9889, 0.9891	1.890, 1.843
randomly initialized network	–	0.747 ± 0.006	–
randomly initialized network, BatchNorm recomputed	–	0.837 ± 0.006	–

pretraining	QCD $b\bar{b}$ vs $c\bar{c}$		QCD with b vs light
	AUC	ρ	AUC
188 classes	0.9871 ± 0.0001	1.00	$0.9882 \pm <0.0001$
162 classes	0.9860 ± 0.0003	1.085 ± 0.030	0.9876 ± 0.0001
43 classes	0.9855 ± 0.0001	1.121 ± 0.017	0.9875 ± 0.0001
17 classes	0.9701 ± 0.0008	2.31 ± 0.04	0.9842 ± 0.0002
random partition, no flavor boundary	0.9680, 0.9647	2.50, 2.72	0.9845, 0.9847
random partition, b-quark boundary	0.9817, 0.9821	1.432, 1.372	0.9863, 0.9865
random partition, random flavor boundary	0.9776, 0.9775	1.745, 1.725	0.9858, 0.9859
randomly initialized network	0.7228 ± 0.0031	–	0.8185 ± 0.0019
randomly initialized network, BatchNorm recomputed	0.822 ± 0.005	–	0.8879 ± 0.0033

pretraining	QCD with b vs light	QCD with c, no b, vs light	
	ρ	AUC	ρ
188 classes	1.00	0.9329 ± 0.0001	1.00
162 classes	1.050 ± 0.004	0.9303 ± 0.0002	1.039 ± 0.005
43 classes	1.055 ± 0.005	0.9292 ± 0.0003	1.055 ± 0.006
17 classes	1.337 ± 0.021	0.9145 ± 0.0004	1.274 ± 0.008
random partition, no flavor boundary	1.315, 1.288	0.9174, 0.9186	1.234, 1.211
random partition, b-quark boundary	1.158, 1.140	0.9224, 0.9229	1.159, 1.147
random partition, random flavor boundary	1.201, 1.187	0.9236, 0.9247	1.141, 1.120
randomly initialized network	–	0.682 ± 0.005	–
randomly initialized network, BatchNorm recomputed	–	0.7143 ± 0.0013	–

natural-composition	JetClass-II		ρ on the class pairs
test set	AUC	ρ	$X \to bc$ vs bq, cs
188 classes	0.9767 ± 0.0002	1.00	1.00
162 classes	0.9759 ± 0.0003	1.033 ± 0.016	1.003 ± 0.004
43 classes	0.9736 ± 0.0005	1.135 ± 0.018	1.135 ± 0.017
17 classes	0.9682 ± 0.0003	1.364 ± 0.008	1.76 ± 0.04
162 classes, mass regression	0.9752, 0.9757	1.061, 1.051	1.057, 1.018
17 classes, mass regression	0.9674, 0.9679	1.393, 1.391	1.839, 1.797
random partition 1, seed 1	0.9736	1.13	1.20
random partition 2, seed 2	0.9736	1.14	1.17
random partition 3, seed 1	0.9737	1.12	1.10
random partition 4, seed 2	0.9738	1.13	1.09
random partition 5, seed 2	0.9735	1.15	1.21

natural-composition	ρ on the class pairs		
test set	$X \to b\bar{b}$ vs $c\bar{c}$	$X \to YY \to b\bar{b}qq$ vs $c\bar{c}qq$	QCD $b\bar{b}$ vs $c\bar{c}$
188 classes	1.00	1.00	1.00
162 classes	1.00 ± 0.09	0.97 ± 0.08	1.08 ± 0.09
43 classes	1.112 ± 0.009	1.02 ± 0.06	1.13 ± 0.05
17 classes	3.26 ± 0.10	3.34 ± 0.29	2.71 ± 0.13
162 classes, mass regression	1.21, 1.09	1.17, 1.04	1.30, 1.16
17 classes, mass regression	3.65, 3.11	3.74, 3.34	3.26, 2.88
random partition 1, seed 1	1.44	1.35	1.53
random partition 2, seed 2	1.35	1.30	1.47
random partition 3, seed 1	1.44	1.32	1.45
random partition 4, seed 2	1.25	1.25	1.37
random partition 5, seed 2	1.56	1.30	1.53

natural-composition	ρ on the class pairs	
test set	QCD with b vs light	QCD with c, no b, vs light
188 classes	1.00	1.00
162 classes	1.06 ± 0.06	1.041 ± 0.021
43 classes	1.05 ± 0.09	1.057 ± 0.016
17 classes	1.32 ± 0.18	1.303 ± 0.032
162 classes, mass regression	1.027, 0.938	1.071, 1.045
17 classes, mass regression	1.38, 1.26	1.326, 1.323
random partition 1, seed 1	1.05	1.13
random partition 2, seed 2	1.00	1.10
random partition 3, seed 1	1.16	1.12
random partition 4, seed 2	1.16	1.07
random partition 5, seed 2	1.16	1.10

188 classes, linear probe on	JetClass-II AUC	quark/gluon AUC	quark/gluon R_{30}
class token	$0.9801 \pm <0.0001$	0.878 ± 0.006	66 ± 7
512-unit hidden layer	$0.9802 \pm <0.0001$	0.880 ± 0.007	70 ± 6

Table 9: Linear probes on JetClass-II (natural composition, largest training set) with the BatchNorm statistics as accumulated in pretraining and recomputed on training jets: the AUC averaged over the classes and the ratio ρ of $1 - \text{AUC}$ to the 188-class run. The class-balanced probes use the checkpoints as trained, so this check uses the natural-composition test set. Mean $\pm$ standard deviation over the three pretraining runs of each level of the class hierarchy; configurations with two runs give both values, and a random partition has a single run.

	AUC	
pretraining	as trained	recomputed
188 classes	$0.9760 \pm <0.0001$	$0.9760 \pm <0.0001$
162 classes	0.9748 ± 0.0003	0.9748 ± 0.0003
43 classes	0.9719 ± 0.0006	0.9719 ± 0.0006
17 classes	0.9631 ± 0.0006	0.9631 ± 0.0006
162 classes, mass regression	0.9735, 0.9740	0.9735, 0.9740
17 classes, mass regression	0.9623, 0.9632	0.9623, 0.9632
random partition 1, seed 1	0.9719	0.9719
random partition 2, seed 2	0.9719	0.9719
random partition 3, seed 1	0.9724	0.9724
random partition 4, seed 2	0.9723	0.9723
random partition 5, seed 2	0.9720	0.9720
random partition, no flavor boundary	0.9657, 0.9645	0.9657, 0.9645
random partition, b-quark boundary	0.9698, 0.9701	0.9698, 0.9701
random partition, random flavor boundary	0.9698, 0.9698	0.9698, 0.9698
randomly initialized network	0.8058 ± 0.0018	0.8525 ± 0.0019

	ρ	
pretraining	as trained	recomputed
188 classes	1.00	1.00
162 classes	1.049 ± 0.014	1.050 ± 0.014
43 classes	1.171 ± 0.024	1.172 ± 0.025
17 classes	1.538 ± 0.025	1.537 ± 0.026
162 classes, mass regression	1.106, 1.082	1.105, 1.083
17 classes, mass regression	1.571, 1.530	1.572, 1.530
random partition 1, seed 1	1.17	1.17
random partition 2, seed 2	1.17	1.17
random partition 3, seed 1	1.15	1.15
random partition 4, seed 2	1.15	1.15
random partition 5, seed 2	1.16	1.16
random partition, no flavor boundary	1.432, 1.477	1.432, 1.477
random partition, b-quark boundary	1.260, 1.245	1.260, 1.245
random partition, random flavor boundary	1.259, 1.254	1.260, 1.254

Table 10: Uncertainty of the label-set comparisons at the largest training set, in per cent: the standard deviation of the logarithm of the ratio over the pretraining runs, then that from the finite test sample. JetClass-II: the ratio of $1 - \mathrm{AUC}$, the test-sample part from a bootstrap of the class-balanced test jets (200 resamples within each class). Top and quark/gluon: the ratio of R_{30}, the test-sample part from the number of background jets passing the cut.

	linear probe		
ratio	JetClass-II, $1 - \mathrm{AUC}$	top, R_{30}	quark/gluon, R_{30}
162 classes / 188 classes	0.45 / 0.15	25 / 2.6	24 / 3.5
43 classes / 188 classes	0.70 / 0.16	29 / 2.4	20 / 3.6
17 classes / 188 classes	1.3 / 0.21	16 / 2.4	9.8 / 3.6

	fine-tuning		
ratio	JetClass-II, $1 - \mathrm{AUC}$	top, R_{30}	quark/gluon, R_{30}
162 classes / 188 classes	1.1 / 0.13	3.2 / 13	2.6 / 4.6
43 classes / 188 classes	0.20 / 0.13	7.3 / 13	2.8 / 4.6
17 classes / 188 classes	0.29 / 0.15	9.9 / 13	2.6 / 4.6

Table 11: JetClass-II linear probes with 10^6 training jets on the natural-composition test set and on the class-balanced one: AUC averaged over the classes and the ratio ρ of $1 - \mathrm{AUC}$ to the 188-class run. Mean $\pm$ standard deviation over the three pretraining runs of each level of the class hierarchy; configurations with two runs give both values, and a random partition has a single run.

	natural composition	
pretraining	AUC	ρ
188 classes	$0.9760 \pm <0.0001$	1.00
162 classes	0.9748 ± 0.0003	1.049 ± 0.014
43 classes	0.9719 ± 0.0006	1.171 ± 0.024
17 classes	0.9631 ± 0.0006	1.538 ± 0.025
162 classes, mass regression	0.9735, 0.9740	1.106, 1.082
17 classes, mass regression	0.9623, 0.9632	1.571, 1.530
random partition 1, seed 1	0.9719	1.17
random partition 2, seed 2	0.9719	1.17
random partition 3, seed 1	0.9724	1.15
random partition 4, seed 2	0.9723	1.15
random partition 5, seed 2	0.9720	1.16
random partition, no flavor boundary	0.9657, 0.9645	1.432, 1.477
random partition, b-quark boundary	0.9698, 0.9701	1.260, 1.245
random partition, random flavor boundary	0.9698, 0.9698	1.259, 1.254
randomly initialized network, BatchNorm recomputed	0.8525 ± 0.0019	–

	class-balanced	
pretraining	AUC	ρ
188 classes	$0.9801 \pm <0.0001$	1.00
162 classes	0.9796 ± 0.0001	1.026 ± 0.005
43 classes	0.9756 ± 0.0002	1.227 ± 0.009
17 classes	0.9691 ± 0.0004	1.557 ± 0.019
162 classes, mass regression	0.9791, 0.9794	1.053, 1.037
17 classes, mass regression	0.9679, 0.9696	1.617, 1.528
random partition 1, seed 1	0.9776	1.13
random partition 2, seed 2	0.9776	1.13
random partition 3, seed 1	0.9779	1.11
random partition 4, seed 2	0.9778	1.12
random partition 5, seed 2	0.9775	1.13
random partition, no flavor boundary	0.9717, 0.9710	1.424, 1.462
random partition, b-quark boundary	0.9758, 0.9759	1.218, 1.212
random partition, random flavor boundary	0.9760, 0.9760	1.206, 1.208
randomly initialized network, BatchNorm recomputed	0.866 ± 0.004	–

Table 12: ParT's top-tagging training script on our cluster, then with Sophon's output head, then also with Sophon's rescaling of every jet's four-momenta to $p_\mathrm{T} = 500$ GeV (cumulative, one run each), beside ParT's published row and our network trained from scratch, which uses the rescaling.

	accuracy	AUC	R_{50}
ParT's script, unchanged	0.9398	0.9858	424
+ Sophon's output head	0.9403	0.9859	416
+ Sophon's jet-p_T rescaling	0.9385	0.9850	357
ParT, published	0.940	0.9858	413 ± 16
our network from scratch	0.9387 ± 0.0002	0.9852 ± 0.0001	358 ± 11

	R_{30}	background jets passing at R_{30}
ParT's script, unchanged	1,711	118
+ Sophon's output head	1,577	128
+ Sophon's jet-p_T rescaling	1,224	165
ParT, published	$1{,}602 \pm 81$	–
our network from scratch	$1{,}412 \pm 20$	–

Table 13: Linear probes on the frozen features at the largest training set of each task. Upper part: JetClass-II (class-balanced test set) accuracy and AUC averaged over its 188 classes each against the rest, over the classes with the full 5,000 test jets, over the resonant and over the QCD classes; JetClass accuracy and AUC. Lower part: top tagging and quark/gluon discrimination (also on Herwig jets) with the missing inputs substituted as chosen on validation for every model (random draws for top, the training mean for quark/gluon; column headings): accuracy, AUC and the background rejection $1/\varepsilon_B$ at 50% and 30% signal efficiency. The ratio column is ρ, $(1 - \text{AUC})$ over that of the 188-class run of the same seed (a random partition: of its seed). Mean ± standard deviation over the three pretraining runs of each level of the class hierarchy; configurations with two runs give both values, and a random partition has a single run. The 188-class model with kinematic inputs only, a diagnostic of the missing inputs, has one run and no input missing, and is evaluated on top tagging only. AUC: area under the curve of signal efficiency against background efficiency.

	JetClass-II		
pretraining	accuracy	AUC	AUC, full count
188 classes	0.4348 ± 0.0003	$0.9801 \pm <0.0001$	$0.9813 \pm <0.0001$
162 classes	0.4328 ± 0.0008	0.9796 ± 0.0001	0.9810 ± 0.0001
43 classes	0.361 ± 0.005	0.9756 ± 0.0002	0.9769 ± 0.0002
17 classes	0.334 ± 0.007	0.9691 ± 0.0004	0.9708 ± 0.0005
162 classes, mass regression	0.4271, 0.4305	0.9791, 0.9794	0.9805, 0.9808
17 classes, mass regression	0.312, 0.350	0.9679, 0.9696	0.9695, 0.9714
random partition 1, seed 1	0.4201	0.9776	0.9792
random partition 2, seed 2	0.4213	0.9776	0.9791
random partition 3, seed 1	0.4231	0.9779	0.9795
random partition 4, seed 2	0.4228	0.9778	0.9794
random partition 5, seed 2	0.4204	0.9775	0.9791
random partition, no flavor boundary	0.3769, 0.3723	0.9717, 0.9710	0.9736, 0.9729
random partition, b-quark boundary	0.3945, 0.3963	0.9758, 0.9759	0.9774, 0.9775
random partition, random flavor boundary	0.4071, 0.4089	0.9760, 0.9760	0.9777, 0.9777
randomly initialized network	0.0724 ± 0.0015	0.8121 ± 0.0019	0.8123 ± 0.0021
randomly initialized network, BatchNorm recomputed	0.1017 ± 0.0017	0.866 ± 0.004	0.869 ± 0.004

	JetClass-II		
pretraining	AUC, resonant	AUC, QCD	ρ
188 classes	$0.9830 \pm <0.0001$	$0.9632 \pm <0.0001$	1.00
162 classes	0.9829 ± 0.0001	0.9600 ± 0.0002	1.026 ± 0.005
43 classes	0.9785 ± 0.0002	0.9583 ± 0.0003	1.227 ± 0.009
17 classes	0.9729 ± 0.0005	0.9460 ± 0.0006	1.557 ± 0.019
162 classes, mass regression	0.9824, 0.9827	0.9592, 0.9598	1.053, 1.037
17 classes, mass regression	0.9716, 0.9737	0.9458, 0.9457	1.617, 1.528
random partition 1, seed 1	0.9813	0.9556	1.13
random partition 2, seed 2	0.9812	0.9558	1.13
random partition 3, seed 1	0.9815	0.9561	1.11
random partition 4, seed 2	0.9814	0.9564	1.12
random partition 5, seed 2	0.9812	0.9554	1.13
random partition, no flavor boundary	0.9760, 0.9753	0.9460, 0.9449	1.424, 1.462
random partition, b-quark boundary	0.9794, 0.9795	0.9543, 0.9545	1.218, 1.212
random partition, random flavor boundary	0.9798, 0.9798	0.9536, 0.9535	1.206, 1.208
randomly initialized network	0.8152 ± 0.0024	0.794 ± 0.005	–
randomly initialized network, BatchNorm recomputed	0.874 ± 0.004	0.8232 ± 0.0012	–

	JetClass	
pretraining	accuracy	AUC
188 classes	0.7709 ± 0.0023	0.9682 ± 0.0006
162 classes	0.7709 ± 0.0005	0.9683 ± 0.0002
43 classes	0.7695 ± 0.0014	0.9680 ± 0.0004
17 classes	0.7605 ± 0.0027	0.9650 ± 0.0008
162 classes, mass regression	0.7718, 0.7760	0.9684, 0.9697
17 classes, mass regression	0.7624, 0.7612	0.9655, 0.9651
random partition 1, seed 1	0.7528	0.9636
random partition 2, seed 2	0.7553	0.9641
random partition 3, seed 1	0.7546	0.9638
random partition 4, seed 2	0.7537	0.9636
random partition 5, seed 2	0.7538	0.9639
random partition, no flavor boundary	0.7535, 0.7486	0.9633, 0.9622
random partition, b-quark boundary	0.7615, 0.7644	0.9661, 0.9665
random partition, random flavor boundary	0.7593, 0.7564	0.9654, 0.9645
randomly initialized network	0.4077 ± 0.0013	0.8280 ± 0.0011
randomly initialized network, BatchNorm recomputed	0.475 ± 0.005	0.866 ± 0.004

	top missing inputs: random draws	
pretraining	accuracy	AUC
188 classes	0.9112 ± 0.0015	0.9686 ± 0.0007
162 classes	0.9098 ± 0.0031	0.9671 ± 0.0020
43 classes	0.9101 ± 0.0019	0.9681 ± 0.0013
17 classes	0.9152 ± 0.0010	0.9710 ± 0.0006
162 classes, mass regression	0.9166, 0.9123	0.9715, 0.9702
17 classes, mass regression	0.9186, 0.9206	0.9734, 0.9748
random partition 1, seed 1	0.9065	0.9645
random partition 2, seed 2	0.9015	0.9615
random partition 3, seed 1	0.9051	0.9635
random partition 4, seed 2	0.8976	0.9598
random partition 5, seed 2	0.9040	0.9634
random partition, no flavor boundary	0.9145, 0.9103	0.9708, 0.9683
random partition, b-quark boundary	0.9088, 0.9063	0.9673, 0.9657
random partition, random flavor boundary	0.9091, 0.9086	0.9671, 0.9674
randomly initialized network	0.8867 ± 0.0013	0.9408 ± 0.0012
randomly initialized network, BatchNorm recomputed	0.8783 ± 0.0021	0.9354 ± 0.0017
188 classes, kinematic inputs only	0.9163	0.9713

	top missing inputs: random draws	
pretraining	R_{50}	R_{30}
188 classes	87 ± 4	228 ± 25
162 classes	79 ± 9	199 ± 25
43 classes	87.7 ± 3.5	231 ± 8
17 classes	97 ± 4	255 ± 17
162 classes, mass regression	96.8, 103.2	260, 306
17 classes, mass regression	112, 124	306, 378
random partition 1, seed 1	69.4	174
random partition 2, seed 2	62.3	151
random partition 3, seed 1	66.0	164
random partition 4, seed 2	58.7	152
random partition 5, seed 2	65.6	173
random partition, no flavor boundary	98, 87	262, 243
random partition, b-quark boundary	84.0, 75.1	202.5, 211.6
random partition, random flavor boundary	83.1, 84.9	214, 233
randomly initialized network	25.0 ± 0.8	56.8 ± 2.7
randomly initialized network, BatchNorm recomputed	22.6 ± 1.0	48.6 ± 3.0
188 classes, kinematic inputs only	94.2	231

	quark/gluon missing inputs: the training mean	
pretraining	accuracy	AUC
188 classes	0.8076 ± 0.0004	0.8835 ± 0.0002
162 classes	0.8051 ± 0.0015	0.8818 ± 0.0018
43 classes	0.8055 ± 0.0009	0.8821 ± 0.0011
17 classes	0.8064 ± 0.0014	0.8821 ± 0.0015
162 classes, mass regression	0.8075, 0.8073	0.8837, 0.8831
17 classes, mass regression	0.8062, 0.8040	0.8805, 0.8798
random partition 1, seed 1	0.8054	0.8810
random partition 2, seed 2	0.8029	0.8791
random partition 3, seed 1	0.8049	0.8803
random partition 4, seed 2	0.7977	0.8747
random partition 5, seed 2	0.8026	0.8790
random partition, no flavor boundary	0.8056, 0.8006	0.8823, 0.8774
random partition, b-quark boundary	0.8043, 0.8025	0.8813, 0.8797
random partition, random flavor boundary	0.8058, 0.8043	0.8829, 0.8807
randomly initialized network	0.7851 ± 0.0018	0.8612 ± 0.0016
randomly initialized network, BatchNorm recomputed	0.7845 ± 0.0013	0.8600 ± 0.0009

	quark/gluon missing inputs: the training mean	
pretraining	R_{50}	R_{30}
188 classes	26.17 ± 0.35	72.7 ± 2.4
162 classes	25.6 ± 0.4	70.99 ± 0.31
43 classes	25.7 ± 0.6	70.8 ± 1.8
17 classes	25.8 ± 0.7	74.3 ± 1.9
162 classes, mass regression	25.69, 25.53	73.9, 70.7
17 classes, mass regression	25.82, 25.85	68.4, 72.2
random partition 1, seed 1	25.0	69.6
random partition 2, seed 2	24.5	69.3
random partition 3, seed 1	24.0	68.4
random partition 4, seed 2	22.8	64.7
random partition 5, seed 2	24.1	69.6
random partition, no flavor boundary	25.55, 24.64	73.9, 71.7
random partition, b-quark boundary	24.97, 24.58	72.52, 72.20
random partition, random flavor boundary	25.19, 25.50	71.2, 72.8
randomly initialized network	19.40 ± 0.28	56.4 ± 1.0
randomly initialized network, BatchNorm recomputed	18.70 ± 0.18	52.2 ± 0.8

	quark/gluon, Herwig missing inputs: the training mean	
pretraining	accuracy	AUC
188 classes	0.7229 ± 0.0033	0.8075 ± 0.0006
162 classes	0.7234 ± 0.0022	0.8068 ± 0.0022
43 classes	0.7206 ± 0.0005	0.8058 ± 0.0011
17 classes	0.7201 ± 0.0028	0.8053 ± 0.0020
162 classes, mass regression	0.7227, 0.7218	0.8064, 0.8078
17 classes, mass regression	0.7256, 0.7240	0.8079, 0.8078
random partition 1, seed 1	0.7262	0.8086
random partition 2, seed 2	0.7171	0.8029
random partition 3, seed 1	0.7194	0.8039
random partition 4, seed 2	0.7224	0.8052
random partition 5, seed 2	0.7208	0.8023
random partition, no flavor boundary	0.7200, 0.7213	0.8058, 0.8055
random partition, b-quark boundary	0.7218, 0.7209	0.8057, 0.8060
random partition, random flavor boundary	0.7201, 0.7232	0.8065, 0.8060
randomly initialized network	0.7205 ± 0.0021	0.7965 ± 0.0009
randomly initialized network, BatchNorm recomputed	0.7194 ± 0.0034	0.7972 ± 0.0019

	quark/gluon, Herwig missing inputs: the training mean	
pretraining	R_{50}	R_{30}
188 classes	11.81 ± 0.10	35.0 ± 2.6
162 classes	11.6 ± 0.4	35.9 ± 1.7
43 classes	11.52 ± 0.23	34.4 ± 1.7
17 classes	11.50 ± 0.16	35.5 ± 1.0
162 classes, mass regression	11.71, 11.94	34.93, 35.60
17 classes, mass regression	11.93, 11.91	34.7, 36.3
random partition 1, seed 1	11.8	36.2
random partition 2, seed 2	11.2	34.0
random partition 3, seed 1	10.9	32.1
random partition 4, seed 2	11.5	33.8
random partition 5, seed 2	11.2	34.1
random partition, no flavor boundary	11.36, 11.59	34.32, 34.93
random partition, b-quark boundary	11.66, 11.47	35.27, 34.60
random partition, random flavor boundary	11.60, 11.72	34.99, 35.51
randomly initialized network	10.58 ± 0.11	31.6 ± 0.8
randomly initialized network, BatchNorm recomputed	10.48 ± 0.21	30.7 ± 0.5

Table 14: Linear probes against the number of training jets (column headers): JetClass-II AUC averaged over the classes, JetClass accuracy, and R_{30} on top, quark/gluon and Herwig jets with the missing inputs substituted as chosen on validation (random draws for top, the training mean for quark/gluon). Mean $\pm$ standard deviation over the three pretraining runs of each level of the class hierarchy; configurations with two runs give both values, and a random partition has a single run. Here the five random partitions are pooled into one row, mean $\pm$ standard deviation over their runs. The 188-class model with kinematic inputs only, a diagnostic of the missing inputs, has one run and no input missing, and is evaluated on top tagging only. *: a probe that did not converge.

JetClass-II, AUC training jets	1,000	10,000	100,000	1,000,000
188 classes	0.9497 ± 0.0001	0.9701 ± 0.0001	0.9783 ± 0.0001	$0.9801 \pm <0.0001$
162 classes	0.9466 ± 0.0006	0.9692 ± 0.0002	0.9778 ± 0.0001	0.9796 ± 0.0001
43 classes	0.9324 ± 0.0029	0.9633 ± 0.0011	0.9735 ± 0.0003	0.9756 ± 0.0002
17 classes	0.9046 ± 0.0021	0.9508 ± 0.0006	0.9660 ± 0.0004	0.9691 ± 0.0004
162 classes, mass regression	0.9445, 0.9435	0.9684, 0.9684	0.9772, 0.9775	0.9791, 0.9794
17 classes, mass regression	0.8999, 0.9044	0.9483, 0.9518	0.9646, 0.9665	0.9679, 0.9696
random partition, no flavor boundary	0.9227, 0.9206	0.9562, 0.9544	0.9691, 0.9682	0.9717, 0.9710
random partition, b-quark boundary	0.9373, 0.9375	0.9647, 0.9648	0.9735, 0.9737	0.9758, 0.9759
random partition, random flavor boundary	0.9346, 0.9338	0.9642, 0.9645	0.9737, 0.9737	0.9760, 0.9760
random partitions (all five)	0.9401 ± 0.0020	0.9669 ± 0.0003	0.9757 ± 0.0001	0.9777 ± 0.0002
randomly initialized network	0.618 ± 0.004	0.718 ± 0.004	0.7946 ± 0.0023	0.8121 ± 0.0019
randomly initialized network, BatchNorm recomputed	0.67 ± 0.04	0.800 ± 0.013	0.855 ± 0.004	0.866 ± 0.004
JetClass, accuracy training jets	1,000	10,000	100,000	1,000,000
188 classes	0.711 ± 0.006	0.7529 ± 0.0008	0.7686 ± 0.0022	0.7709 ± 0.0023
162 classes	0.714 ± 0.008	0.7537 ± 0.0018	0.7687 ± 0.0002	0.7709 ± 0.0005

continued

43 classes	0.7109 ± 0.0005	0.7521 ± 0.0004	0.7679 ± 0.0012	0.7695 ± 0.0014
17 classes	0.689 ± 0.006	0.742 ± 0.004	0.7586 ± 0.0026	0.7605 ± 0.0027
162 classes, mass regression	0.7083, 0.7093	0.7543, 0.7574	0.7694, 0.7741	0.7718, 0.7760
17 classes, mass regression	0.6863, 0.6947	0.7447, 0.7426	0.7606, 0.7595	0.7624, 0.7612
random partition, no flavor boundary	0.6795, 0.6738	0.7339, 0.7303	0.7517, 0.7470	0.7535, 0.7486
random partition, b-quark boundary	0.6995, 0.7020	0.7443, 0.7455	0.7597, 0.7628	0.7615, 0.7644
random partition, random flavor boundary	0.6920, 0.6930	0.7404, 0.7391	0.7575, 0.7546	0.7593, 0.7564
random partitions (all five)	0.6960 ± 0.0027	0.7380 ± 0.0006	0.7524 ± 0.0010	0.7541 ± 0.0009
randomly initialized network	0.306 ± 0.006	0.3854 ± 0.0016	0.4049 ± 0.0016	0.4077 ± 0.0013
randomly initialized network, BatchNorm recomputed	0.363 ± 0.009	0.450 ± 0.005	0.472 ± 0.006	0.475 ± 0.005

top, R_{30} (missing inputs: random draws)

training jets	1,000	10,000	100,000	1,200,000
188 classes	162 ± 9	227 ± 18	228 ± 23	228 ± 25
162 classes	130 ± 8	199 ± 21	203 ± 26	199 ± 25
43 classes	146 ± 32	233 ± 18	236 ± 10	231 ± 8
17 classes	154 ± 5	252 ± 19	259 ± 20	255 ± 17
162 classes, mass regression	199, 186	248, 306	266, 312	260, 306
17 classes, mass regression	193.0, 191.0	286, 341	306, 379	306, 378
random partition, no flavor boundary	158, 138	265, 230	262, 248	262, 243
random partition, b-quark boundary	139.3, 139.6	209.2, 218.0	204, 217	202.5, 211.6
random partition, random flavor boundary	138.7, 148.5	204, 222	218, 242	214, 233
random partitions (all five)	101 ± 10	165 ± 11	165 ± 12	163 ± 11
randomly initialized network	43.1 ± 1.1	55.3 ± 3.1	57.5 ± 3.0	56.8 ± 2.7
randomly initialized network, BatchNorm recomputed	36.9 ± 2.6	49.0 ± 3.3	49.2 ± 3.1	48.6 ± 3.0
188 classes, kinematic inputs only	133	208	222	231

quark/gluon, R_{30} (missing inputs: the training mean)

training jets	1,000	10,000	100,000	1,600,000
188 classes	66.5 ± 1.7	70.5 ± 1.8	73.4 ± 2.0	72.7 ± 2.4
162 classes	60.2 ± 1.6	68.9 ± 0.9	71.5 ± 0.8	70.99 ± 0.31
43 classes	63.4 ± 2.7	66.9 ± 0.6	70.8 ± 1.4	70.8 ± 1.8
17 classes	62 ± 4	72.0 ± 1.8	74.6 ± 1.5	74.3 ± 1.9
162 classes, mass regression	67.8, 60.6	70.1, 66.4	74.3, 71.9	73.9, 70.7
17 classes, mass regression	66.8, 60.7	64.6, 69.3	68.9, 70.9	68.4, 72.2
random partition, no flavor boundary	57.2, 54.7	70.2, 69.1	72.52, 72.15	73.9, 71.7
random partition, b-quark boundary	64.6, 56.4	68.6, 67.2	73.4, 70.8	72.52, 72.20
random partition, random flavor boundary	59.3, 64.2	68.6, 70.4	71.5, 72.6	71.2, 72.8
random partitions (all five)	55 ± 5	64.8 ± 2.3	68.3 ± 2.2	68.3 ± 2.1
randomly initialized network	43.0 ± 2.5	55.6 ± 1.2	56.7 ± 0.5	56.4 ± 1.0
randomly initialized network, BatchNorm recomputed	42.0 ± 2.3	51.1 ± 1.4	52.2 ± 1.3	52.2 ± 0.8

quark/gluon, Herwig, R_{30} (missing inputs: the training mean)

training jets	1,000	10,000	100,000	1,600,000
188 classes	32.99 ± 0.13	34.5 ± 1.2	34.5 ± 2.4	35.0 ± 2.6
162 classes	31.3 ± 2.5	35.0 ± 2.2	35.8 ± 1.9	35.9 ± 1.7
43 classes	31.8 ± 0.8	32.4 ± 1.7	34.6 ± 1.8	34.4 ± 1.7
17 classes	31.0 ± 2.8	34.5 ± 1.1	35.3 ± 1.0	35.5 ± 1.0
162 classes, mass regression	34.7, 31.7	33.98, 33.85	34.55, 35.47	34.93, 35.60
17 classes, mass regression	33.7, 32.4	33.46, 33.94	34.5, 35.6	34.7, 36.3
random partition, no flavor boundary	26.39, 27.23	34.19, 34.20	34.01, 35.00	34.32, 34.93
random partition, b-quark boundary	32.6, 28.3	33.68, 32.86	34.82, 34.11	35.27, 34.60
random partition, random flavor boundary	29.8, 31.6	33.3, 34.8	34.79, 35.16	34.99, 35.51
random partitions (all five)	28.9 ± 1.8	33.0 ± 1.4	33.7 ± 1.5	34.0 ± 1.4
randomly initialized network	25.2 ± 1.0	30.8 ± 0.4	31.9 ± 1.2	31.6 ± 0.8

continued

randomly initialized network, BatchNorm recomputed	24.7 ± 2.0	29.7 ± 0.5	30.5 ± 0.6	30.7 ± 0.5

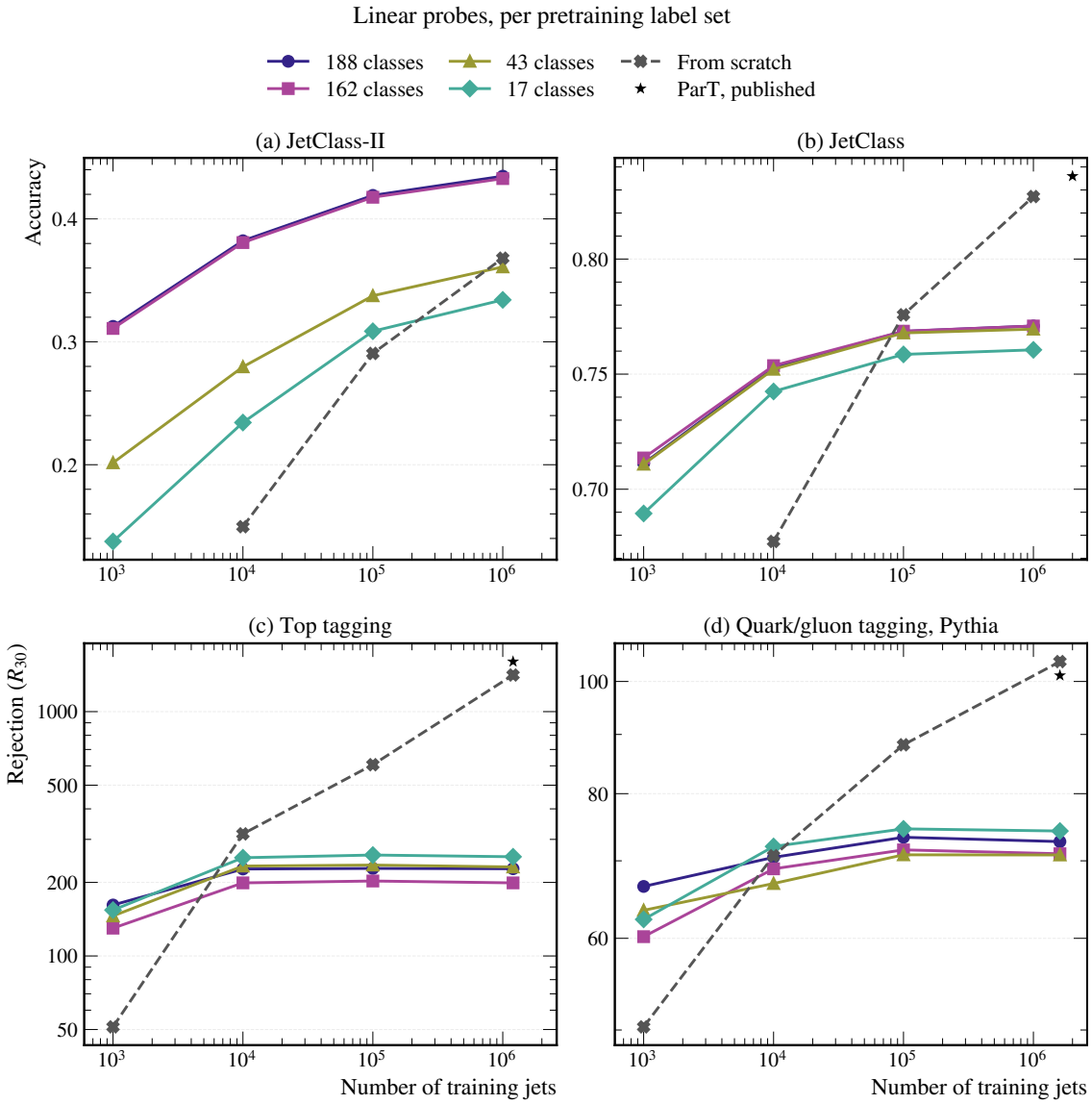


Figure 15: Linear probes of the four label sets against the number of training jets: accuracy on class-balanced JetClass-II and on JetClass, and rejection R_{30} at 30% signal efficiency on top and quark/gluon tagging, with a network trained from scratch dashed. Points are means over three runs; overlapping markers agree within their size. Stars are published ParT: same inputs on JetClass, kinematic only on top, kinematic and particle type on quark/gluon.

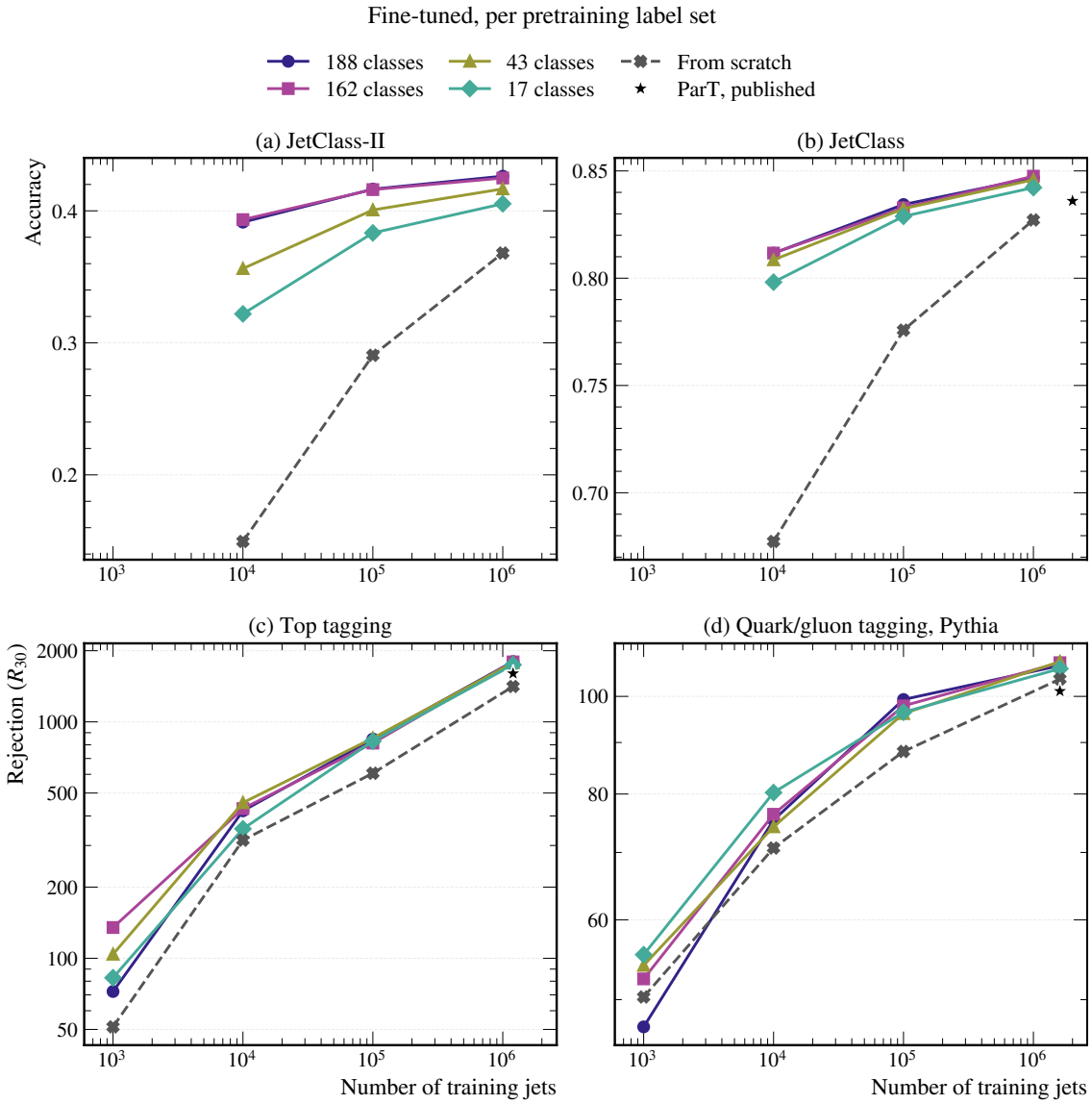


Figure 16: Fine-tuned models of the four label sets against the number of training jets: accuracy on class-balanced JetClass-II and on JetClass, and rejection R_{30} at 30% signal efficiency on top and quark/gluon tagging, with a network trained from scratch dashed. Points are means over three runs; overlapping markers agree within their size. Fine-tuning on JetClass-II and JetClass starts at 10^4 jets. Stars are published ParT: same inputs on JetClass, kinematic only on top, kinematic and particle type on quark/gluon.

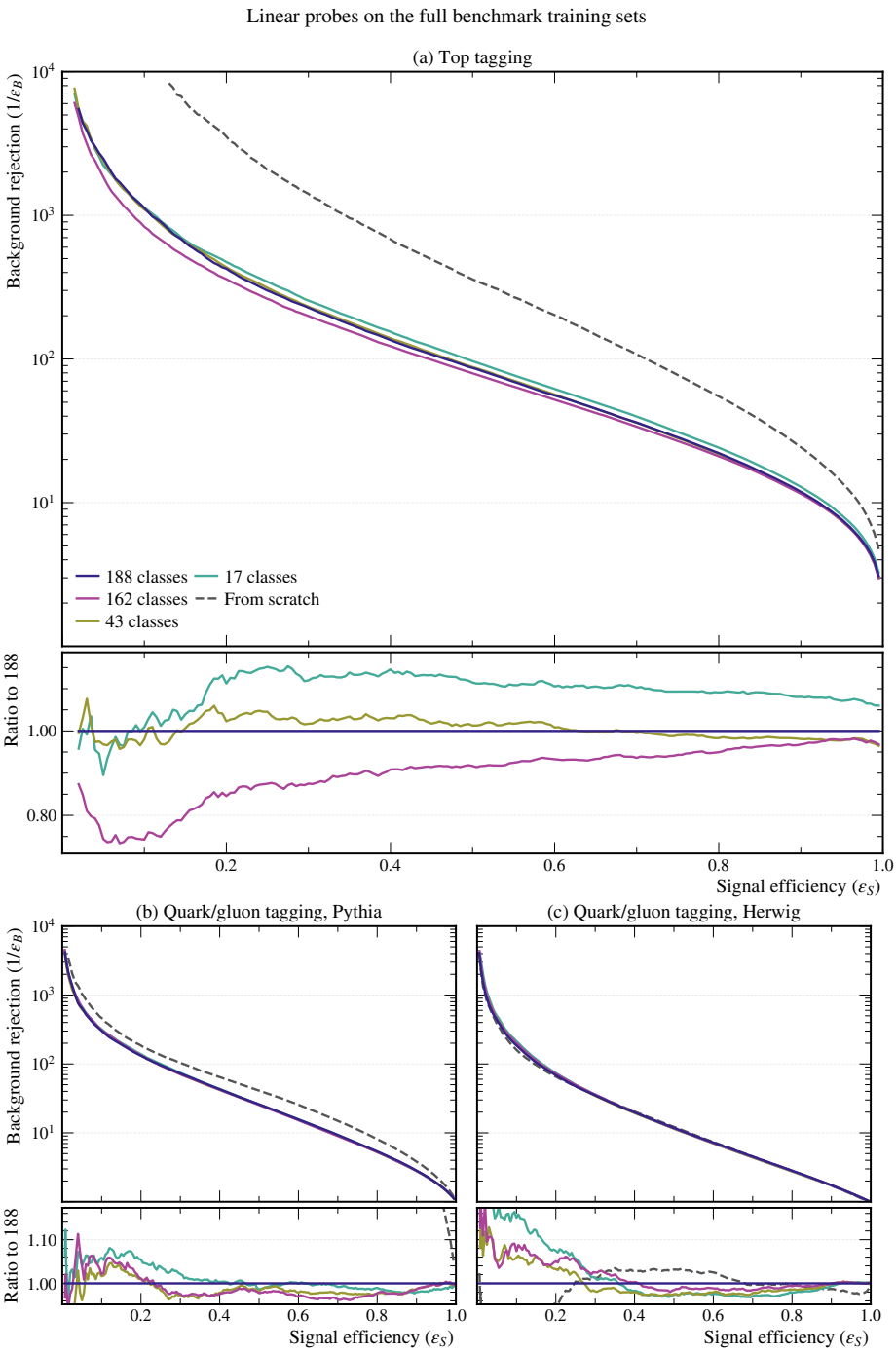


Figure 17: Background rejection against signal efficiency from linear probes on the full training sets (missing inputs filled by random draws on top tagging, by the training mean on quark/gluon), on networks pretrained with 188, 162, 43 and 17 classes, and for reference the network fully trained from scratch (three runs each). Curves end where fewer than 20 background jets remain. Lower panels: ratio to the 188-class probe; from scratch falls outside it in (a), (b) and partly (c).

Table 15: Fine-tuning against the number of training jets (column headers), in each task's metric: AUC averaged over the classes on class-balanced JetClass-II (all classes and the QCD classes), accuracy on JetClass, and R_{30} on top, quark/gluon and Herwig jets. Mean ± standard deviation over the three pretraining runs of each level of the class hierarchy; configurations with two runs give both values, and a random partition has a single run. Each pretrained run is fine-tuned once; from scratch: three training seeds, on the same jets for top and quark/gluon and for JetClass below 10^6 jets, and on 10^6 JetClass jets each on its own training subset. The last column of the JetClass-II block is ρ at 10^6 training jets, $1 - \text{AUC}$ over that of the 188-class run of the same seed. A dash: not run at that size (the random partitions are fine-tuned on JetClass-II at 10^6 jets only). The 188-class model with kinematic inputs only, a diagnostic of the missing inputs, has one run and no input missing, and is evaluated on top tagging only.

JetClass-II, AUC training jets	10,000	100,000	1,000,000	ρ, 10^6 jets
188 classes	0.9722 ± 0.0002	0.9778 ± 0.0002	0.9797 ± 0.0001	1.00
162 classes	0.9725 ± 0.0001	0.9775 ± 0.0001	0.9794 ± 0.0002	1.014 ± 0.011
43 classes	0.9669 ± 0.0006	0.9763 ± 0.0001	0.9788 ± 0.0001	1.043 ± 0.002
17 classes	0.9623 ± 0.0003	0.9743 ± 0.0002	0.9772 ± 0.0001	1.123 ± 0.003
162 classes, mass regression	0.9721, 0.9724	0.9769, 0.9771	0.9790, 0.9792	1.036, 1.024
17 classes, mass regression	0.9625, 0.9627	0.9737, 0.9737	0.9766, 0.9769	1.152, 1.139
random partition 1, seed 1	–	–	0.9783	1.07
random partition 2, seed 2	–	–	0.9783	1.07
random partition 3, seed 1	–	–	0.9786	1.06
random partition 4, seed 2	–	–	0.9786	1.05
random partition 5, seed 2	–	–	0.9783	1.07
random partition, no flavor boundary	–	–	0.9776, 0.9775	1.105, 1.108
random partition, b-quark boundary	–	–	0.9778, 0.9782	1.094, 1.072
random partition, random flavor boundary	–	–	0.9779, 0.9781	1.089, 1.076
from scratch	0.9196 ± 0.0005	0.9611 ± 0.0011	0.9732 ± 0.0004	–

JetClass-II, QCD classes, AUC over the QCD classes training jets	10,000	100,000	1,000,000
188 classes	0.9498 ± 0.0003	0.9598 ± 0.0002	$0.9631 \pm <0.0001$
162 classes	0.9479 ± 0.0001	0.9574 ± 0.0002	0.9618 ± 0.0004
43 classes	0.9416 ± 0.0011	0.9573 ± 0.0001	0.9613 ± 0.0002
17 classes	0.9368 ± 0.0014	0.9543 ± 0.0003	0.9586 ± 0.0001
162 classes, mass regression	0.9463, 0.9481	0.9566, 0.9571	0.9610, 0.9612
17 classes, mass regression	0.9335, 0.9349	0.9533, 0.9531	0.9580, 0.9581
random partition 1, seed 1	–	–	0.9598
random partition 2, seed 2	–	–	0.9598
random partition 3, seed 1	–	–	0.9609
random partition 4, seed 2	–	–	0.9608
random partition 5, seed 2	–	–	0.9602
random partition, no flavor boundary	–	–	0.9592, 0.9589
random partition, b-quark boundary	–	–	0.9593, 0.9607
random partition, random flavor boundary	–	–	0.9595, 0.9603
from scratch	0.8873 ± 0.0010	0.9354 ± 0.0019	0.9530 ± 0.0006

JetClass, accuracy training jets	10,000	100,000	1,000,000
188 classes	0.8117 ± 0.0031	0.8343 ± 0.0035	0.8468 ± 0.0006
162 classes	0.8117 ± 0.0013	0.8328 ± 0.0005	0.8475 ± 0.0009
43 classes	0.8086 ± 0.0031	0.8324 ± 0.0004	0.8459 ± 0.0002
17 classes	0.7982 ± 0.0023	0.8288 ± 0.0005	0.8422 ± 0.0005
162 classes, mass regression	0.8107, 0.8098	0.8326, 0.8348	0.8459, 0.8471
17 classes, mass regression	0.7944, 0.7990	0.8253, 0.8262	0.8400, 0.8398
random partition 1, seed 1	0.8015	0.8304	0.8435
random partition 2, seed 2	0.8055	0.8311	0.8436
random partition 3, seed 1	0.7991	0.8309	0.8448
random partition 4, seed 2	0.8016	0.8291	0.8449
random partition 5, seed 2	0.8004	0.8313	0.8448
random partition, no flavor boundary	0.8000, 0.7993	0.8283, 0.8294	0.8434, 0.8431

continued

random partition, b-quark boundary	0.8058, 0.8032	0.8317, 0.8301	0.8434, 0.8443	
random partition, random flavor boundary	0.8021, 0.8015	0.8300, 0.8319	0.8432, 0.8436	
from scratch	0.677 ± 0.004	0.776 ± 0.005	0.8271 ± 0.0012	
top, R_{30}				
training jets	1,000	10,000	100,000	1,200,000
188 classes	72 ± 19	420 ± 9	850 ± 90	$1{,}800 \pm 60$
162 classes	135 ± 26	430 ± 60	810 ± 60	$1{,}790 \pm 60$
43 classes	104 ± 20	455 ± 30	850 ± 80	$1{,}780 \pm 150$
17 classes	83 ± 25	350 ± 40	830 ± 60	$1{,}740 \pm 140$
162 classes, mass regression	105.9, 112.7	373.2, 376.7	848, 848	1,941, 1,590
17 classes, mass regression	59.1, 63.9	365.8, 359.9	856, 774	1,683, 1,628
random partition 1, seed 1	102	330	711	1,660
random partition 2, seed 2	82.4	394	935	1,900
random partition 3, seed 1	80.1	370	789	1,640
random partition 4, seed 2	80.3	370	742	1,820
random partition 5, seed 2	62.1	291	952	1,740
random partition, no flavor boundary	107, 167	401, 380	841, 777	1,726, 1,852
random partition, b-quark boundary	116, 100	454, 316	878, 926	1,590, 2,060
random partition, random flavor boundary	56, 85	388, 347	783, 756	1,565, 1,655
from scratch	51 ± 5	320 ± 50	610 ± 80	$1{,}412 \pm 20$
188 classes, kinematic inputs only	92.4	397	1,070	1,800
quark/gluon, R_{30}				
training jets	1,000	10,000	100,000	1,600,000
188 classes	47 ± 10	75 ± 7	99.3 ± 1.4	107.3 ± 2.2
162 classes	52.4 ± 3.1	76.4 ± 0.6	97.9 ± 3.3	108.0 ± 1.4
43 classes	54 ± 6	74 ± 5	96.1 ± 1.8	108.3 ± 1.0
17 classes	55 ± 5	80.3 ± 1.1	96.4 ± 1.0	106.5 ± 0.8
162 classes, mass regression	59.3, 57.1	80.1, 72.6	95.6, 97.6	107.0, 106.6
17 classes, mass regression	54.2, 57.7	79.5, 77.9	97.85, 97.94	104.9, 105.6
random partition 1, seed 1	45.1	80.4	98.9	107
random partition 2, seed 2	52.0	81.4	99.6	109
random partition 3, seed 1	43.8	76.6	96.5	106
random partition 4, seed 2	52.5	79.5	99.1	107
random partition 5, seed 2	44.7	58.6	95.2	106
random partition, no flavor boundary	50.9, 58.5	79.2, 86.1	98.91, 99.90	108.9, 109.2
random partition, b-quark boundary	56.4, 55.3	84.2, 81.8	100.0, 96.8	106.5, 106.8
random partition, random flavor boundary	46.4, 51.9	68, 79	97.3, 98.7	108.2, 104.4
from scratch	50.3 ± 2.1	70.7 ± 2.8	88.2 ± 1.1	104.0 ± 1.0
quark/gluon, Herwig, R_{30}				
training jets	1,000	10,000	100,000	1,600,000
188 classes	21 ± 6	27.9 ± 2.8	34.3 ± 1.0	35.2 ± 2.1
162 classes	26.2 ± 0.9	27.4 ± 0.5	35.5 ± 0.5	34.6 ± 2.0
43 classes	26.9 ± 2.7	28.4 ± 2.1	35.5 ± 2.1	36.1 ± 2.7
17 classes	25.2 ± 2.5	32.7 ± 1.8	34.9 ± 1.1	36.9 ± 0.9
162 classes, mass regression	28.0, 29.0	34.3, 28.5	34.0, 36.5	37.8, 34.4
17 classes, mass regression	25.8, 27.4	34.4, 31.2	35.9, 39.0	37.7, 32.9
random partition 1, seed 1	23.9	30.5	34.2	33.4
random partition 2, seed 2	25.2	34.1	35.2	35.2
random partition 3, seed 1	23.5	30.2	34.0	34.8
random partition 4, seed 2	24.6	34.4	33.9	37.5
random partition 5, seed 2	19.7	23.2	36.3	35.3
random partition, no flavor boundary	24.2, 27.7	31.6, 33.3	37.0, 35.1	34.1, 39.3
random partition, b-quark boundary	27.61, 26.89	35.2, 33.5	34.9, 37.0	38.1, 34.7
random partition, random flavor boundary	21.2, 25.3	24.7, 34.4	32.9, 35.1	34.9, 37.4
from scratch	27.3 ± 1.2	32.6 ± 2.9	37.3 ± 1.7	35.5 ± 1.2

Table 16: Fine-tuning on the full training sets of top and quark/gluon (also scored on Herwig jets), every configuration: accuracy, AUC and $1/\varepsilon_B$ at 50% and 30% signal efficiency. Mean ± standard deviation over the three pretraining runs of each level of the class hierarchy; configurations with two runs give both values, and a random partition has a single run. Every rejection rests on at least 98 background jets passing the cut.

pretraining	top			
	accuracy	AUC	R_{50}	R_{30}
188 classes	0.9400 ± 0.0002	0.9861 ± 0.0001	461 ± 4	$1{,}800 \pm 60$
162 classes	0.9401 ± 0.0002	0.9861 ± 0.0001	443 ± 17	$1{,}790 \pm 60$
43 classes	0.9402 ± 0.0004	0.9861 ± 0.0001	450 ± 13	$1{,}780 \pm 150$
17 classes	0.9398 ± 0.0001	0.9859 ± 0.0001	429 ± 10	$1{,}740 \pm 140$
162 classes, mass regression	0.9406, 0.9402	0.9860, 0.9860	420, 436	1,941, 1,590
17 classes, mass regression	0.9397, 0.9399	0.9858, 0.9859	435, 421	1,683, 1,628
random partition 1, seed 1	0.9402	0.9859	432	1,660
random partition 2, seed 2	0.9404	0.9861	430	1,900
random partition 3, seed 1	0.9402	0.9860	421	1,640
random partition 4, seed 2	0.9397	0.9859	458	1,820
random partition 5, seed 2	0.9402	0.9860	431	1,740
random partition, no flavor boundary	0.9402, 0.9401	0.9860, 0.9861	427, 463	1,726, 1,852
random partition, b-quark boundary	0.9401, 0.9405	0.9860, 0.9861	432, 444	1,590, 2,060
random partition, random flavor boundary	0.9397, 0.9406	0.9858, 0.9861	394, 447	1,565, 1,655
from scratch	0.9387 ± 0.0002	0.9852 ± 0.0001	358 ± 11	$1{,}412 \pm 20$
188 classes, kinematic inputs only	0.9407	0.9862	443	1,800

pretraining	quark/gluon	
	accuracy	AUC
188 classes	0.8433 ± 0.0002	0.9148 ± 0.0001
162 classes	$0.8432 \pm <0.0001$	0.9148 ± 0.0001
43 classes	0.8430 ± 0.0005	0.9146 ± 0.0003
17 classes	0.8427 ± 0.0003	$0.9144 \pm <0.0001$
162 classes, mass regression	0.8425, 0.8427	0.9147, 0.9145
17 classes, mass regression	0.8425, 0.8424	0.9142, 0.9140
random partition 1, seed 1	0.8434	0.9146
random partition 2, seed 2	0.8435	0.9147
random partition 3, seed 1	0.8428	0.9145
random partition 4, seed 2	0.8431	0.9146
random partition 5, seed 2	0.8431	0.9146
random partition, no flavor boundary	0.8423, 0.8434	0.9144, 0.9146
random partition, b-quark boundary	0.8430, 0.8431	0.9147, 0.9147
random partition, random flavor boundary	0.8427, 0.8428	0.9145, 0.9147
from scratch	0.8397 ± 0.0007	0.9121 ± 0.0002

pretraining	quark/gluon		quark/gluon, Herwig
	R_{50}	R_{30}	accuracy
188 classes	42.56 ± 0.10	107.3 ± 2.2	0.712 ± 0.005
162 classes	42.39 ± 0.22	108.0 ± 1.4	0.7100 ± 0.0017
43 classes	42.2 ± 0.8	108.3 ± 1.0	0.713 ± 0.006
17 classes	41.97 ± 0.25	106.5 ± 0.8	0.7175 ± 0.0033
162 classes, mass regression	42.46, 42.21	107.0, 106.6	0.7178, 0.7121
17 classes, mass regression	41.77, 41.39	104.9, 105.6	0.7160, 0.7107
random partition 1, seed 1	42.1	107	0.7120
random partition 2, seed 2	42.5	109	0.7121
random partition 3, seed 1	42.1	106	0.7106
random partition 4, seed 2	41.8	107	0.7141
random partition 5, seed 2	42.1	106	0.7132
random partition, no flavor boundary	41.67, 41.95	108.9, 109.2	0.7091, 0.7121
random partition, b-quark boundary	42.43, 42.77	106.5, 106.8	0.7153, 0.7110
random partition, random flavor boundary	42.12, 42.09	108.2, 104.4	0.7114, 0.7171
from scratch	41.47 ± 0.29	104.0 ± 1.0	0.7153 ± 0.0024

pretraining	quark/gluon, Herwig		
	AUC	R_{50}	R_{30}
188 classes	0.8078 ± 0.0023	12.3 ± 0.4	35.2 ± 2.1
162 classes	0.8074 ± 0.0010	12.33 ± 0.28	34.6 ± 2.0
43 classes	0.809 ± 0.005	12.5 ± 0.6	36.1 ± 2.7
17 classes	0.8103 ± 0.0018	12.79 ± 0.20	36.9 ± 0.9
162 classes, mass regression	0.8109, 0.8074	12.99, 12.21	37.8, 34.4
17 classes, mass regression	0.8101, 0.8052	12.58, 11.96	37.7, 32.9
random partition 1, seed 1	0.8072	12.2	33.4
random partition 2, seed 2	0.8078	12.3	35.2
random partition 3, seed 1	0.8072	12.2	34.8
random partition 4, seed 2	0.8090	12.7	37.5
random partition 5, seed 2	0.8076	12.2	35.3
random partition, no flavor boundary	0.8045, 0.8094	12.07, 12.92	34.1, 39.3
random partition, b-quark boundary	0.8109, 0.8066	12.93, 12.19	38.1, 34.7
random partition, random flavor boundary	0.8047, 0.8106	12.18, 12.76	34.9, 37.4
from scratch	0.8061 ± 0.0007	12.18 ± 0.11	35.5 ± 1.2

Table 17: Every pretraining run's values at the largest training set: linear probe and fine-tuning. Linear probes on top and quark/gluon with the missing inputs substituted as chosen on validation (random draws for top, the training mean for quark/gluon); fine-tuning with them set to zero.

run	linear probe				fine-tuning				
	JetClass-II AUC	JetClass AUC	top R_{30}	q/g R_{30}	JetClass-II AUC	JetClass accuracy	top R_{30}	q/g R_{30}	Herwig R_{30}
188 classes, seed 1	0.9801	0.9683	203	75.5	0.9797	0.8475	1,740	108	33.7
188 classes, seed 2	0.9801	0.9687	226	71.7	0.9797	0.8463	1,850	109	37.5
188 classes, seed 3	0.9801	0.9675	253	71.0	0.9796	0.8467	1,800	105	34.2
162 classes, seed 1	0.9795	0.9682	177	71.3	0.9792	0.8479	1,740	110	33.3
162 classes, seed 2	0.9797	0.9683	194	70.9	0.9795	0.8482	1,790	107	36.9
162 classes, seed 3	0.9796	0.9685	226	70.7	0.9794	0.8465	1,850	108	33.5
43 classes, seed 1	0.9758	0.9678	223	72.0	0.9788	0.8459	1,760	107	33.1
43 classes, seed 2	0.9755	0.9678	235	68.8	0.9789	0.8461	1,940	108	38.2
43 classes, seed 3	0.9755	0.9685	237	71.7	0.9787	0.8457	1,640	109	37.1
17 classes, seed 1	0.9695	0.9658	236	72.1	0.9772	0.8418	1,870	106	37.5
17 classes, seed 2	0.9689	0.9643	260	75.8	0.9772	0.8421	1,770	106	37.4
17 classes, seed 3	0.9688	0.9651	268	74.9	0.9771	0.8428	1,590	107	35.8
162 classes, mass regression, seed 1	0.9791	0.9684	260	73.9	0.9790	0.8459	1,940	107	37.8
162 classes, mass regression, seed 2	0.9794	0.9697	306	70.7	0.9792	0.8471	1,590	107	34.4
17 classes, mass regression, seed 1	0.9679	0.9655	306	68.4	0.9766	0.8400	1,680	105	37.7
17 classes, mass regression, seed 2	0.9696	0.9651	378	72.2	0.9769	0.8398	1,630	106	32.9
random partition 1, seed 1	0.9776	0.9636	174	69.6	0.9783	0.8435	1,660	107	33.4
random partition 2, seed 2	0.9776	0.9641	151	69.3	0.9783	0.8436	1,900	109	35.2
random partition 3, seed 1	0.9779	0.9638	164	68.4	0.9786	0.8448	1,640	106	34.8
random partition 4, seed 2	0.9778	0.9636	152	64.7	0.9786	0.8449	1,820	107	37.5
random partition 5, seed 2	0.9775	0.9639	173	69.6	0.9783	0.8448	1,740	106	35.3
random partition, no flavor boundary, seed 1	0.9717	0.9633	262	73.9	0.9776	0.8434	1,730	109	34.1
random partition, no flavor boundary, seed 2	0.9710	0.9622	243	71.7	0.9775	0.8431	1,850	109	39.3
random partition, *b*-quark boundary, seed 1	0.9758	0.9661	203	72.5	0.9778	0.8434	1,590	106	38.1
random partition, *b*-quark boundary, seed 2	0.9759	0.9665	212	72.2	0.9782	0.8443	2,060	107	34.7
random partition, random flavor boundary, seed 1	0.9760	0.9654	214	71.2	0.9779	0.8432	1,570	108	34.9
random partition, random flavor boundary, seed 2	0.9760	0.9645	233	72.8	0.9781	0.8436	1,660	104	37.4
188 classes, kinematic inputs only	–	–	231	–	–	–	1,800	–	–
from scratch, seed 1	–	–	–	–	0.9736	0.8277	1,410	103	34.9
from scratch, seed 2	–	–	–	–	0.9729	0.8280	1,430	105	36.9
from scratch, seed 3	–	–	–	–	0.9731	0.8258	1,390	104	34.7

Table 18: JetClass fine-tuning as table 1, every configuration. Mean ± standard deviation over the three pretraining runs of each level of the class hierarchy; configurations with two runs give both values, and a random partition has a single run. The row of all five random partitions gives the mean ± standard deviation over them. Bold and † as there.

	jets	accuracy	AUC
from scratch	1 M	0.8271 ± 0.0012	0.9819 ± 0.0002
188 classes	1 M	$\mathbf{0.8468 \pm 0.0006}$	0.9854 ± 0.0001
162 classes	1 M	$\mathbf{0.8475 \pm 0.0009}$	$\mathbf{0.9856 \pm <0.0001}$
43 classes	1 M	0.8459 ± 0.0002	0.9853 ± 0.0001
17 classes	1 M	0.8422 ± 0.0005	0.9847 ± 0.0001
162 classes, mass regression	1 M	0.8459, 0.8471	0.9852, 0.9854
17 classes, mass regression	1 M	0.8400, 0.8398	0.9844, 0.9843
random partitions (all five)	1 M	0.8443 ± 0.0007	0.9851 ± 0.0001
random partition, no flavor boundary	1 M	0.8434, 0.8431	0.9849, 0.9848
random partition, b-quark boundary	1 M	0.8434, 0.8443	0.9849, 0.9850
random partition, random flavor boundary	1 M	0.8432, 0.8436	0.9849, 0.9849

		rejection of QCD		
	jets	$H \to b\bar{b}$ R_{50}	$H \to c\bar{c}$ R_{50}	$H \to gg$ R_{50}
from scratch	1 M	$4{,}920 \pm 190$	$1{,}580 \pm 60$	88.3 ± 3.2
188 classes	1 M	$\mathbf{6{,}600 \pm 330}$	$\mathbf{2{,}650 \pm 260}$	$\mathbf{104 \pm 4}$
162 classes	1 M	$6{,}060 \pm 0$	$\mathbf{2{,}820 \pm 40}$	101.8 ± 1.5
43 classes	1 M	$\mathbf{6{,}500 \pm 500}$	$2{,}610 \pm 160$	103.5 ± 1.3
17 classes	1 M	$\mathbf{6{,}670 \pm 220}$	$2{,}390 \pm 40$	$\mathbf{101 \pm 6}$
162 classes, mass regression	1 M	**6,452, 6,667**	2,532, 2,299	105.3, 101.9
17 classes, mass regression	1 M	6,452, 5,556	2,299, 2,020	105.5, 102.7
random partitions (all five)	1 M	$\mathbf{6{,}100 \pm 900}$	$2{,}580 \pm 160$	102.6 ± 1.6
random partition, no flavor boundary	1 M	6,060, 6,060	2,151, 2,174	104.7, 106.3
random partition, b-quark boundary	1 M	**5,714, 7,407**	2,353, 2,410	**104.9, 108.0**
random partition, random flavor boundary	1 M	6,452, 5,882	2,857, 2,632	106.0, 96.8

		rejection of QCD		
	jets	$H \to 4q$ R_{50}	$H \to \ell\nu qq'$ R_{99}	$t \to bqq'$ R_{50}
from scratch	1 M	670 ± 40	$1{,}270 \pm 100$	$6{,}100 \pm 600$
188 classes	1 M	$\mathbf{1{,}091 \pm 15}$	$\mathbf{3{,}100 \pm 400}$	$12{,}900 \pm 1{,}600^\dagger$
162 classes	1 M	$\mathbf{1{,}150 \pm 70}$	$\mathbf{3{,}070 \pm 330}$	$14{,}100 \pm 3{,}500^\dagger$
43 classes	1 M	$1{,}030 \pm 40$	$\mathbf{3{,}160 \pm 280}$	$14{,}200 \pm 2{,}100^\dagger$
17 classes	1 M	$1{,}049 \pm 17$	$2{,}800 \pm 200$	$10{,}600 \pm 2{,}700^\dagger$
162 classes, mass regression	1 M	**1,099, 1,130**	2,778, 2,941	11,111, 14,286†
17 classes, mass regression	1 M	966, 935	**2,941, 2,899**	10,526, 14,286†
random partitions (all five)	1 M	$1{,}006 \pm 31$	$2{,}690 \pm 300$	$10{,}900 \pm 2{,}000^\dagger$
random partition, no flavor boundary	1 M	**1,176, 1,036**	2,820, 2,820	13,333, 16,667†
random partition, b-quark boundary	1 M	1,042, 1,031	2,564, 2,597	12,500, 14,286†
random partition, random flavor boundary	1 M	1,000, 1,047	2,817, 2,667	11,765, 13,333†

		rejection of QCD		
	jets	$t \to b\ell\nu$ $R_{99.5}$	$W \to qq'$ R_{50}	$Z \to q\bar{q}$ R_{50}
from scratch	1 M	$3{,}940 \pm 320$	272 ± 6	215 ± 8
188 classes	1 M	$9{,}400 \pm 1{,}500^\dagger$	374 ± 8	$\mathbf{312 \pm 6}$
162 classes	1 M	$7{,}800 \pm 340$	$\mathbf{389 \pm 9}$	$\mathbf{307.7 \pm 1.2}$
43 classes	1 M	$8{,}450 \pm 210$	376 ± 6	$\mathbf{307 \pm 7}$
17 classes	1 M	$8{,}600 \pm 1{,}300$	$\mathbf{378 \pm 22}$	285 ± 11
162 classes, mass regression	1 M	8,333, 6,897	**392, 380**	302.6, 300.8
17 classes, mass regression	1 M	8,000, 11,111†	363, 353	289, 276
random partitions (all five)	1 M	$7{,}700 \pm 1{,}300$	$\mathbf{379 \pm 18}$	299 ± 7
random partition, no flavor boundary	1 M	6,667, 7,143	351, 351	291.5, 292.0
random partition, b-quark boundary	1 M	7,140, 7,140	366, 347	292.4, 297.6
random partition, random flavor boundary	1 M	7,143, 7,407	371.7, 371.1	295.9, 301.2

Table 19: JetClass linear probes on the frozen features, largest training set, in the columns of table 1 (AUC one against one where the fit records it). Mean ± standard deviation over the three pretraining runs of each level of the class hierarchy; configurations with two runs give both values, and a random partition has a single run.

pretraining	accuracy	AUC	rejection of QCD
			$H \to b\bar{b}$
			R_{50}
188 classes	0.7709 ± 0.0023	0.9682 ± 0.0006	840 ± 90
162 classes	0.7709 ± 0.0005	0.9683 ± 0.0002	800 ± 130
43 classes	0.7695 ± 0.0014	0.9680 ± 0.0004	880 ± 180
17 classes	0.7605 ± 0.0027	0.9650 ± 0.0008	831 ± 16
162 classes, mass regression	0.7718, 0.7760	0.9684, 0.9697	823, 806
17 classes, mass regression	0.7624, 0.7612	0.9655, 0.9651	637, 926
random partition 1, seed 1	0.7528	0.9636	712
random partition 2, seed 2	0.7553	0.9641	658
random partition 3, seed 1	0.7546	0.9638	877
random partition 4, seed 2	0.7537	0.9636	797
random partition 5, seed 2	0.7538	0.9639	749
random partition, no flavor boundary	0.7535, 0.7486	0.9633, 0.9622	775, 508
random partition, b-quark boundary	0.7615, 0.7644	0.9661, 0.9665	990, 1,099
random partition, random flavor boundary	0.7593, 0.7564	0.9654, 0.9645	758, 662
randomly initialized network	0.4077 ± 0.0013	0.8280 ± 0.0011	25.3 ± 1.1
randomly initialized network, BatchNorm recomputed	0.475 ± 0.005	0.866 ± 0.004	46.3 ± 2.3

pretraining	rejection of QCD			
	$H \to c\bar{c}$ R_{50}	$H \to gg$ R_{50}	$H \to 4q$ R_{50}	$H \to \ell\nu qq'$ R_{99}
188 classes	382 ± 34	20.0 ± 1.0	200 ± 15	920 ± 40
162 classes	410 ± 50	20.7 ± 0.8	207 ± 15	930 ± 50
43 classes	440 ± 31	20.8 ± 0.7	207 ± 13	920 ± 70
17 classes	294 ± 20	21.6 ± 0.7	183 ± 12	810 ± 110
162 classes, mass regression	377, 413	19.9, 22.5	210, 243	901, 957
17 classes, mass regression	287, 261	21.90, 22.58	169.6, 171.2	781, 800
random partition 1, seed 1	323	18.3	161	557
random partition 2, seed 2	357	18.7	186	667
random partition 3, seed 1	382	18.5	180	617
random partition 4, seed 2	348	17.7	154	662
random partition 5, seed 2	353	16.5	164	610
random partition, no flavor boundary	233, 244	21.5, 19.2	157.4, 154.2	683, 712
random partition, b-quark boundary	398, 301	20.6, 18.0	205, 195	722, 690
random partition, random flavor boundary	332, 318	18.86, 18.80	184.7, 174.7	660, 685
randomly initialized network	20.4 ± 1.6	17.9 ± 0.8	15.8 ± 1.2	1.452 ± 0.035
randomly initialized network, BatchNorm recomputed	13.7 ± 1.2	15.4 ± 0.5	13.6 ± 0.8	2.1 ± 0.5

pretraining	rejection of QCD			
	$t \to bqq'$ R_{50}	$t \to b\ell\nu$ $R_{99.5}$	$W \to qq'$ R_{50}	$Z \to q\bar{q}$ R_{50}
188 classes	$1{,}100 \pm 80$	$2{,}690 \pm 310$	76.1 ± 1.7	63.1 ± 2.5
162 classes	$1{,}220 \pm 150$	$3{,}400 \pm 600$	81.6 ± 2.1	65.8 ± 3.2
43 classes	$1{,}300 \pm 200$	$3{,}700 \pm 500$	79.1 ± 1.6	69.5 ± 0.9
17 classes	$1{,}180 \pm 310$	$3{,}090 \pm 280$	71 ± 5	68.9 ± 2.8
162 classes, mass regression	980, 1,183	5,000, 4,878	77.9, 72.9	69.1, 67.6
17 classes, mass regression	873, 813	1,923, 2,381	69.7, 65.8	70.9, 68.1
random partition 1, seed 1	847	2,250	67.2	56.1
random partition 2, seed 2	664	1,550	71.3	57.7
random partition 3, seed 1	1,120	2,020	72.4	58.7
random partition 4, seed 2	858	2,500	70.1	56.5
random partition 5, seed 2	840	2,330	69.0	56.0
random partition, no flavor boundary	889, 909	1,449, 2,667	65.5, 69.2	69.2, 68.2
random partition, b-quark boundary	837, 1,058	1,724, 2,222	75.0, 70.6	67.8, 66.1
random partition, random flavor boundary	1,117, 758	2,083, 1,818	72.3, 75.0	66.29, 65.64
randomly initialized network	20.0 ± 1.4	1.204 ± 0.015	17.3 ± 1.3	14.3 ± 1.0
randomly initialized network, BatchNorm recomputed	23.0 ± 1.5	2.3 ± 0.4	13.47 ± 0.28	11.0 ± 0.6

Table 20: Area under the curve of each JetClass class against the other nine at one million training jets, for every configuration. Upper part: after fine-tuning, with the network trained from scratch. Lower part: linear probes on the frozen network, with the randomly initialized network whose BatchNorm statistics are recomputed. Mean ± standard deviation over the three pretraining runs of each level of the class hierarchy; configurations with two runs give both values, and a random partition has a single run.

pretraining	QCD	$H \to b\bar{b}$
fine-tuned		
188 classes	0.9753 ± 0.0001	0.9964 ± 0.0001
162 classes	$0.9753 \pm <0.0001$	0.9964 ± 0.0001
43 classes	$0.9752 \pm <0.0001$	0.9962 ± 0.0001
17 classes	0.9745 ± 0.0004	0.9955 ± 0.0001
162 classes, mass regression	0.9751, 0.9753	0.9961, 0.9963
17 classes, mass regression	0.9744, 0.9745	0.9951, 0.9954
random partition 1, seed 1	0.9749	0.9958
random partition 2, seed 2	0.9749	0.9958
random partition 3, seed 1	0.9751	0.9961
random partition 4, seed 2	0.9748	0.9960
random partition 5, seed 2	0.9748	0.9959
random partition, no flavor boundary	0.9749, 0.9751	0.9957, 0.9957
random partition, b-quark boundary	0.9750, 0.9752	0.9959, 0.9961
random partition, random flavor boundary	0.9750, 0.9746	0.9957, 0.9957
from scratch	0.9714 ± 0.0002	0.9939 ± 0.0001

pretraining	$H \to c\bar{c}$	$H \to gg$	$H \to 4q$
fine-tuned			
188 classes	0.9853 ± 0.0001	0.9774 ± 0.0001	0.9865 ± 0.0001
162 classes	0.9855 ± 0.0002	0.9776 ± 0.0001	0.9867 ± 0.0001
43 classes	0.9850 ± 0.0001	0.9773 ± 0.0001	0.9863 ± 0.0001
17 classes	0.9831 ± 0.0001	0.9764 ± 0.0002	$0.9859 \pm <0.0001$
162 classes, mass regression	0.9845, 0.9851	0.9773, 0.9776	0.9863, 0.9866
17 classes, mass regression	0.9824, 0.9826	0.9759, 0.9761	0.9854, 0.9854
random partition 1, seed 1	0.9838	0.9769	0.9860
random partition 2, seed 2	0.9842	0.9771	0.9862
random partition 3, seed 1	0.9845	0.9768	0.9861
random partition 4, seed 2	0.9841	0.9770	0.9861
random partition 5, seed 2	0.9842	0.9770	0.9862
random partition, no flavor boundary	0.9835, 0.9835	0.9772, 0.9768	0.9863, 0.9859
random partition, b-quark boundary	0.9840, 0.9842	0.9770, 0.9769	0.9861, 0.9863
random partition, random flavor boundary	0.9837, 0.9836	0.9768, 0.9766	0.9861, 0.9860
from scratch	0.9780 ± 0.0001	0.9727 ± 0.0003	0.9822 ± 0.0003

pretraining	$H \to \ell\nu qq'$	$t \to bqq'$
fine-tuned		
188 classes	$0.9993 \pm <0.0001$	0.9977 ± 0.0001
162 classes	$0.9993 \pm <0.0001$	$0.9978 \pm <0.0001$
43 classes	$0.9993 \pm <0.0001$	$0.9977 \pm <0.0001$
17 classes	$0.9992 \pm <0.0001$	$0.9975 \pm <0.0001$
162 classes, mass regression	0.9993, 0.9992	0.9977, 0.9978
17 classes, mass regression	0.9992, 0.9992	0.9972, 0.9974
random partition 1, seed 1	0.9993	0.9975
random partition 2, seed 2	0.9992	0.9975
random partition 3, seed 1	0.9993	0.9976
random partition 4, seed 2	0.9992	0.9975
random partition 5, seed 2	0.9992	0.9975
random partition, no flavor boundary	0.9992, 0.9993	0.9975, 0.9976
random partition, b-quark boundary	0.9993, 0.9993	0.9976, 0.9977
random partition, random flavor boundary	0.9992, 0.9992	0.9974, 0.9975
from scratch	$0.9990 \pm <0.0001$	0.9965 ± 0.0001

pretraining	$t \to b\ell\nu$	$W \to qq'$	$Z \to q\bar{q}$
fine-tuned			
188 classes	$0.9998 \pm <0.0001$	0.9762 ± 0.0003	0.9603 ± 0.0003
162 classes	$0.9998 \pm <0.0001$	0.9766 ± 0.0001	0.9610 ± 0.0002
43 classes	$0.9998 \pm <0.0001$	0.9762 ± 0.0002	0.9604 ± 0.0002
17 classes	$0.9998 \pm <0.0001$	0.9758 ± 0.0002	0.9595 ± 0.0005
162 classes, mass regression	0.9997, 0.9998	0.9764, 0.9764	0.9608, 0.9603
17 classes, mass regression	0.9997, 0.9997	0.9754, 0.9751	0.9592, 0.9578
random partition 1, seed 1	0.9998	0.9761	0.9600
random partition 2, seed 2	0.9998	0.9762	0.9602
random partition 3, seed 1	0.9998	0.9758	0.9594
random partition 4, seed 2	0.9998	0.9764	0.9605
random partition 5, seed 2	0.9998	0.9763	0.9604
random partition, no flavor boundary	0.9997, 0.9997	0.9755, 0.9755	0.9593, 0.9594
random partition, b-quark boundary	0.9998, 0.9998	0.9756, 0.9756	0.9591, 0.9592
random partition, random flavor boundary	0.9998, 0.9998	0.9758, 0.9761	0.9600, 0.9602
from scratch	$0.9996 \pm <0.0001$	0.9726 ± 0.0005	0.9541 ± 0.0011

pretraining	QCD	$H \to b\bar{b}$
linear probe, frozen		
188 classes	0.9474 ± 0.0005	0.9912 ± 0.0003
162 classes	0.9482 ± 0.0010	0.9912 ± 0.0001
43 classes	0.9484 ± 0.0004	0.9909 ± 0.0002
17 classes	0.9467 ± 0.0010	0.9873 ± 0.0003
162 classes, mass regression	0.9473, 0.9492	0.9912, 0.9912
17 classes, mass regression	0.9476, 0.9460	0.9865, 0.9873
random partition 1, seed 1	0.9449	0.9890
random partition 2, seed 2	0.9446	0.9889
random partition 3, seed 1	0.9456	0.9889
random partition 4, seed 2	0.9433	0.9891
random partition 5, seed 2	0.9432	0.9890
random partition, no flavor boundary	0.9462, 0.9453	0.9850, 0.9812
random partition, b-quark boundary	0.9476, 0.9445	0.9889, 0.9898
random partition, random flavor boundary	0.9450, 0.9448	0.9882, 0.9876
randomly initialized network, BatchNorm recomputed	0.850 ± 0.007	0.855 ± 0.009

pretraining	$H \to c\bar{c}$	$H \to gg$	$H \to 4q$
linear probe, frozen			
188 classes	0.9719 ± 0.0006	0.9640 ± 0.0003	0.9776 ± 0.0002
162 classes	0.9720 ± 0.0005	0.9642 ± 0.0002	0.9775 ± 0.0002
43 classes	0.9719 ± 0.0003	0.9632 ± 0.0004	0.9769 ± 0.0002
17 classes	0.9568 ± 0.0020	0.9622 ± 0.0007	0.9746 ± 0.0009
162 classes, mass regression	0.9720, 0.9732	0.9638, 0.9651	0.9775, 0.9782
17 classes, mass regression	0.9573, 0.9559	0.9622, 0.9621	0.9753, 0.9746
random partition 1, seed 1	0.9653	0.9606	0.9735
random partition 2, seed 2	0.9661	0.9612	0.9741
random partition 3, seed 1	0.9671	0.9599	0.9746
random partition 4, seed 2	0.9666	0.9610	0.9743
random partition 5, seed 2	0.9656	0.9611	0.9750
random partition, no flavor boundary	0.9564, 0.9495	0.9619, 0.9624	0.9739, 0.9736
random partition, b-quark boundary	0.9649, 0.9657	0.9633, 0.9635	0.9766, 0.9771
random partition, random flavor boundary	0.9644, 0.9630	0.9629, 0.9626	0.9755, 0.9749
randomly initialized network, BatchNorm recomputed	0.7793 ± 0.0018	0.883 ± 0.004	0.837 ± 0.005

pretraining	$H \to \ell\nu qq'$	$t \to bqq'$
linear probe, frozen		
188 classes	0.9988 ± 0.0001	0.9925 ± 0.0003
162 classes	$0.9988 \pm <0.0001$	0.9925 ± 0.0001
43 classes	$0.9988 \pm <0.0001$	0.9924 ± 0.0002
17 classes	$0.9985 \pm <0.0001$	0.9905 ± 0.0008
162 classes, mass regression	0.9988, 0.9988	0.9925, 0.9929
17 classes, mass regression	0.9983, 0.9984	0.9909, 0.9900
random partition 1, seed 1	0.9985	0.9898
random partition 2, seed 2	0.9985	0.9899
random partition 3, seed 1	0.9984	0.9896
random partition 4, seed 2	0.9985	0.9905
random partition 5, seed 2	0.9985	0.9894
random partition, no flavor boundary	0.9982, 0.9981	0.9894, 0.9896
random partition, b-quark boundary	0.9985, 0.9985	0.9907, 0.9914
random partition, random flavor boundary	0.9983, 0.9981	0.9901, 0.9897
randomly initialized network, BatchNorm recomputed	0.906 ± 0.017	0.9298 ± 0.0009

pretraining	$t \to b\ell\nu$	$W \to qq'$	$Z \to q\bar{q}$
linear probe, frozen			
188 classes	$0.9995 \pm <0.0001$	0.9428 ± 0.0012	0.8958 ± 0.0029
162 classes	$0.9995 \pm <0.0001$	0.9431 ± 0.0005	0.8962 ± 0.0002
43 classes	$0.9995 \pm <0.0001$	0.9424 ± 0.0014	0.8959 ± 0.0023
17 classes	0.9993 ± 0.0001	0.9409 ± 0.0006	0.8937 ± 0.0025
162 classes, mass regression	0.9995, 0.9995	0.9428, 0.9461	0.8987, 0.9021
17 classes, mass regression	0.9992, 0.9991	0.9423, 0.9421	0.8954, 0.8958
random partition 1, seed 1	0.9993	0.9334	0.8818
random partition 2, seed 2	0.9993	0.9351	0.8831
random partition 3, seed 1	0.9993	0.9345	0.8804
random partition 4, seed 2	0.9993	0.9335	0.8796
random partition 5, seed 2	0.9992	0.9348	0.8831
random partition, no flavor boundary	0.9991, 0.9990	0.9363, 0.9367	0.8867, 0.8867
random partition, b-quark boundary	0.9993, 0.9993	0.9403, 0.9404	0.8907, 0.8948
random partition, random flavor boundary	0.9992, 0.9992	0.9386, 0.9366	0.8918, 0.8884
randomly initialized network, BatchNorm recomputed	0.954 ± 0.006	0.8703 ± 0.0031	0.791 ± 0.008

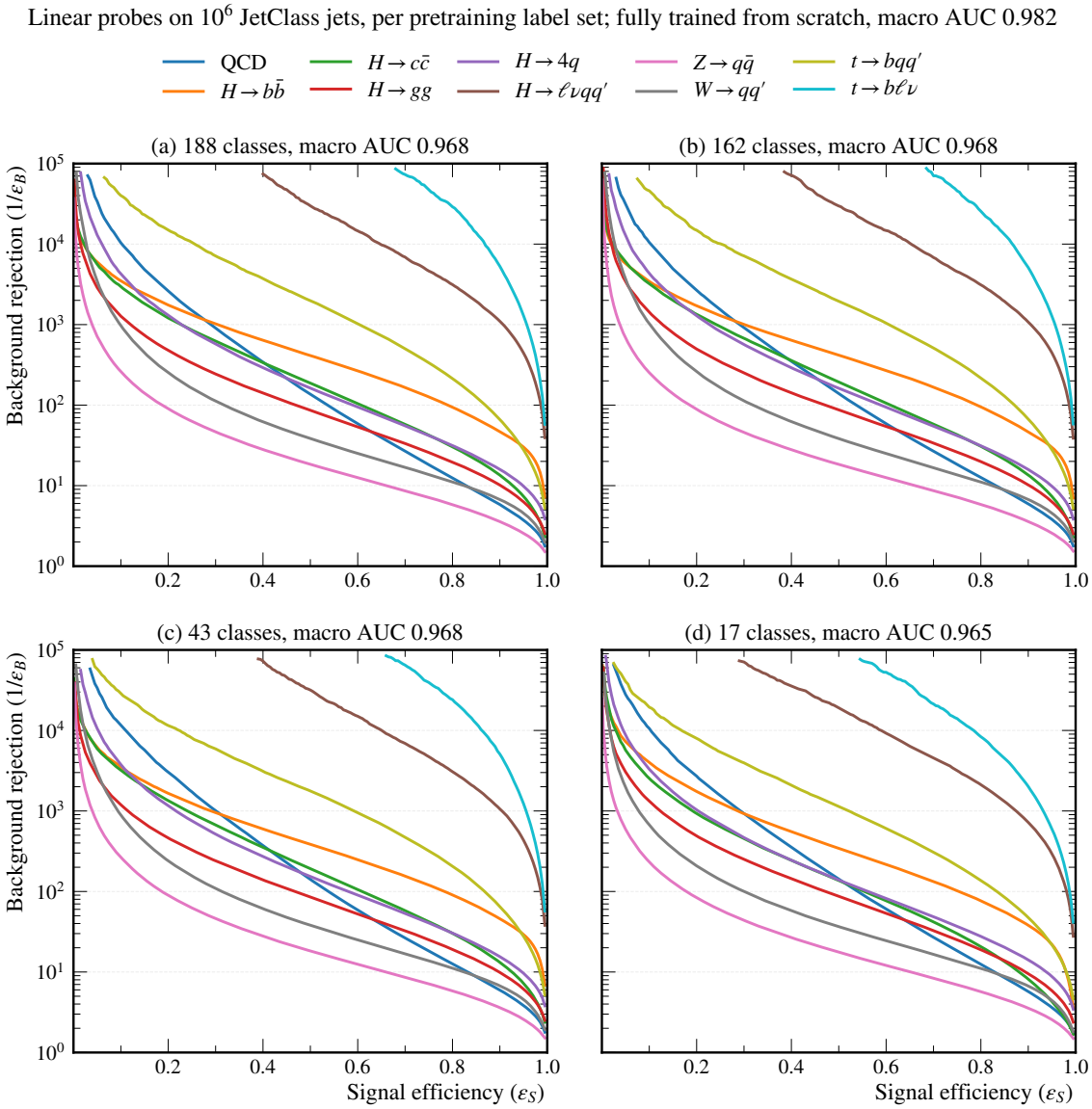


Figure 18: Background rejection against signal efficiency on JetClass from linear probes, trained on one million jets, on frozen pretrained networks. Each curve takes one of the ten classes as signal and the other nine as background, and is the mean over three pretraining runs. Curves begin where 20 background jets remain. Titles give the mean area under the curve; the heading gives the fully trained network from scratch for reference.

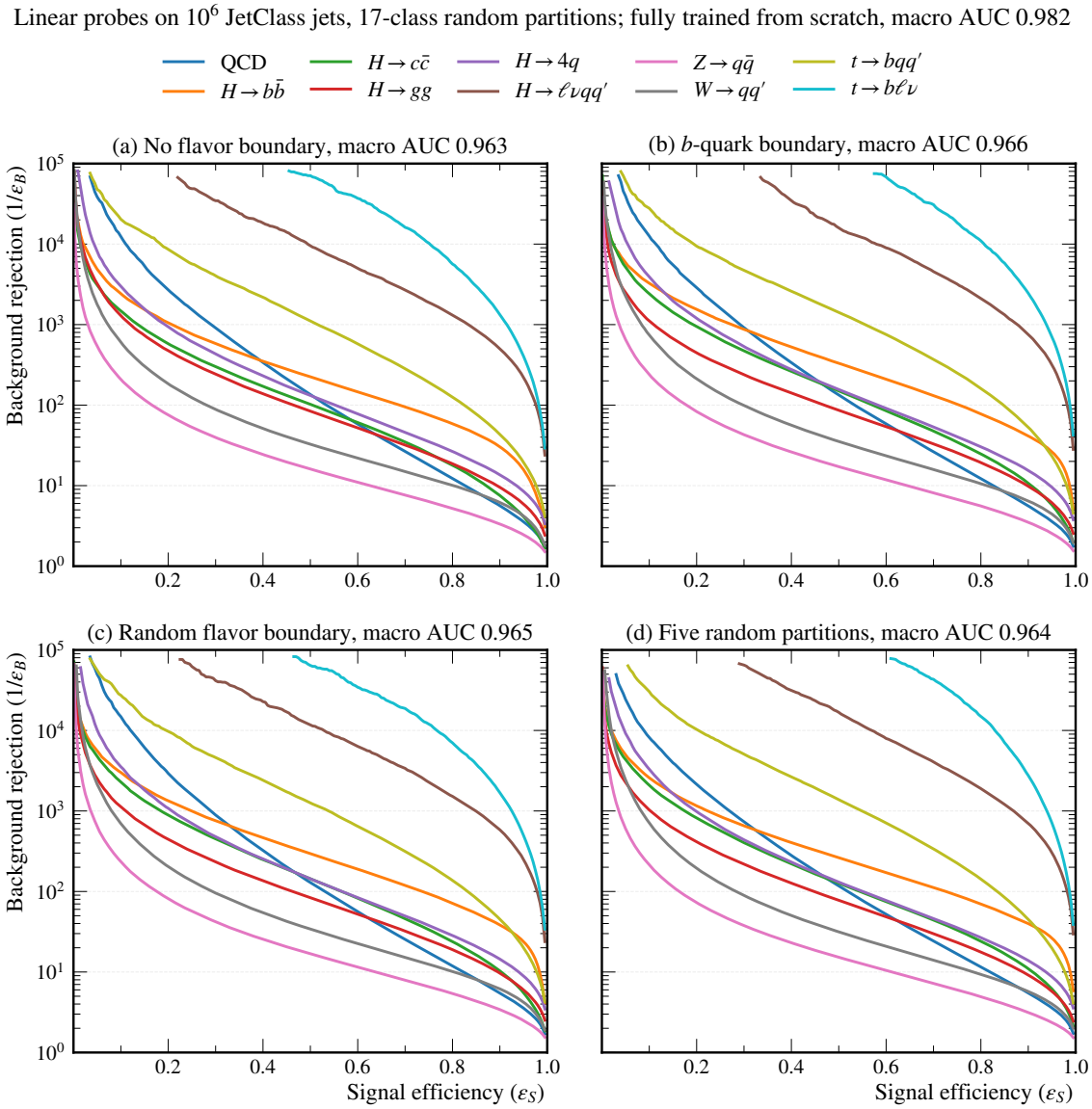


Figure 19: Background rejection against signal efficiency on JetClass from linear probes trained on one million jets, on frozen networks pretrained on 17-class random partitions: with no flavor boundary, a b-quark boundary or a random flavor boundary (two runs each), and five plain ones (averaged). Each curve takes one class as signal against the other nine and begins where 20 background jets remain. Titles give the mean area under the curve; the heading, from scratch.

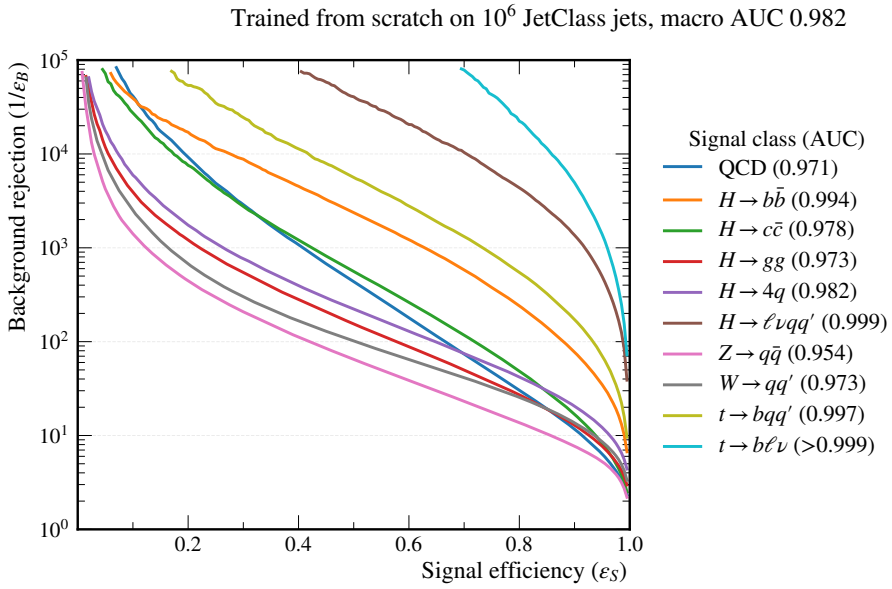


Figure 20: Background rejection against signal efficiency on JetClass for the network trained from scratch on one million jets, with no pretraining. Each curve takes one of the ten JetClass classes as signal and the other nine as background, and is the mean over three trainings. Curves begin where 20 background jets remain. The legend gives each class's area under the curve; the heading gives their mean.

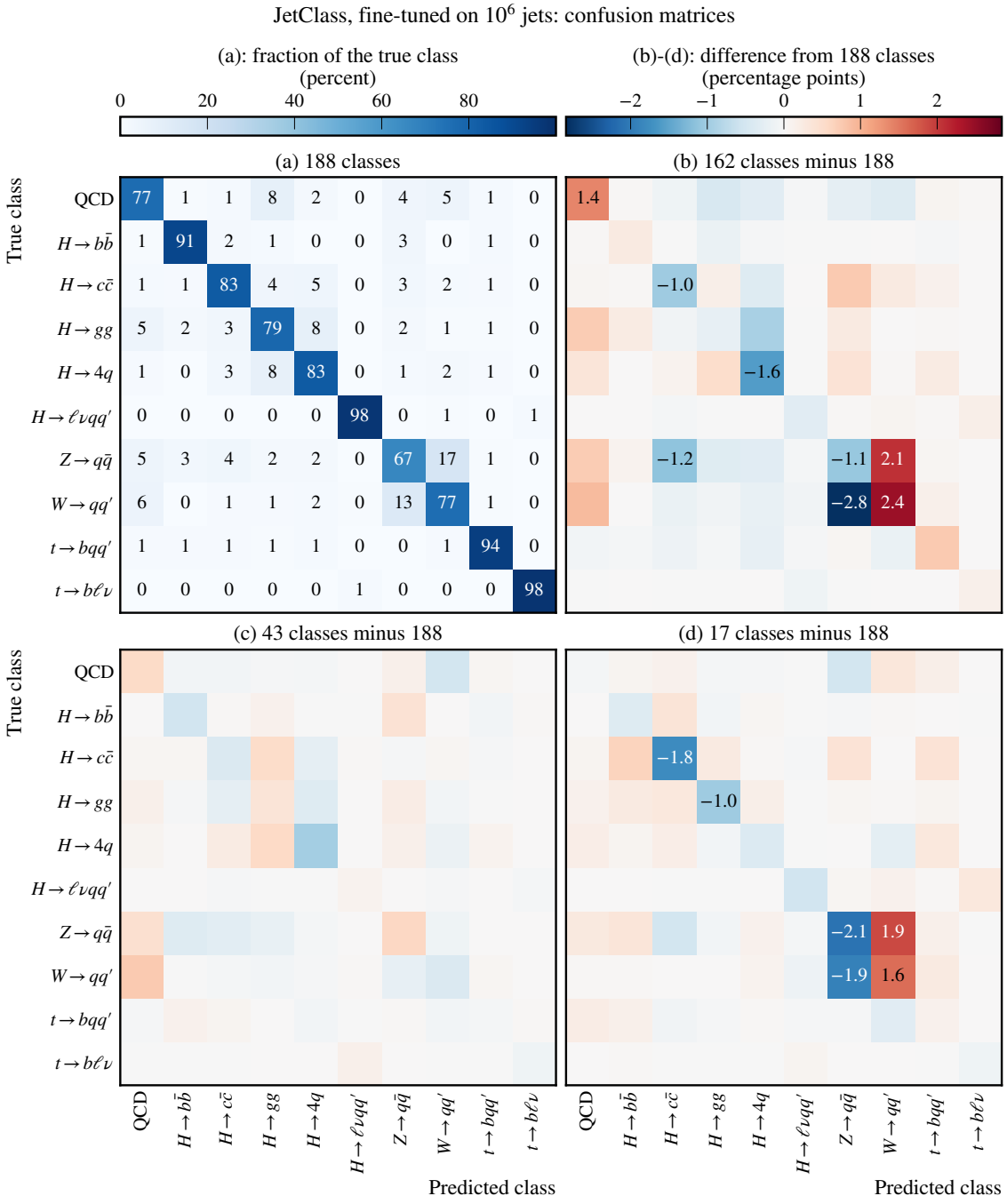


Figure 21: Confusion matrices on JetClass after fine-tuning on one million jets: (a) the percentage of each true class assigned to each predicted class by the model pretrained on 188 classes; (b)–(d) the change in that percentage for the models pretrained on 162, 43 and 17 classes, each against the 188-class model of the same seed, printed where it reaches one percentage point, which no change in (c) does. Values are means over three runs.

Table 21: QCD jets passing each signal class's working point after fine-tuning, the fewest over the runs of each configuration: the counts behind the rejections of tables 1 and 18.

fine-tuned from	$H \to b\bar{b}$	$H \to c\bar{c}$	$H \to gg$	$H \to 4q$	$H \to \ell\nu qq'$
from scratch	39	123	2,216	286	145
188 classes	29	68	1,884	181	57
162 classes	33	70	1,933	166	58
43 classes	29	72	1,914	187	58
17 classes	29	82	1,873	188	66
162 classes, mass regression	30	79	1,899	177	68
17 classes, mass regression	31	87	1,896	207	68
random partitions (all five)	26	71	1,921	194	64
random partition, no flavor boundary	33	92	1,882	170	71
random partition, b-quark boundary	27	83	1,851	192	77
random partition, random flavor boundary	31	70	1,886	191	71

fine-tuned from	$t \to bqq'$	$t \to b\ell\nu$	$W \to qq'$	$Z \to q\bar{q}$
from scratch	30	48	724	897
188 classes	14	19	524	628
162 classes	11	25	508	648
43 classes	13	23	525	636
17 classes	15	21	503	679
162 classes, mass regression	14	24	510	661
17 classes, mass regression	14	18	551	691
random partitions (all five)	15	22	492	649
random partition, no flavor boundary	12	28	569	685
random partition, b-quark boundary	14	28	546	672
random partition, random flavor boundary	15	27	538	664

Table 22: The random partitions on JetClass: none keeps a boundary between two, three and four prongs (every one has a group holding classes of each two of these prong counts); the groups that mix prong counts, and its fine-tuned rejection of QCD per signal class over that of the 188-class run of its seed. †: fewer than 20 QCD jets pass in one of the two runs.

		rejection over 188 classes			
partition	mixed groups	$H \to b\bar{b}$	$H \to c\bar{c}$	$H \to gg$	$H \to 4q$
random partition 1, seed 1	12	0.842	1.04	0.984	0.862
random partition 2, seed 2	12	0.853	1.10	0.944	0.935
random partition 3, seed 1	13	0.889	1.09	0.966	0.928
random partition 4, seed 2	12	0.906	0.940	0.966	0.959
random partition 5, seed 2	11	1.12	0.963	0.981	0.944

	rejection over 188 classes				
partition	$H \to \ell\nu qq'$	$t \to bqq'$	$t \to b\ell\nu$	$W \to qq'$	$Z \to q\bar{q}$
random partition 1, seed 1	1.12	$0.652^\dagger$	1.13	0.991	0.929
random partition 2, seed 2	0.958	$0.875^\dagger$	0.606	1.03	0.929
random partition 3, seed 1	0.935	$1.00^\dagger$	1.18	0.934	0.939
random partition 4, seed 2	0.908	$0.636^\dagger$	0.667	1.11	0.971
random partition 5, seed 2	0.793	$0.778^\dagger$	0.800	1.03	0.991

Table 23: Binary linear probes between pairs of JetClass-II classes: the AUC and the ratio ρ of $1 - \mathrm{AUC}$ to the 188-class run of the same seed. †: the label set merges the two sides into one class; ‡: it merges the first side with part of the second. Mean $\pm$ standard deviation over the three pretraining runs of each level of the class hierarchy; configurations with two runs give both values, and a random partition has a single run. The b-quark boundary and the random flavor boundary lie only in the four-quark decays; every label set but the 188-class one merges all QCD classes. The randomly initialized network, as initialized and with its BatchNorm statistics recomputed, has no label set and no ρ. The $X \to ee$ vs $\mu\mu$ probe uses the jets of the two classes in the class-balanced sets (9,571 training jets, 128,833–135,245 test jets per side). A dash in a ratio column: the 188-class run of that seed separates the test jets almost completely ($1 - \mathrm{AUC} < 10^{-4}$), so a ratio to it would mean nothing.

pretraining	$X \to bc$ vs bq, cs	
	AUC	ρ
188 classes	0.9409 ± 0.0003	1.00
162 classes	0.9410 ± 0.0002	0.998 ± 0.007
43 classes	$0.9319 \pm 0.0014^{\ddagger}$	1.151 ± 0.029
17 classes	$0.885 \pm 0.004^{\dagger}$	1.95 ± 0.06
162 classes, mass regression	0.9375, 0.9400	1.057, 1.010
17 classes, mass regression	0.8808, $0.8833^{\dagger}$	2.018, 1.964
random partition 1, seed 1	$0.9311^{\dagger}$	1.17
random partition 2, seed 2	$0.9298^{\ddagger}$	1.18
random partition 3, seed 1	0.9369	1.07
random partition 4, seed 2	0.9361	1.08
random partition 5, seed 2	$0.9266^{\dagger}$	1.24
random partition, no flavor boundary	0.882, $0.870^{\dagger}$	2.00, 2.19
random partition, b-quark boundary	0.9103, $0.9091^{\dagger}$	1.518, 1.530
random partition, random flavor boundary	0.9096, $0.9100^{\dagger}$	1.530, 1.515
randomly initialized network	0.6011 ± 0.0028	–
randomly initialized network, BatchNorm recomputed	0.653 ± 0.009	–

pretraining	$X \to b\bar{b}$ vs $c\bar{c}$	
	AUC	ρ
188 classes	$0.9977 \pm <0.0001$	1.00
162 classes	0.9976 ± 0.0001	1.015 ± 0.027
43 classes	$0.9974 \pm <0.0001$	1.102 ± 0.013
17 classes	$0.9902 \pm 0.0007^{\dagger}$	4.22 ± 0.29
162 classes, mass regression	0.9973, 0.9975	1.184, 1.087
17 classes, mass regression	0.9891, $0.9904^{\dagger}$	4.70, 4.13
random partition 1, seed 1	$0.9965^{\dagger}$	1.51
random partition 2, seed 2	0.9964	1.54
random partition 3, seed 1	$0.9965^{\dagger}$	1.53
random partition 4, seed 2	0.9968	1.37
random partition 5, seed 2	$0.9961^{\dagger}$	1.67
random partition, no flavor boundary	0.986, $0.969^{\dagger}$	5.9, 13.4
random partition, b-quark boundary	0.9957, $0.9959^{\dagger}$	1.837, 1.748
random partition, random flavor boundary	0.9940, $0.9924^{\dagger}$	2.58, 3.25
randomly initialized network	0.701 ± 0.012	–
randomly initialized network, BatchNorm recomputed	0.817 ± 0.020	–

pretraining	$X \to YY \to b\bar{b}qq$ vs $c\bar{c}qq$		$X \to ee$ vs $\mu\mu$	
	AUC	ρ	AUC	ρ
188 classes	$0.9939 \pm <0.0001$	1.00	$1.0000 \pm <0.0001$	–
162 classes	0.9939 ± 0.0002	1.005 ± 0.032	$1.0000 \pm <0.0001$	–
43 classes	$0.9932 \pm <0.0001$	1.117 ± 0.008	$0.9610 \pm 0.0030^{\dagger}$	–
17 classes	$0.9716 \pm 0.0033^{\dagger}$	4.7 ± 0.5	$0.960 \pm 0.020^{\dagger}$	–
162 classes, mass regression	0.9930, 0.9935	1.164, 1.070	1.0000, 1.0000	–
17 classes, mass regression	0.9676, $0.9706^{\dagger}$	5.36, 4.82	0.969, $0.998^{\dagger}$	–
random partition 1, seed 1	$0.9914^{\dagger}$	1.43	$1.0000^{\dagger}$	–
random partition 2, seed 2	0.9911	1.46	$1.0000^{\dagger}$	–
random partition 3, seed 1	$0.9914^{\dagger}$	1.43	1.0000	–
random partition 4, seed 2	0.9919	1.32	$1.0000^{\dagger}$	–
random partition 5, seed 2	0.9910	1.48	1.0000	–
random partition, no flavor boundary	0.961, $0.943^{\dagger}$	6.4, 9.4	1.0000, 1.0000	–
random partition, b-quark boundary	0.9909, 0.9912	1.510, 1.447	1.0000, 1.0000	–
random partition, random flavor boundary	0.9863, $0.9860^{\dagger}$	2.272, 2.281	1.0000, 1.0000	–
randomly initialized network	0.680 ± 0.007	–	0.9995 ± 0.0003	–
randomly initialized network, BatchNorm recomputed	0.782 ± 0.019	–	$1.0000 \pm <0.0001$	–

pretraining	QCD $b\bar{b}$ vs $c\bar{c}$		QCD with b vs light
	AUC	ρ	AUC
188 classes	0.9867 ± 0.0001	1.00	0.9880 ± 0.0001
162 classes	$0.9848 \pm 0.0004^\dagger$	1.15 ± 0.04	$0.9870 \pm 0.0001^\dagger$
43 classes	$0.9846 \pm 0.0001^\dagger$	1.163 ± 0.016	$0.9870 \pm 0.0001^\dagger$
17 classes	$0.9563 \pm 0.0003^\dagger$	3.294 ± 0.033	$0.9815 \pm 0.0007^\dagger$
162 classes, mass regression	0.9829, $0.9842^\dagger$	1.30, 1.19	0.9868, $0.9870^\dagger$
17 classes, mass regression	0.9481, $0.9531^\dagger$	3.95, 3.51	0.9804, $0.9800^\dagger$
random partition 1, seed 1	$0.9795^\dagger$	1.56	$0.9857^\dagger$
random partition 2, seed 2	$0.9789^\dagger$	1.58	$0.9854^\dagger$
random partition 3, seed 1	$0.9802^\dagger$	1.51	$0.9856^\dagger$
random partition 4, seed 2	$0.9809^\dagger$	1.43	$0.9860^\dagger$
random partition 5, seed 2	$0.9786^\dagger$	1.60	$0.9857^\dagger$
random partition, no flavor boundary	0.944, $0.930^\dagger$	4.24, 5.21	0.9814, $0.9825^\dagger$
random partition, b-quark boundary	0.9795, $0.9795^\dagger$	1.561, 1.532	0.9855, $0.9854^\dagger$
random partition, random flavor boundary	0.9721, $0.9707^\dagger$	2.124, 2.189	0.9848, $0.9845^\dagger$
randomly initialized network	0.655 ± 0.005	–	0.742 ± 0.008
randomly initialized network, BatchNorm recomputed	0.765 ± 0.016	–	0.829 ± 0.012

pretraining	QCD with b vs light	QCD with c, no b, vs light	
	ρ	AUC	ρ
188 classes	1.00	0.9324 ± 0.0003	1.00
162 classes	1.085 ± 0.004	$0.9282 \pm 0.0001^\dagger$	1.061 ± 0.005
43 classes	1.081 ± 0.008	$0.9275 \pm 0.0003^\dagger$	1.072 ± 0.008
17 classes	1.54 ± 0.07	$0.9048 \pm 0.0010^\dagger$	1.407 ± 0.021
162 classes, mass regression	1.107, 1.078	0.9262, $0.9273^\dagger$	1.093, 1.070
17 classes, mass regression	1.638, 1.662	0.9046, $0.9002^\dagger$	1.414, 1.470
random partition 1, seed 1	1.20	$0.9219^\dagger$	1.16
random partition 2, seed 2	1.21	$0.9210^\dagger$	1.16
random partition 3, seed 1	1.21	$0.9229^\dagger$	1.14
random partition 4, seed 2	1.16	$0.9240^\dagger$	1.12
random partition 5, seed 2	1.19	$0.9223^\dagger$	1.14
random partition, no flavor boundary	1.56, 1.45	0.9088, $0.9122^\dagger$	1.352, 1.293
random partition, b-quark boundary	1.212, 1.213	0.9166, $0.9171^\dagger$	1.237, 1.220
random partition, random flavor boundary	1.277, 1.286	0.9204, $0.9211^\dagger$	1.179, 1.162
randomly initialized network	–	0.656 ± 0.005	–
randomly initialized network, BatchNorm recomputed	–	0.6821 ± 0.0003	–

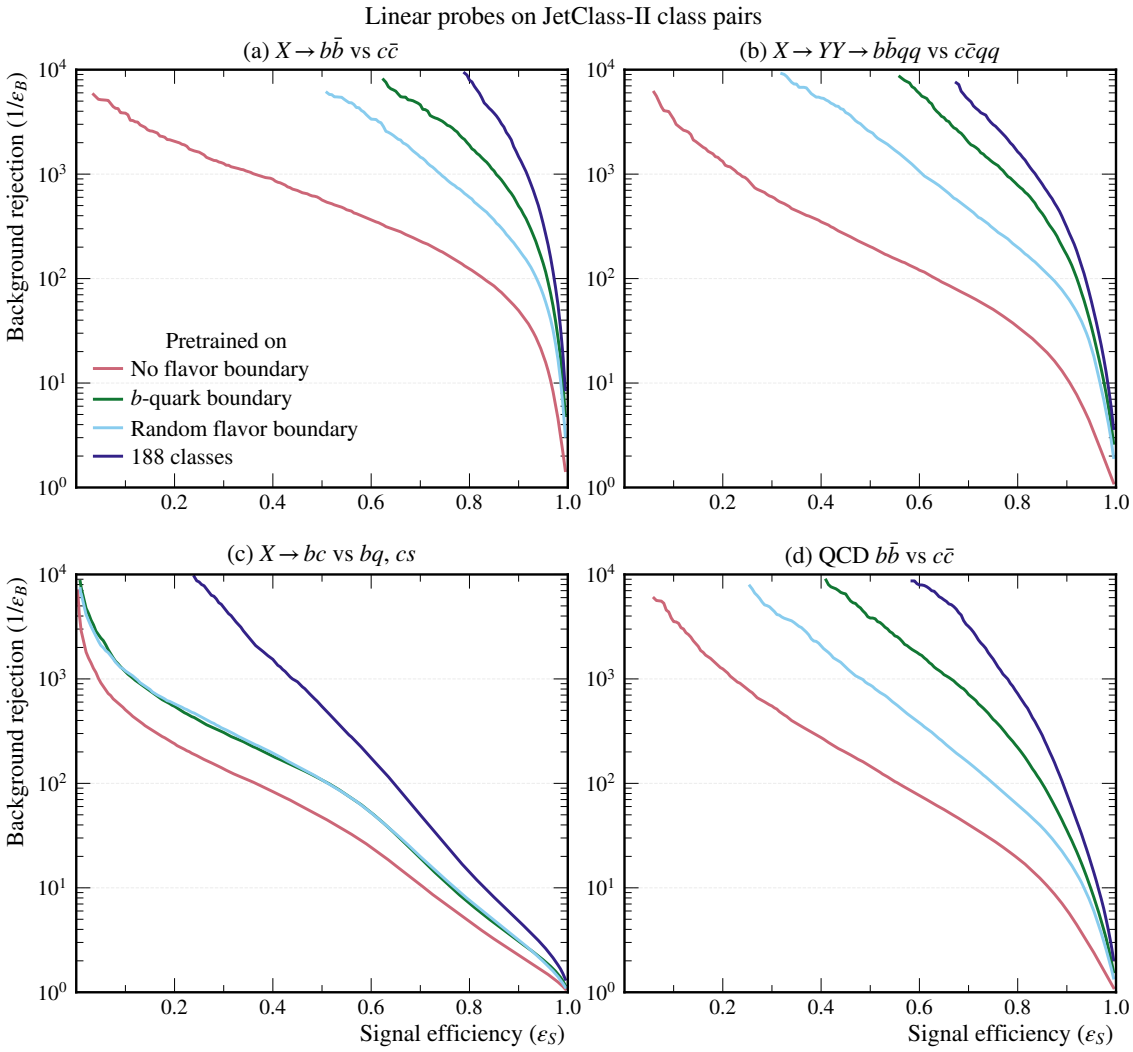


Figure 22: Background rejection against signal efficiency for four pairs of JetClass-II classes that differ only in flavor, (d) of QCD jets, from linear probes on networks pretrained on 17-class random partitions with no flavor boundary, a b-quark boundary or a random flavor boundary, and on 188 classes for reference. The first class named is the signal. Curves are means over two runs (188 classes: three) and end where fewer than 20 background jets remain.

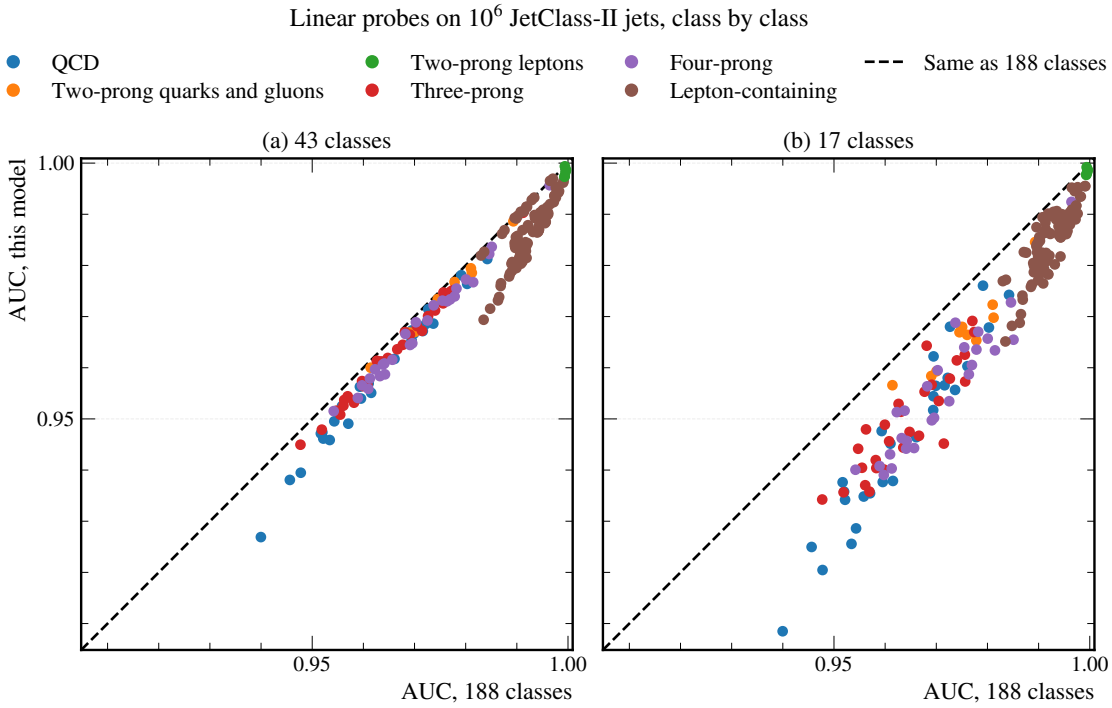


Figure 23: Each of the 188 classes of class-balanced JetClass-II, from linear probes trained on one million jets: the class's area under the curve for the 43- and 17-class label sets, against the same for the 188-class model. On the dashed line the two agree; below it the class does worse than with 188 classes. Colors are decay families. Values are means over runs.

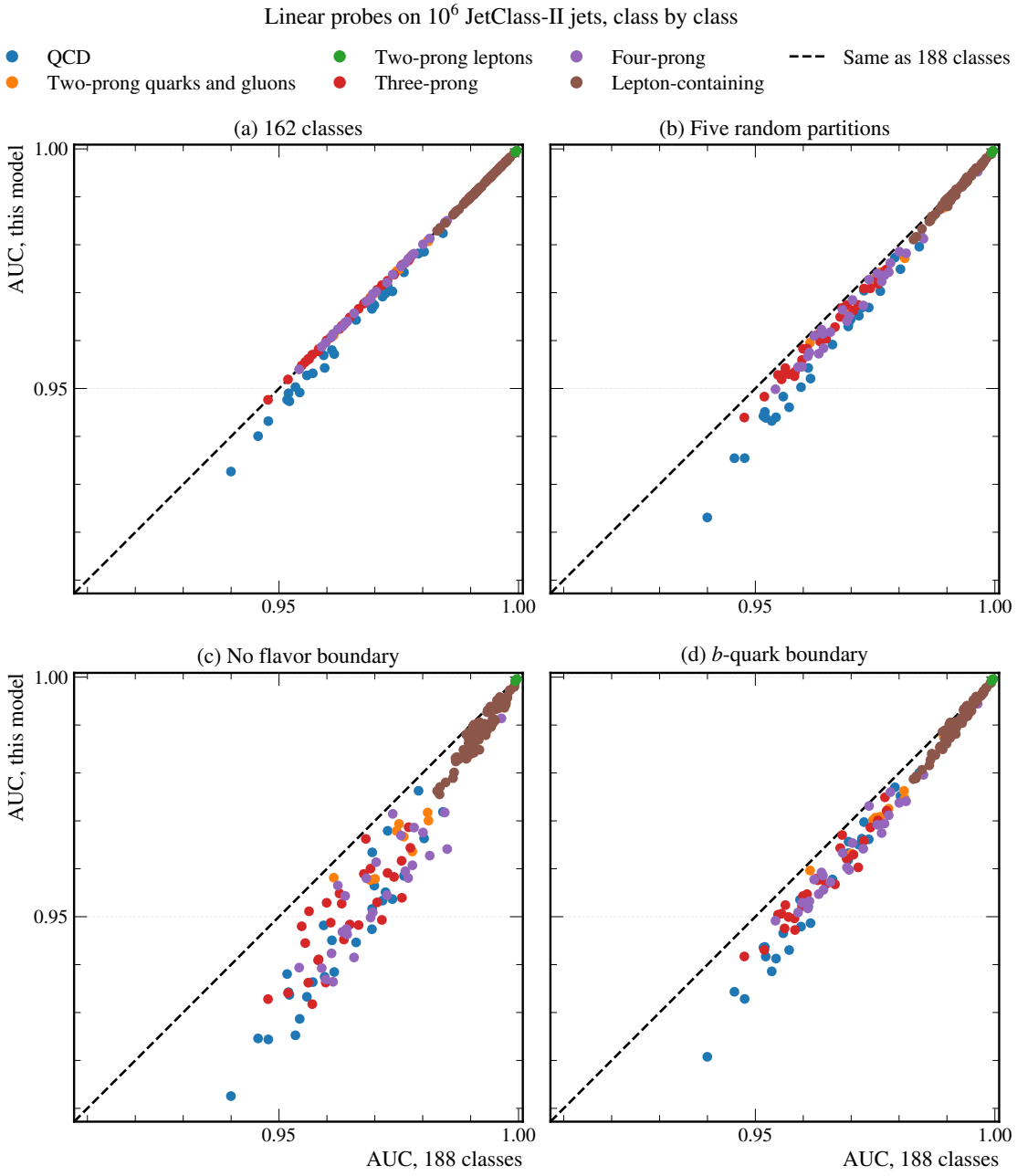


Figure 24: Each class of class-balanced JetClass-II, from linear probes trained on one million jets: its area under the curve for (a) the 162-class label set, (b) the five plain random partitions averaged, (c) the no-flavor-boundary and (d) the b-quark-boundary partitions, against the same for the 188-class model. Below the dashed line the class does worse than with 188 classes. Colors are decay families; values are means over runs.

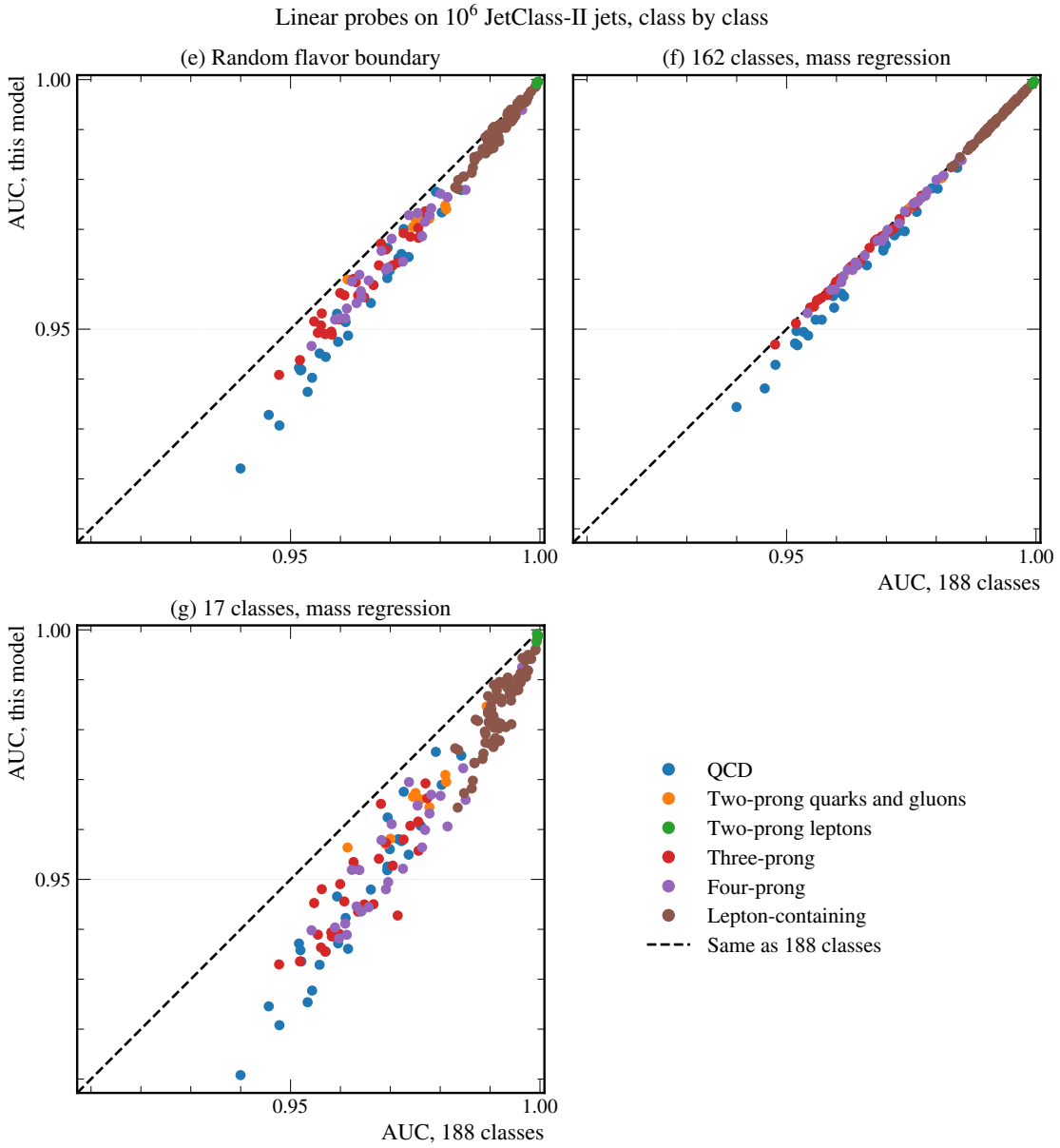


Figure 25: Continued from the previous figure: each class's area under the curve for (e) the random-flavor-boundary partition and the (f) 162- and (g) 17-class label sets pretrained with an auxiliary regression of the jet mass, against the same for the 188-class model, from linear probes trained on one million jets on class-balanced JetClass-II. Colors are decay families; values are means over runs.

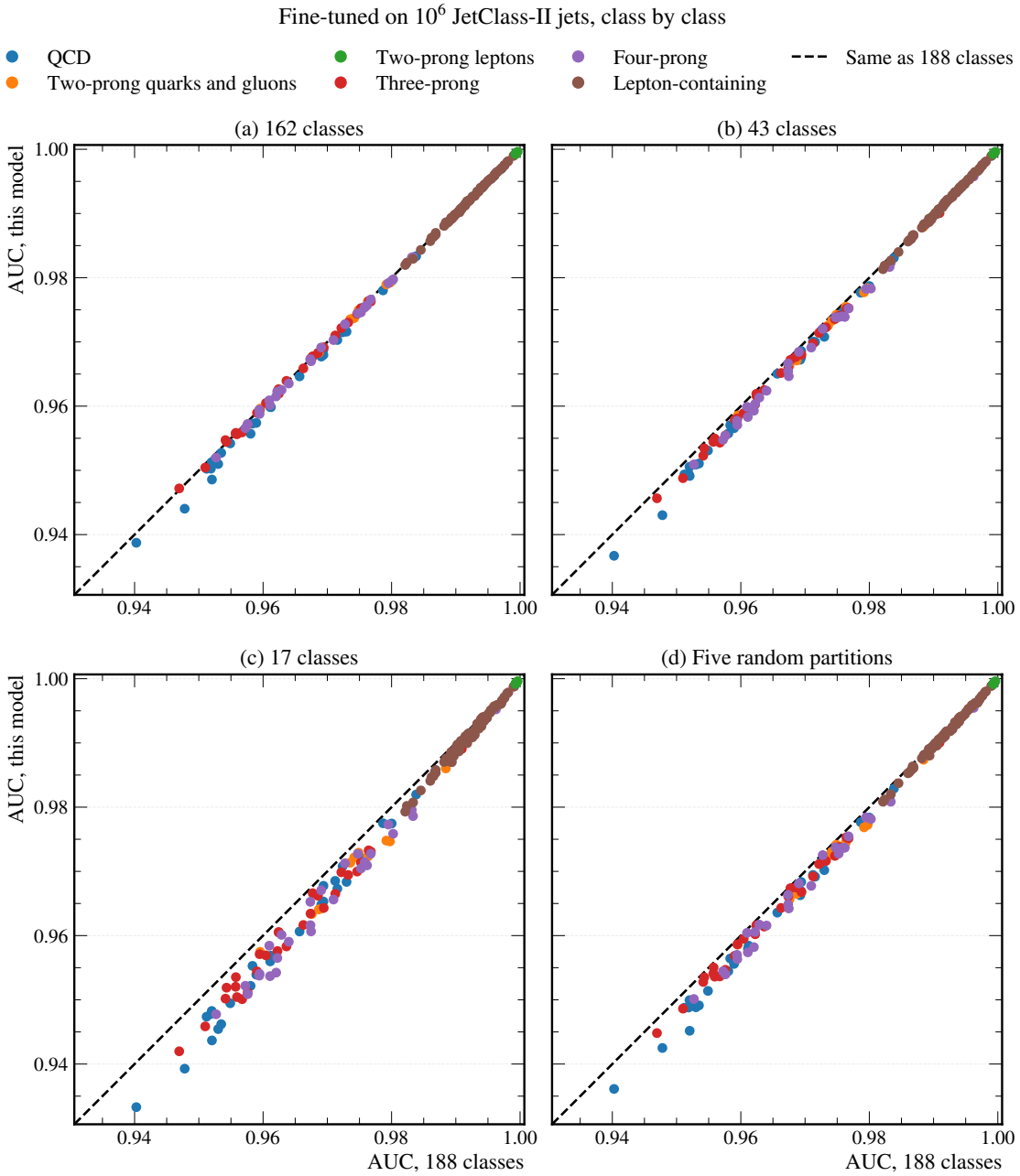


Figure 26: Each class of class-balanced JetClass-II after fine-tuning on one million jets: its area under the curve for the (a) 162-, (b) 43- and (c) 17-class label sets and (d) the five plain random partitions averaged, against the same for the 188-class model. Below the dashed line the class does worse than with 188 classes. Colors are decay families; values are means over three runs, in (d) over the five partitions, one run each.

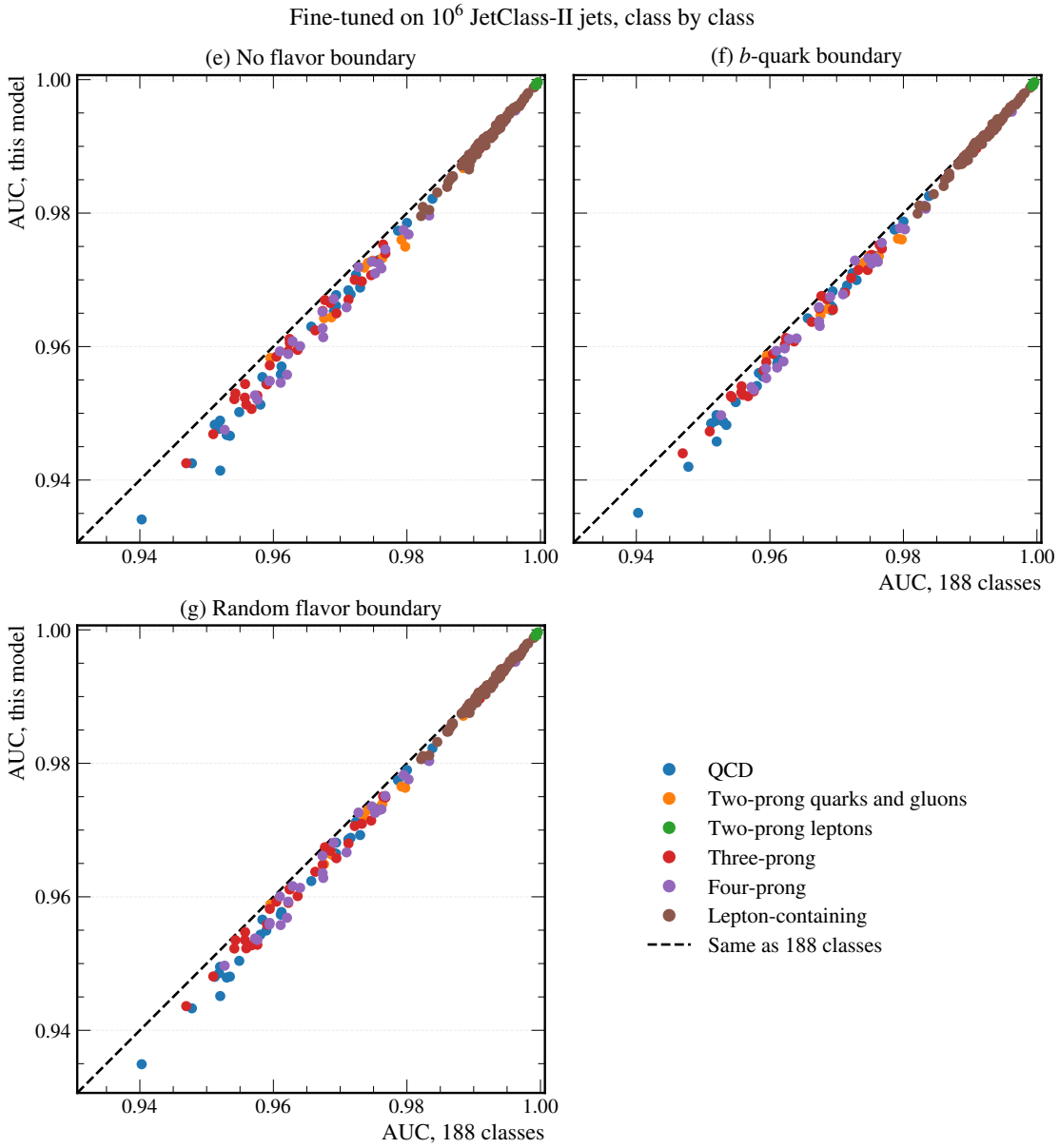


Figure 27: Continued from the previous figure: each class's area under the curve after fine-tuning on one million class-balanced JetClass-II jets, for the partitions with (e) no flavor boundary, (f) a b-quark boundary and (g) a random flavor boundary, against the same for the 188-class model. Colors are decay families; values are means over two runs.

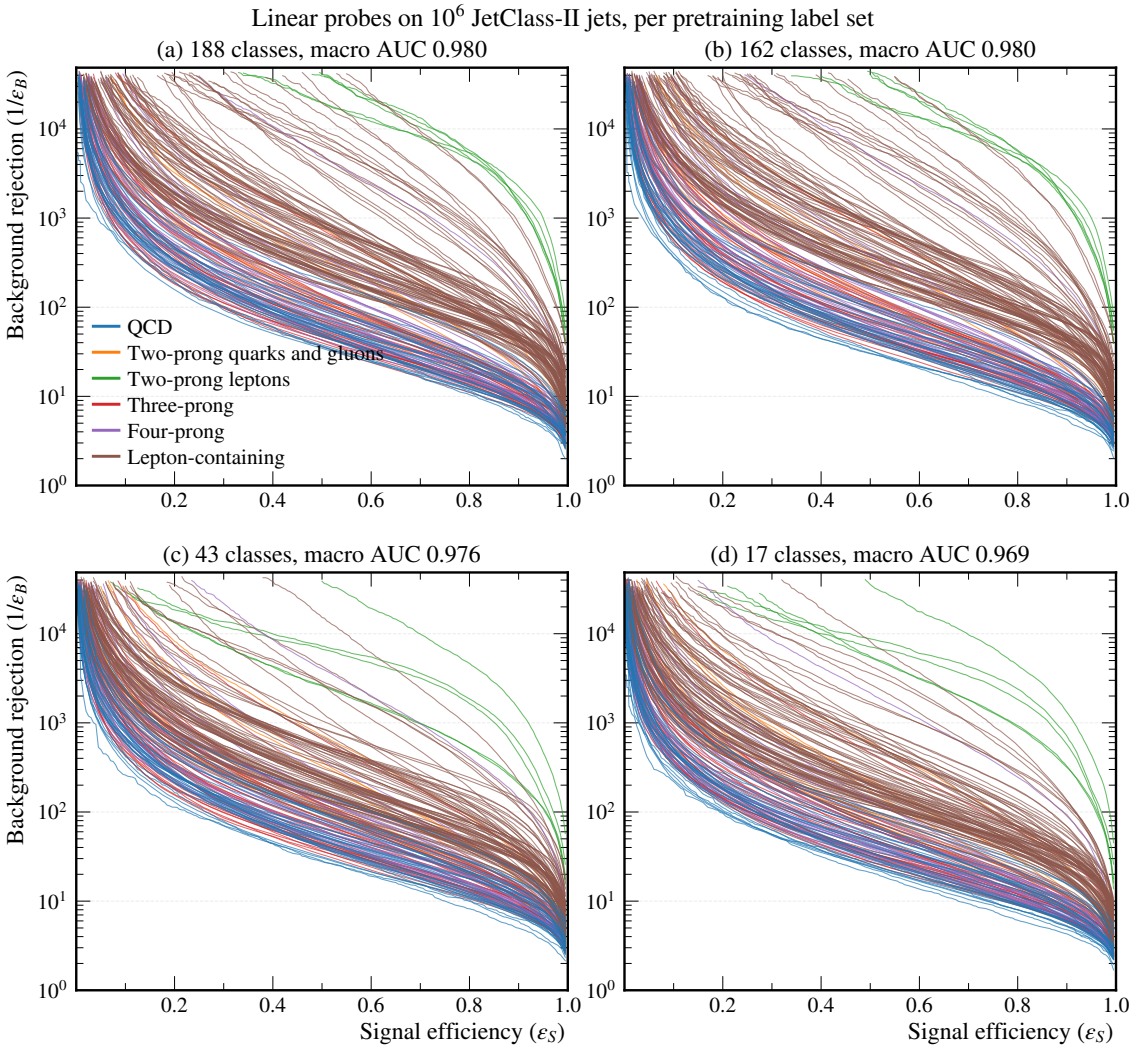


Figure 28: Background rejection against signal efficiency for each of the 188 JetClass-II classes against the rest of the class-balanced test set, from linear probes on frozen networks pretrained with 188, 162, 43 and 17 classes. Thin curves are means over three pretraining runs, colored by decay family, and end where fewer than 20 background jets remain. Each title gives the mean area under the curve over all classes.

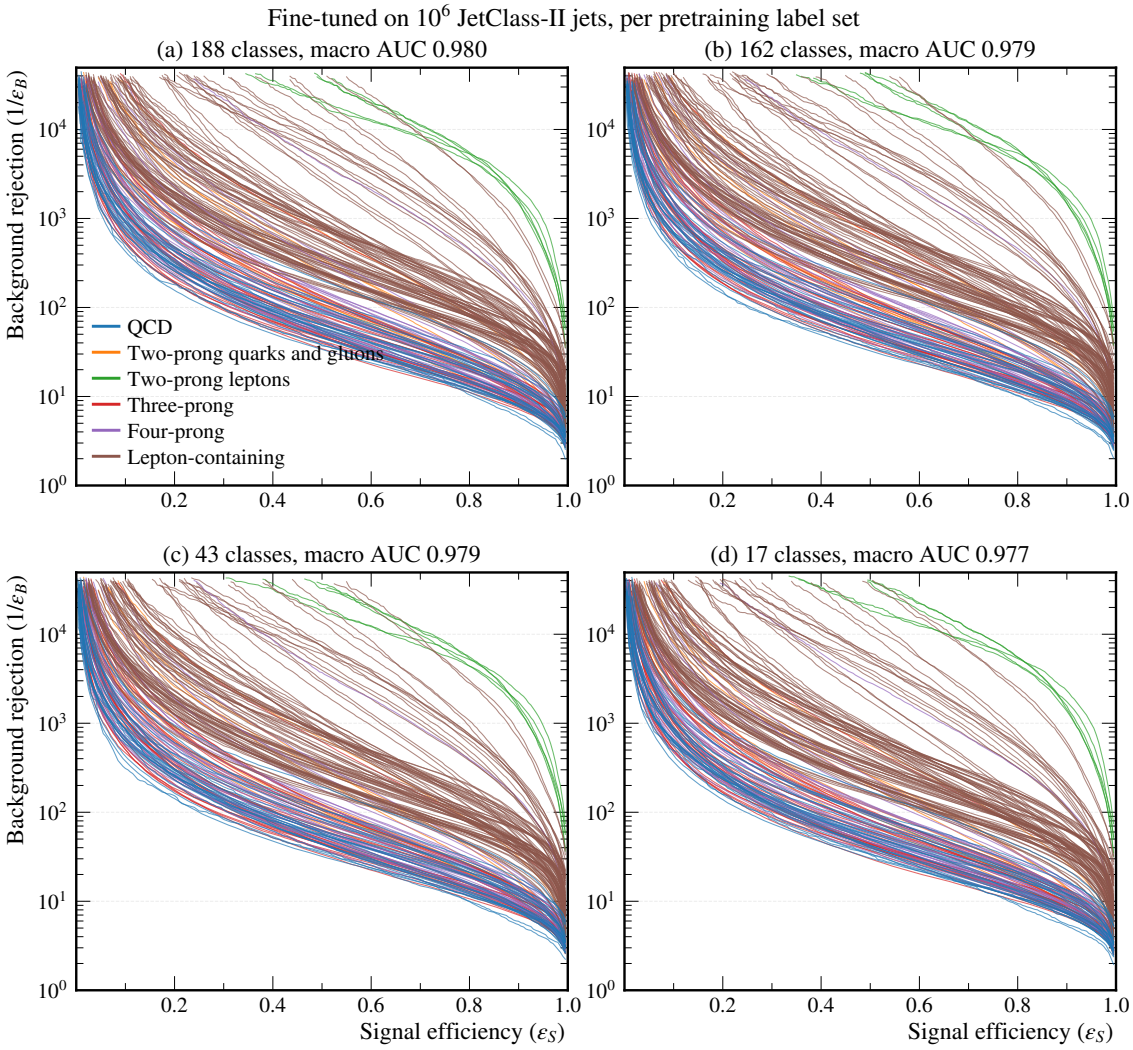


Figure 29: Background rejection against signal efficiency for each of the 188 JetClass-II classes against the rest of the class-balanced test set, after fine-tuning on one million jets, for networks pretrained with 188, 162, 43 and 17 classes. Thin curves are means over runs, colored by decay family, and end where fewer than 20 background jets remain. Each title gives the mean area under the curve over all classes.

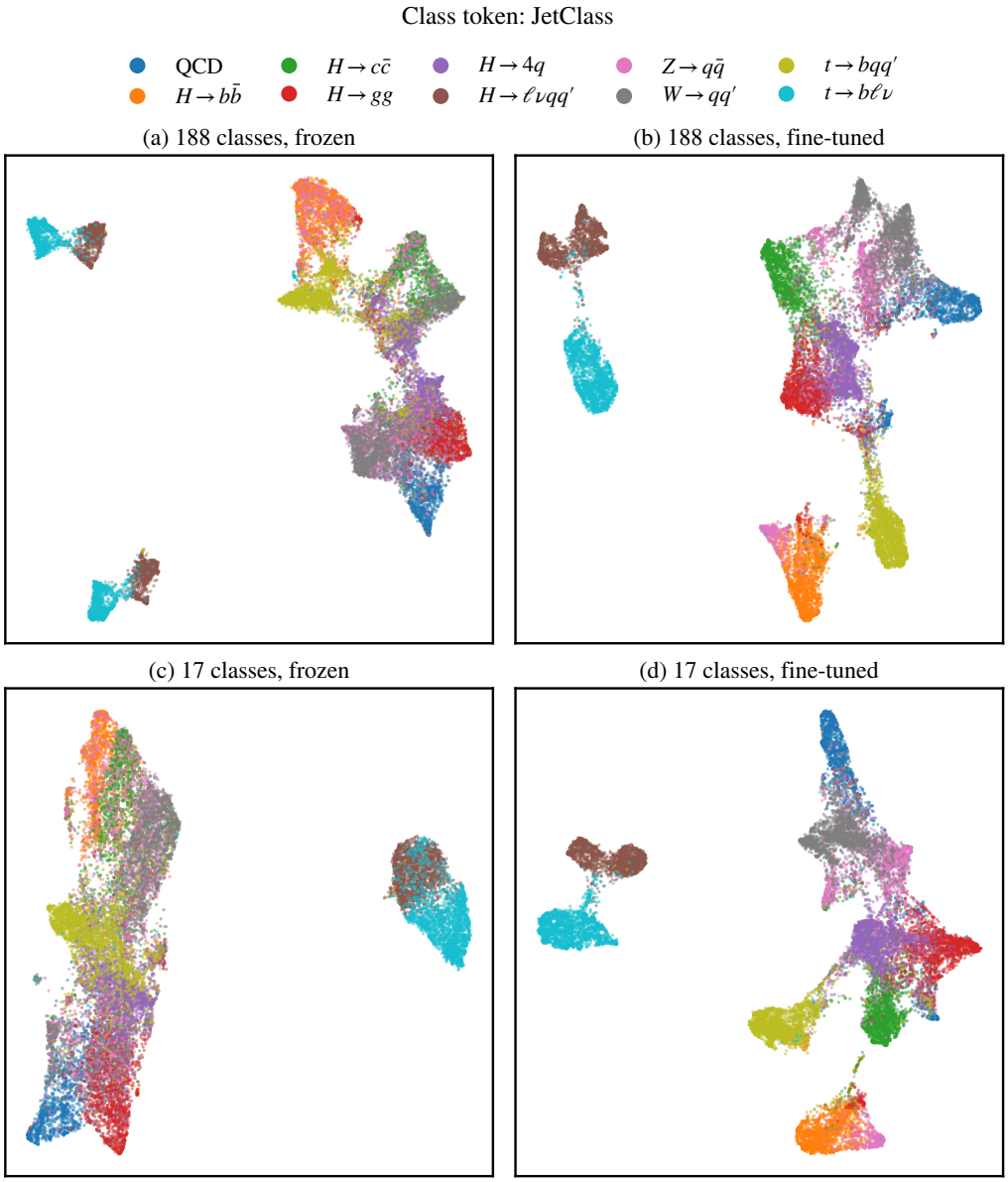


Figure 30: UMAP of the class token of a network pretrained on 188 classes (upper) and one on 17 (lower), frozen (left) and fine-tuned on one million JetClass jets (right), for 2,000 test jets of each JetClass class in every panel; the axes carry no scale. Both models' frozen features set the leptonic decays apart, in two islands for 188 classes; fine-tuning separates most hadronic classes, $Z \to q\bar{q}$ partly mixed with $H \to b\bar{b}$ and $W \to qq'$.

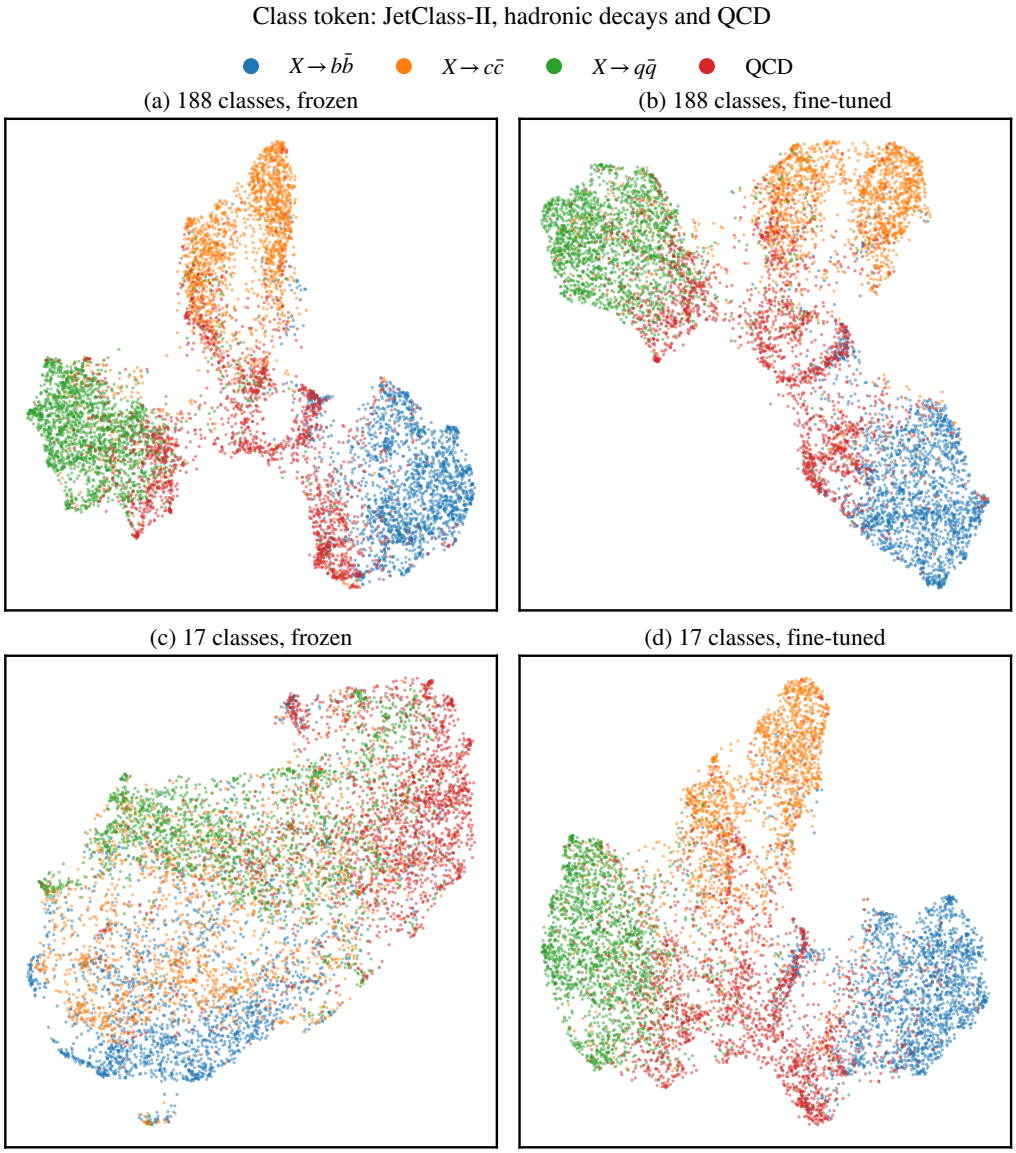


Figure 31: UMAP of the class token of a network pretrained on 188 classes (upper) and one on 17 (lower), frozen (left) and fine-tuned on one million class-balanced JetClass-II jets (right), for 2,000 test jets each of $X \to b\bar{b}$, $X \to c\bar{c}$, $X \to q\bar{q}$ and QCD in every panel; the axes carry no scale. The 188-class frozen features separate the three decays; the 17-class ones only order them, $X \to b\bar{b}$ at one edge and $X \to q\bar{q}$ near QCD at the other; fine-tuning separates them.

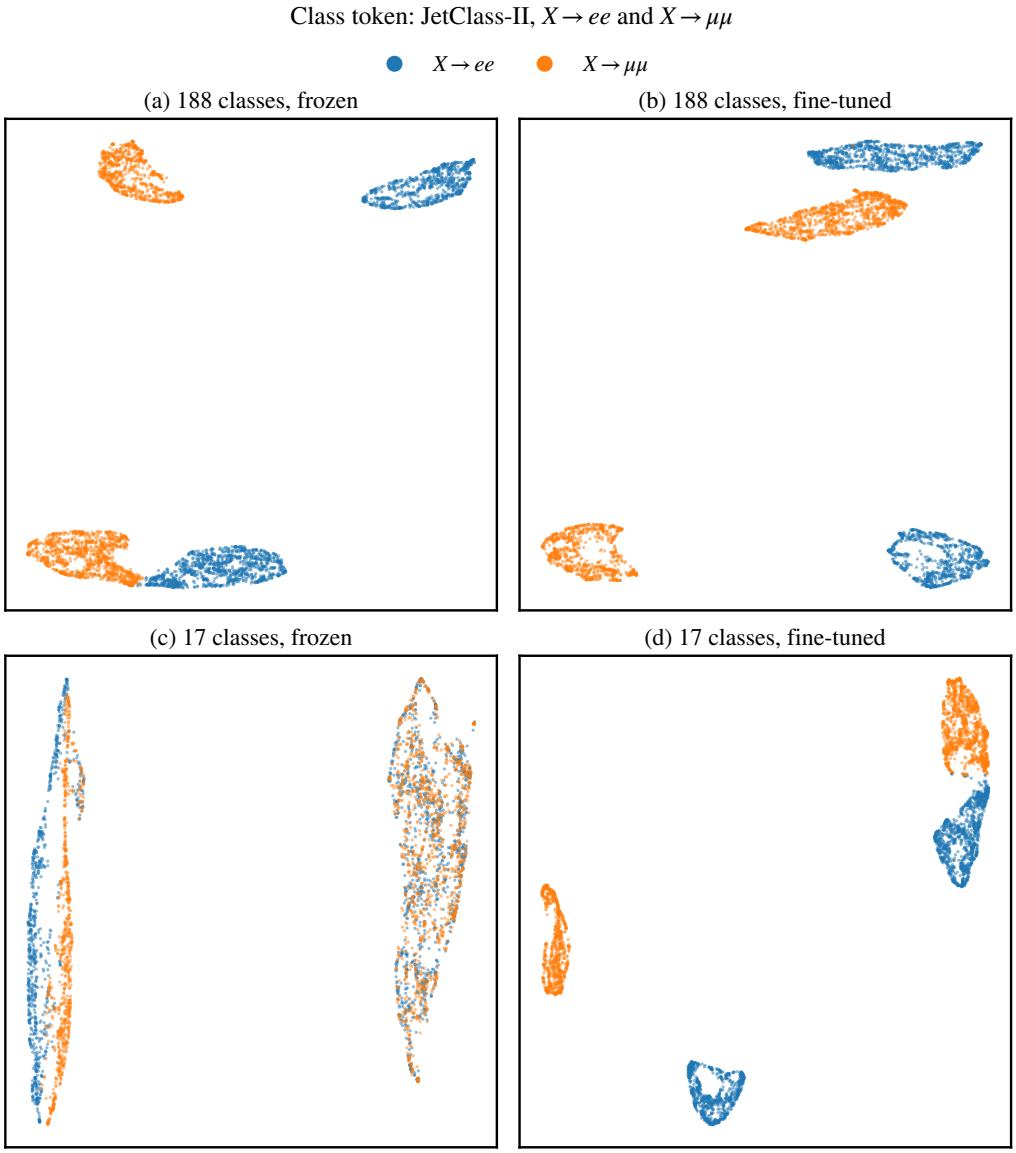


Figure 32: UMAP of the class token of a network pretrained on 188 classes (upper) and one on 17 (lower), frozen (left) and fine-tuned on one million class-balanced JetClass-II jets (right), for 2,000 test jets each of $X \to ee$ and $X \to \mu\mu$ in every panel; the axes carry no scale. Each class falls in two islands in every panel; the 188-class frozen features keep electrons and muons apart, the 17-class ones mix them, and fine-tuning separates them.

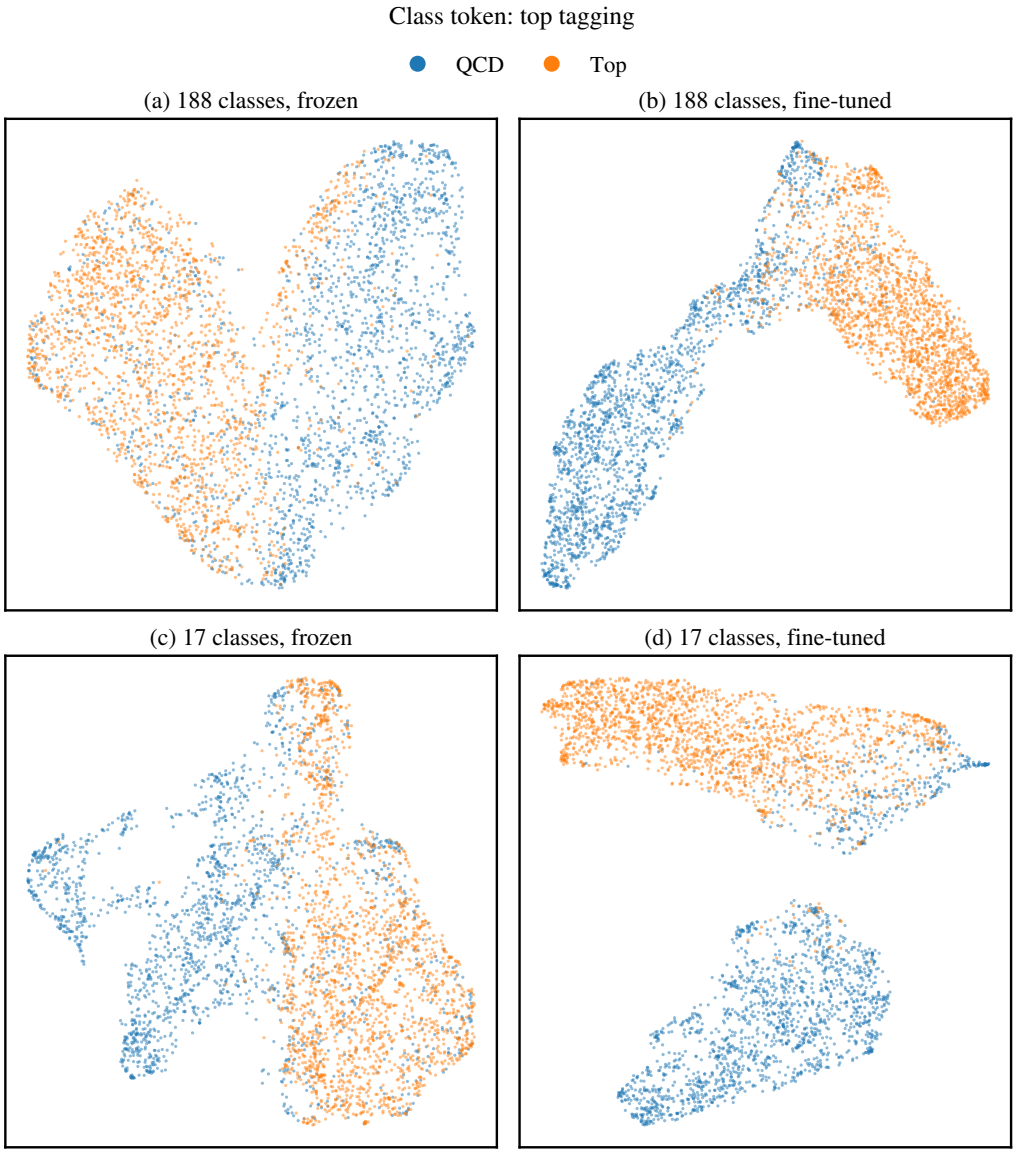


Figure 33: UMAP of the class token of a network pretrained on 188 classes (upper) and one on 17 (lower), frozen (left) and fine-tuned on the full top-tagging training set (right), for 2,000 top and 2,000 QCD test jets in every panel; the axes carry no scale. Top and QCD jets lie mostly apart in both models' frozen features; fine-tuned, the 17-class model gives two islands, some QCD jets in the top one; the 188-class model joins them by mixed jets.

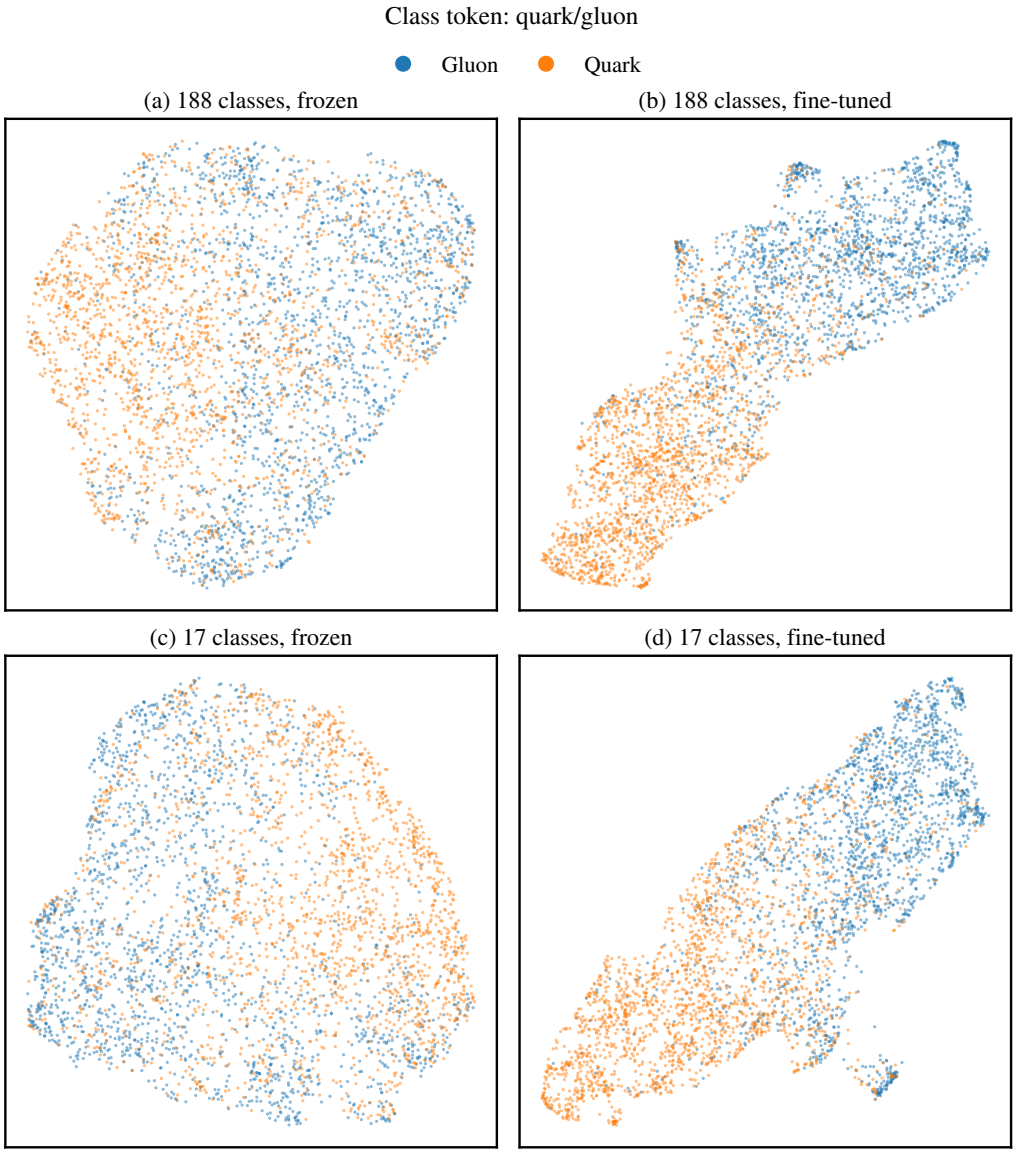


Figure 34: UMAP of the class token of a network pretrained on 188 classes (upper) and one on 17 (lower), frozen (left) and fine-tuned on the full quark/gluon training set (right), for 2,000 quark and 2,000 gluon test jets (Pythia) in every panel; the axes carry no scale. Quark and gluon jets overlap in the frozen features of both models, with only a gradient between them, and separate partly after fine-tuning.

Table 24: Jet-mass regression in detail, over all test jets and over resonant and QCD jets. Upper part: resolution and median response of the regression output of the models pretrained with the mass regression, and of ridge and multilayer perceptron regressions on frozen features. Lower parts: resolution and median response of the soft-drop mass and of the regression output against jet p_T and against the generator soft-drop mass; every bin holds at least 100 jets. Mean ± standard deviation over the three pretraining runs; the models with the mass regression have two runs, and both values are given.

		resolution
estimator	pretraining	all
regression output	162 classes, mass regression	0.1471, 0.1463
regression output	17 classes, mass regression	0.1450, 0.1461
ridge, frozen features	randomly initialized network	0.312 ± 0.007
ridge, frozen features	188 classes	0.3211 ± 0.0034
ridge, frozen features	162 classes	0.316 ± 0.009
ridge, frozen features	43 classes	0.304 ± 0.016
ridge, frozen features	17 classes	0.293 ± 0.012
ridge, frozen features	162 classes, mass regression	0.1885, 0.1922
ridge, frozen features	17 classes, mass regression	0.1842, 0.1824
MLP, frozen features	randomly initialized network	0.2421 ± 0.0018
MLP, frozen features	188 classes	0.2200 ± 0.0014
MLP, frozen features	162 classes	0.2195 ± 0.0017
MLP, frozen features	43 classes	0.2158 ± 0.0029
MLP, frozen features	17 classes	0.2098 ± 0.0031
MLP, frozen features	162 classes, mass regression	0.1591, 0.1586
MLP, frozen features	17 classes, mass regression	0.1583, 0.1588

		resolution	
estimator	pretraining	resonant	QCD
regression output	162 classes, mass regression	0.1332, 0.1320	0.2263, 0.2256
regression output	17 classes, mass regression	0.1312, 0.1320	0.2239, 0.2240
ridge, frozen features	randomly initialized network	0.286 ± 0.006	0.913 ± 0.031
ridge, frozen features	188 classes	0.2921 ± 0.0026	0.559 ± 0.013
ridge, frozen features	162 classes	0.288 ± 0.008	0.548 ± 0.021
ridge, frozen features	43 classes	0.280 ± 0.013	0.49 ± 0.06
ridge, frozen features	17 classes	0.271 ± 0.012	0.448 ± 0.030
ridge, frozen features	162 classes, mass regression	0.1751, 0.1785	0.2674, 0.2701
ridge, frozen features	17 classes, mass regression	0.1696, 0.1687	0.2630, 0.2601
MLP, frozen features	randomly initialized network	0.2214 ± 0.0017	0.502 ± 0.014
MLP, frozen features	188 classes	0.1990 ± 0.0016	0.3487 ± 0.0029
MLP, frozen features	162 classes	0.1989 ± 0.0020	0.3425 ± 0.0031
MLP, frozen features	43 classes	0.1949 ± 0.0023	0.338 ± 0.007
MLP, frozen features	17 classes	0.1894 ± 0.0035	0.327 ± 0.006
MLP, frozen features	162 classes, mass regression	0.1436, 0.1430	0.2473, 0.2452
MLP, frozen features	17 classes, mass regression	0.1420, 0.1431	0.2465, 0.2463

		median response
estimator	pretraining	all
regression output	162 classes, mass regression	0.995, 1.003
regression output	17 classes, mass regression	1.005, 1.005
ridge, frozen features	randomly initialized network	0.9406 ± 0.0035
ridge, frozen features	188 classes	0.9641 ± 0.0012
ridge, frozen features	162 classes	0.9655 ± 0.0005
ridge, frozen features	43 classes	0.968 ± 0.005
ridge, frozen features	17 classes	0.973 ± 0.005
ridge, frozen features	162 classes, mass regression	0.9998, 0.9998
ridge, frozen features	17 classes, mass regression	1.002, 1.002
MLP, frozen features	randomly initialized network	0.983 ± 0.005
MLP, frozen features	188 classes	0.991 ± 0.005
MLP, frozen features	162 classes	0.9876 ± 0.0014
MLP, frozen features	43 classes	0.991 ± 0.005
MLP, frozen features	17 classes	0.991 ± 0.004
MLP, frozen features	162 classes, mass regression	1.004, 1.006
MLP, frozen features	17 classes, mass regression	1.009, 1.006

		median response	
estimator	pretraining	resonant	QCD
regression output	162 classes, mass regression	0.996, 1.004	0.9897, 0.9996
regression output	17 classes, mass regression	1.006, 1.006	1.001, 1.000
ridge, frozen features	randomly initialized network	0.936 ± 0.004	0.9649 ± 0.0035
ridge, frozen features	188 classes	0.9754 ± 0.0006	0.905 ± 0.006
ridge, frozen features	162 classes	0.9768 ± 0.0008	0.909 ± 0.007
ridge, frozen features	43 classes	0.977 ± 0.004	0.922 ± 0.012
ridge, frozen features	17 classes	0.980 ± 0.004	0.937 ± 0.010
ridge, frozen features	162 classes, mass regression	1.003, 1.003	0.9826, 0.9830
ridge, frozen features	17 classes, mass regression	1.005, 1.003	0.9888, 0.9935
MLP, frozen features	randomly initialized network	0.984 ± 0.005	0.980 ± 0.007
MLP, frozen features	188 classes	0.994 ± 0.004	0.975 ± 0.007
MLP, frozen features	162 classes	0.9900 ± 0.0024	0.974 ± 0.006
MLP, frozen features	43 classes	0.993 ± 0.004	0.976 ± 0.010
MLP, frozen features	17 classes	0.993 ± 0.004	0.977 ± 0.008
MLP, frozen features	162 classes, mass regression	1.006, 1.008	0.9883, 0.9930
MLP, frozen features	17 classes, mass regression	1.012, 1.008	0.9942, 0.9932

resolution		jet p_{T} [GeV]	
estimator	jets	200–400	400–700
soft-drop mass	all	0.242	0.225
162 classes, mass regression	all	0.1720, 0.1703	0.1483, 0.1468
17 classes, mass regression	all	0.1691, 0.1702	0.1455, 0.1470
soft-drop mass	resonant	0.229	0.215
162 classes, mass regression	resonant	0.1642, 0.1621	0.1412, 0.1393
17 classes, mass regression	resonant	0.1610, 0.1620	0.1382, 0.1396
soft-drop mass	QCD	0.279	0.333
162 classes, mass regression	QCD	0.1945, 0.1928	0.1854, 0.1846
17 classes, mass regression	QCD	0.1918, 0.1947	0.1816, 0.1830

resolution		jet p_T [GeV]	
estimator	jets	700–1000	1000–2500
soft-drop mass	all	0.231	0.327
162 classes, mass regression	all	0.1426, 0.1411	0.1396, 0.1395
17 classes, mass regression	all	0.1403, 0.1414	0.1383, 0.1391
soft-drop mass	resonant	0.216	0.211
162 classes, mass regression	resonant	0.1337, 0.1315	0.1201, 0.1201
17 classes, mass regression	resonant	0.1313, 0.1320	0.1188, 0.1200
soft-drop mass	QCD	1.10	3.61
162 classes, mass regression	QCD	0.2146, 0.2110	0.2801, 0.2831
17 classes, mass regression	QCD	0.2089, 0.2090	0.2800, 0.2775

median response		jet p_T [GeV]	
estimator	jets	200–400	400–700
soft-drop mass	all	0.893	0.914
162 classes, mass regression	all	0.986, 0.997	0.994, 1.003
17 classes, mass regression	all	0.9930, 0.9974	1.001, 1.003
soft-drop mass	resonant	0.904	0.917
162 classes, mass regression	resonant	0.987, 0.998	0.995, 1.004
17 classes, mass regression	resonant	0.9933, 0.9980	1.002, 1.004
soft-drop mass	QCD	0.863	0.895
162 classes, mass regression	QCD	0.984, 0.995	0.986, 0.998
17 classes, mass regression	QCD	0.9921, 0.9955	0.9965, 0.9973

median response		jet p_T [GeV]	
estimator	jets	700–1000	1000–2500
soft-drop mass	all	0.917	0.953
162 classes, mass regression	all	0.997, 1.005	0.997, 1.004
17 classes, mass regression	all	1.006, 1.007	1.010, 1.007
soft-drop mass	resonant	0.915	0.947
162 classes, mass regression	resonant	0.998, 1.005	0.998, 1.004
17 classes, mass regression	resonant	1.006, 1.007	1.010, 1.008
soft-drop mass	QCD	0.931	1.02
162 classes, mass regression	QCD	0.994, 1.003	0.993, 1.001
17 classes, mass regression	QCD	1.004, 1.004	1.008, 1.002

resolution		generator soft-drop mass [GeV]
estimator	jets	20–50
soft-drop mass	all	0.281
162 classes, mass regression	all	0.1751, 0.1756
17 classes, mass regression	all	0.1755, 0.1740
soft-drop mass	resonant	0.275
162 classes, mass regression	resonant	0.1632, 0.1634
17 classes, mass regression	resonant	0.1631, 0.1621
soft-drop mass	QCD	0.292
162 classes, mass regression	QCD	0.2033, 0.2030
17 classes, mass regression	QCD	0.2032, 0.2006

resolution		generator soft-drop mass [GeV]	
estimator	jets	50–100	100–150
soft-drop mass	all	0.206	0.204
162 classes, mass regression	all	0.1425, 0.1416	0.1302, 0.1291
17 classes, mass regression	all	0.1418, 0.1411	0.1284, 0.1288
soft-drop mass	resonant	0.194	0.192
162 classes, mass regression	resonant	0.1366, 0.1359	0.1261, 0.1252
17 classes, mass regression	resonant	0.1357, 0.1351	0.1246, 0.1250
soft-drop mass	QCD	0.239	0.246
162 classes, mass regression	QCD	0.1653, 0.1639	0.1508, 0.1515
17 classes, mass regression	QCD	0.1630, 0.1631	0.1505, 0.1514

resolution		generator soft-drop mass [GeV]	
estimator	jets	150–250	250–500
soft-drop mass	all	0.210	0.225
162 classes, mass regression	all	0.1233, 0.1213	0.1163, 0.1135
17 classes, mass regression	all	0.1210, 0.1220	0.1134, 0.1145
soft-drop mass	resonant	0.201	0.216
162 classes, mass regression	resonant	0.1198, 0.1175	0.1133, 0.1107
17 classes, mass regression	resonant	0.1174, 0.1183	0.1107, 0.1118
soft-drop mass	QCD	0.265	0.278
162 classes, mass regression	QCD	0.1515, 0.1509	0.1511, 0.1470
17 classes, mass regression	QCD	0.1483, 0.1515	0.1434, 0.1482

median response		generator soft-drop mass [GeV]		
estimator	jets	20–50	50–100	100–150
soft-drop mass	all	1.00	0.936	0.908
162 classes, mass regression	all	1.025, 1.037	1.006, 1.014	0.993, 1.001
17 classes, mass regression	all	1.032, 1.041	1.014, 1.016	1.002, 1.002
soft-drop mass	resonant	1.01	0.953	0.922
162 classes, mass regression	resonant	1.032, 1.043	1.012, 1.019	0.997, 1.004
17 classes, mass regression	resonant	1.038, 1.047	1.019, 1.021	1.006, 1.006
soft-drop mass	QCD	0.971	0.864	0.816
162 classes, mass regression	QCD	1.006, 1.019	0.9802, 0.9899	0.9670, 0.9766
17 classes, mass regression	QCD	1.015, 1.021	0.9906, 0.9901	0.9791, 0.9759

median response		generator soft-drop mass [GeV]	
estimator	jets	150–250	250–500
soft-drop mass	all	0.890	0.852
162 classes, mass regression	all	0.9881, 0.9954	0.9710, 0.9772
17 classes, mass regression	all	0.9986, 0.9970	0.9837, 0.9792
soft-drop mass	resonant	0.902	0.862
162 classes, mass regression	resonant	0.9905, 0.9977	0.9722, 0.9783
17 classes, mass regression	resonant	1.001, 0.999	0.9847, 0.9804
soft-drop mass	QCD	0.781	0.738
162 classes, mass regression	QCD	0.9647, 0.9720	0.9565, 0.9631
17 classes, mass regression	QCD	0.9787, 0.9743	0.9704, 0.9641

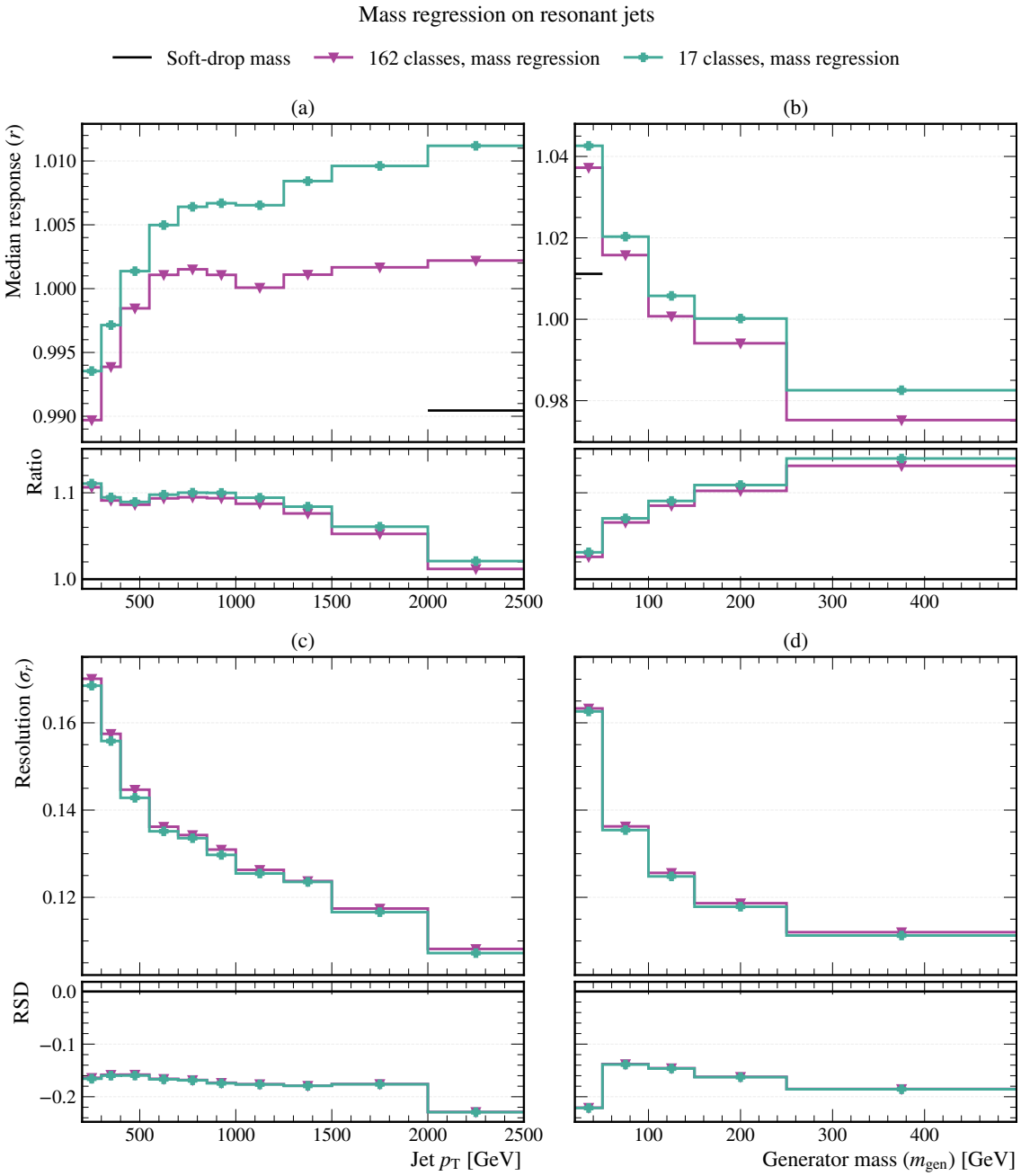


Figure 35: Median mass response r (a, b) and resolution σ_r (c, d), as in figure 11, of resonant JetClass-II test jets against transverse momentum (a, c; all jets) and generator mass (b, d; 20–500 GeV): mass regression of the models pretrained on 162 and 17 classes (two runs). Lower panels: ratio to the soft-drop mass (black where in range) and signed root square difference (RSD), negative where better. Unlabelled axes share the left ones.

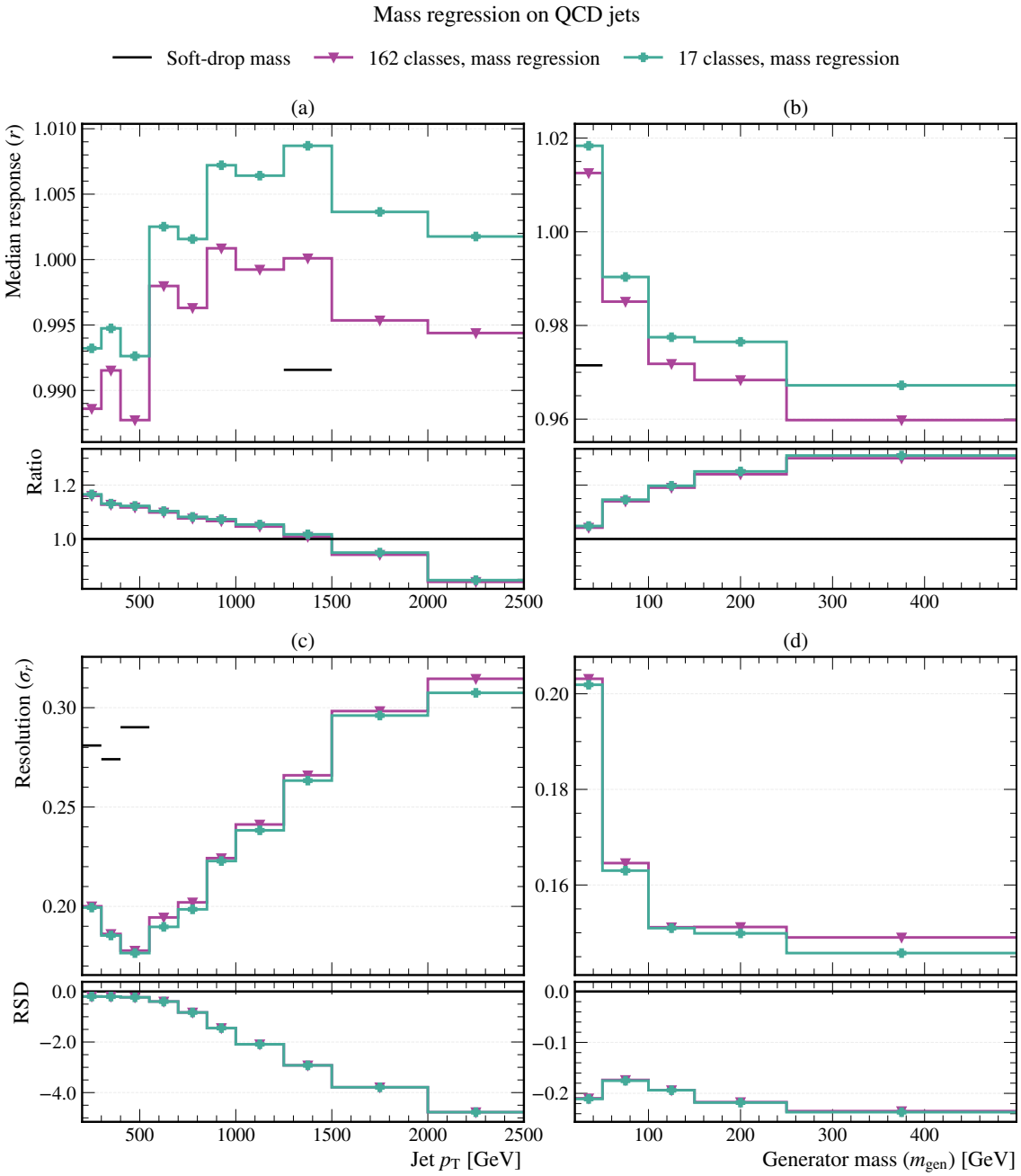


Figure 36: Median mass response r (a, b) and resolution σ_r (c, d), as in figure 11, of QCD JetClass-II test jets against transverse momentum (a, c; all jets) and generator mass (b, d; 20–500 GeV): mass regression of the models pretrained on 162 and 17 classes (two runs). Lower panels: ratio to the soft-drop mass (black where in range) and signed root square difference (RSD), negative where better. Unlabelled axes share the left ones.

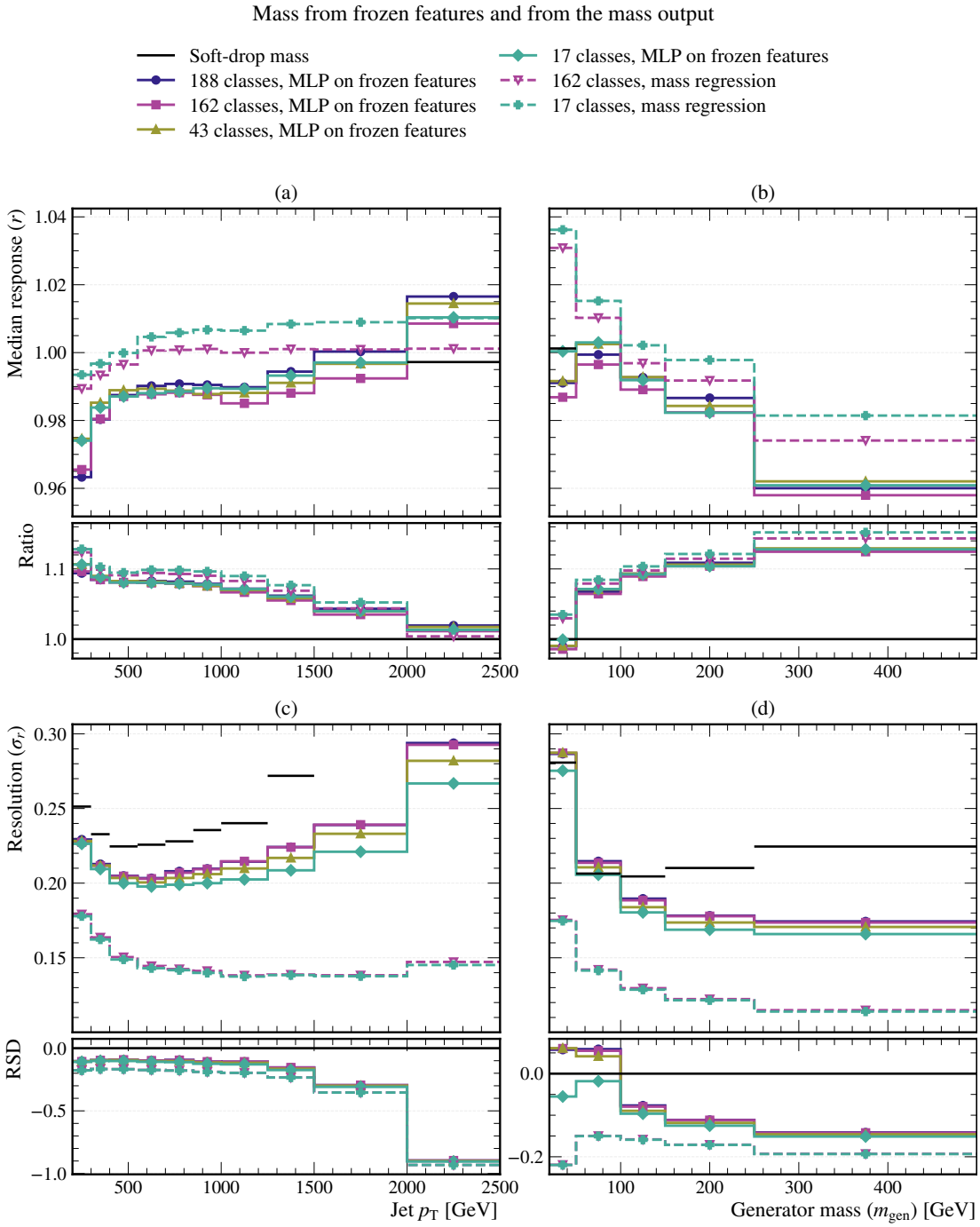


Figure 37: Median mass response r (a, b) and resolution σ_r (c, d) of JetClass-II test jets against transverse momentum (a, c; all jets) and generator mass (b, d; 20–500 GeV): an MLP regressing the mass from the frozen features of the models pretrained on 188, 162, 43 and 17 classes (solid, three runs), and the mass regression of the 162- and 17-class models (dashed, two runs). Sub-panels as in figure 11; overlapping markers agree within their size.

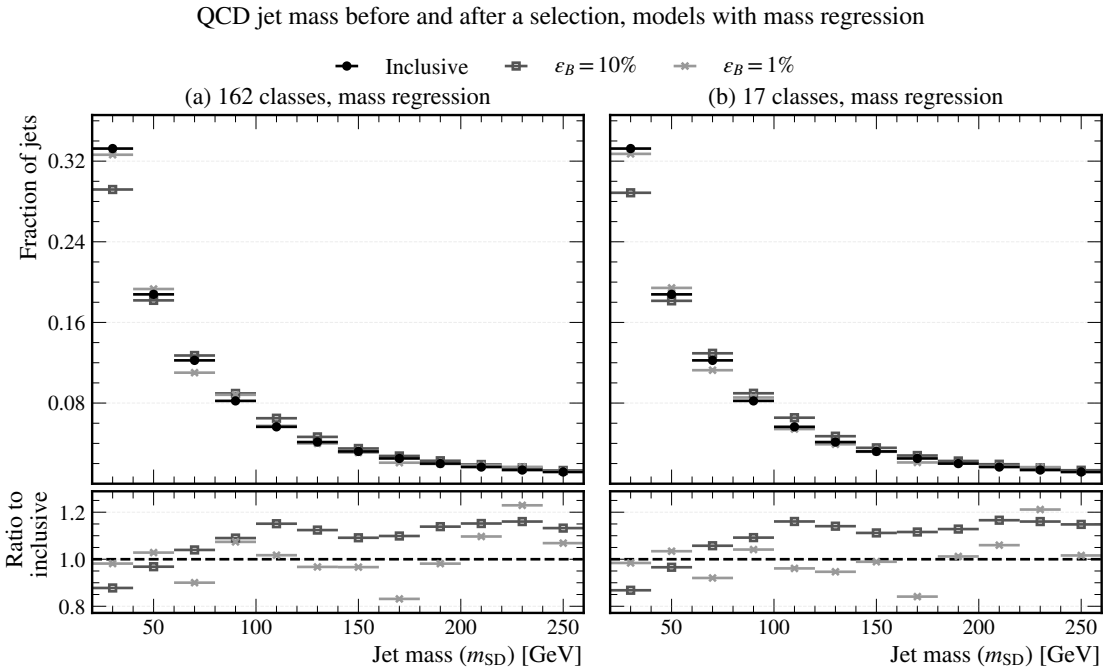


Figure 38: As figure 12, for the models pretrained on 162 and 17 classes with an auxiliary regression of the jet mass: soft-drop mass distribution of QCD test jets of JetClass-II before any selection and after selections on $1 - g_{\mathrm{QCD}}$ at QCD efficiencies ε_B of 10% and 1% (mean of two runs), with the ratio to the inclusive distribution below.

Table 25: Mass sculpting: Jensen–Shannon distance (base 2) $\times 10^2$ between the m_{SD} distributions (24 bins of 20 GeV from 20 to 500 GeV) of all QCD test jets and of those passing a cut on $1 - g_{\mathrm{QCD}}$ at the QCD efficiency ε_B given; lower is less sculpting. The last row is the distance expected for a random selection of as many of the QCD test jets (drawn without replacement). Mean ± standard deviation over the three pretraining runs; the models with the mass regression have two runs, and both values are given.

pretraining	$\varepsilon_B = 10\%$	$\varepsilon_B = 1\%$
188 classes	5.24 ± 0.34	4.39 ± 0.24
162 classes	5.32 ± 0.09	4.29 ± 0.29
43 classes	5.60 ± 0.27	5.1 ± 0.7
17 classes	5.43 ± 0.10	5.61 ± 0.17
162 classes, mass regression	4.22, 4.64	4.24, 4.42
17 classes, mass regression	5.22, 4.23	4.05, 4.52
random selection of as many QCD jets	0.944	3.13